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Tactical Doctrine

EXPEDIENT ROADS AND DRAINAGE METHODS

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This Air Force Tactics, Techniques and Procedures (AFTTP) supports Air Force Instruction (AFI) 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*, and AFI 10-209, *RED HORSE Program*. This publication applies to the Regular Air Force, the Air Force Reserve, and the Air National Guard. This publication does not apply to the United States Space Force. This AFTTP provides Air Force (AF) civil engineer personnel with ideas, options, and methods for developing expedient roads and drainages when standard equipment and materials are not available or when necessary for minimum-essential restoration may be supplemented at any level. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Product*; route DAF Forms 847 from the field through the appropriate functional chain of command. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program,* and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the DAF.

SUMMARY OF CHANGES

This document has been revised and should be completely reviewed. Changes include a publication title change, updated data, terminology, and references, and the removal of expedient protective construction information available in other engineer publications.

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Chapter 1

INTRODUCTION

1.1. General Information. Expedient roads and drainages are often developed in response to immediate needs and usually done quickly and with whatever suitable materials available. Conducting and incorporating a Risk Assessment according to DAFI 90-802, *Risk Management*, should enable engineers to evaluate the best and safest option that conditions allow. At initial contingency locations, roads and drainage systems may be nearly nonexistent or consist primarily of graded dirt surfaces and rudimentary ditches. Earthen roads can usually be improved for increased traffic loads using standard processes very familiar to Air Force civil engineers, such as, aggregate road construction methods, soil stabilization, and expedient surfacing materials. Engineers also use expedient construction to enhance drainage ditches and other structures conveying and diverting stormwater runoff and floodwater away from roadways, airfields, and other critical facilities and infrastructure.

1.2. Overview. This AFTTP addresses expedient methods for roads and drainages when time and materials are limited, or conventional methods are impractical to employ. The expedient methods herein include illustrations of basic material applications and proven field-expedient procedures compiled from multiple sources, including data from Department of War (DOW), Joint, and Service publications as well as best practices and lessons learned from Air Force (AF) Civil Engineers and US Army Corps of Engineers (USACE). Consider using these concepts, methods, and examples singularly, in combination, or modified to suit field conditions for expedient construction. Specific topics include:

1.2.1. Designs for expedient roads and drainage.

1.2.2. Land clearing considerations.

1.2.3. Expedient road surfacing over unstable soils.

1.2.4. Expedient soil classifications and identification.

1.2.5. Soil stabilization for expedient roads.

1.2.6. Expedient drainage methods.

1.3. Designs for Expedient Roads. Generally, expedient roads are either stabilized earth (unpaved) or aggregate-surfaced roads, they are usually short-lived and intended as temporary measures. However, whenever engineers plan and develop expedient roads, they usually consider any future need to construct surfaces that are more durable should the area mature or move towards an enduring location. It may be prudent to design expedient roads that can easily transition from unpaved soil roads to aggregate-surfaced to flexible-pavement surface—this approach is sometimes referred to as stage construction.

1.3.1. Design requirements for both surfaced and unpaved roads are available in Unified Facilities Criteria (UFC) 3-250-01, *Pavement Design for Roads and Parking Areas*. UFC 3-250-01 also provides guidance on using the Pavement-Transportation Computer Assisted Structural Engineering (PCASE) software to determine thickness and compaction requirements for surfaced and unpaved roads. The PCASE software implements Department of War Unified Facility Criteria for the design and evaluation of airfields and roadways. Software is available at: **https://transportation.erdc.dren.mil/triservice/softwarePCASE.aspx**.

1.3.2. Unpaved roads have in-place natural soil or borrow soil as its surface. The condition of the subgrade and other underlying soil layers are important considerations, because any distortions or displacements occurring in the subgrade will be reflected in the base course and continue upward into non-rigid roadway surfaces. Subgrades and base course should be adequate to support the anticipated traffic loads for the time required; in some instances, in-placed soils may need to be compacted and/or stabilized to achieve the desired performance.

1.3.2.1. When the in-place soil is not strong enough for the road design:

1.3.2.1.1. The first preference should be to adjust the moisture content and compact the existing subgrade soil if compaction will achieve sufficient strength.

1.3.2.1.2. The second option would be to compact the in-place soil and then apply an aggregate surface.

1.3.2.1.3. The third would be to mechanically bridge over the poor soil using an expedient surfacing or a combination of geosynthetics and aggregate.

1.3.2.1.4. The last alternative would be to chemically stabilize the in-place soil using mix-in-place techniques with chemical additives such as lime or cement.

1.3.2.2. If the soil is stable, the most practical and used type of expedient road surface consists of the compacted aggregate.

1.3.2.3. Subgrades can be mechanically stabilized by adding granular materials; chemically stabilized by adding chemical admixtures (i.e., lime, Portland cement, and fly ash); or mechanically reinforced using sand-grid, matting, or geosynthetics. Chemical stabilization is generally costly but may prove to be economically feasible, depending on the availability of the chemical stabilization agent in comparison with the availability of granular material.

1.3.2.4. Consider the factors below when determining suitability of a subgrade:

1.3.2.4.1. General characteristics of the subgrade soils.

1.3.2.4.2. Depth to the water table.

1.3.2.4.3. Compaction that can be attained in the subgrade.

1.3.2.4.4. California Bearing Ratio (CBR) values of un-compacted and compacted subgrades.

1.3.2.4.5. Weak or soft layers or organics in the subsoil.

1.3.2.4.6. Detrimental frost action or excessive swell susceptibility.

1.3.3. Aggregate-surfaced roads are similar in design to unpaved roads. In aggregate-surfaced roads, layers of high-quality material cover the natural subgrade to improve its strength (**Figure 1.1**.). Materials used in aggregate roads should have greater strength than the subgrade and be placed so that the higher-quality material is on top of the lower quality material.

Figure 1.1. Placing Rock Aggregate to Improve Roadway.



1.3.3.1. The thickness design of aggregate surfaced roads is similar to the design of flexible pavement as described in UFC 3-250-01. The procedure involves determining the vehicle mix or traffic, subgrade CBR, and using the unpaved design criteria. This method determines total thickness of material to be placed above the subgrade, as well as its strength relative to the CBR value.

1.3.3.2. For non-frost areas, the PCASE computed design thickness may be constructed of compacted granular fill for the total depth over the natural subgrade or in a layered system of granular fill (including subbases) and compacted subgrade for the same total depth. The layered section should be checked to ensure that an adequate thickness of material is used to protect the underlying layer and if it also meets the minimum surface CBR required. For specific road design requirements related to freeze mitigation, unpaved and surfaced roads, and flexible and rigid pavements, see UFC 3-250-01, UFC 3-250-03, *Standard Practice Manual for Flexible Pavements*, and UFC 3-250-04, *Standard Practice for Concrete Pavement Construction*.

1.4. Designs for Expedient Drainage. Drainage for expedient construction is just as important as it is for standard construction operations. Since most expedient structures have a short design life, be sure to consider seasonal factors (inclement weather, flooding, etc.) when designing or examining the need for drainage. Consider the information in the following paragraphs when designing expedient drainage. **Note:** See **Chapter 5** for specific information on building expedient drainage features, including methods for conveying stormwater runoff and floodwater away from roadways, facilities, and other critical areas.

1.4.1. Consider the need for temporary drainage during expedient construction activities. For expedient roads, phased development of the drainage system in the early stages of clearing, grubbing, and stripping is essential to ensure uninterrupted construction. Delays caused by flooding, water ponding on the subgrade, subgrade failures, and heavy mud conditions, and the subsequent immobilization of heavy equipment can be eliminated by providing temporary trenches, drains or ditches necessary to divert or intercept surface water before, or concurrent with other construction. For long-term consolidation of unstable soils, vertical drains may be a good option. Other drainage options to consider:

1.4.1.1. Use the original drainage features as much as possible without disturbing natural grades.

1.4.1.2. Grade drainage ditches downhill.

1.4.1.3. Fill holes left by uprooted trees and stumps with acceptable soil and compact the ground to prevent the accumulation of surface water. Use dozers and graders for this work.

1.4.1.4. Slope the ground toward drainage ditches to prevent ponding on the surface.

1.4.1.5. Backfill existing ditches at the latest possible time to permit the best use of the original drainage.

1.4.2. Ensure designs for expedient drainage and features do not compromise airfield safety. Stormwater elements should be designed to minimize the attraction of hazardous wildlife by reducing open standing water such as retention ponds and stormwater wetlands to the maximum extent practicable. See UFC 3-201-01, *Civil Engineering*, and UFC 3-260-01, *Airfield and Heliport Planning and Design*, for more details on drainage requirements near airfields.

1.5. Land Clearing Considerations. Oftentimes, land clearing precedes construction activities. Land clearing may involve removal and disposal of vegetation, rubbish, and surface boulders embedded in the ground at the site or along the planned route. Temporary drainage may also be necessary during land clearing to limit muddy conditions or damage to subgrade. During hostilities, land clearing may include the removal and disposal of mines, booby traps, and unexploded explosive ordnance. Engineers use heavy equipment, power tools, hand tools, explosives, or even fire to clear and prepare the required acreage. **Note:** Avoid using fire to clear land if suitable equipment and sufficient personnel are available for other methods. Other considerations include:

1.5.1. Climate and Geology. Clearing, grubbing, and stripping operations differ in every climatic zone because each zone has different forest and vegetative types. Soils, altitudes, water tables, and other factors vary widely within each zone. Verify and consider local factors before beginning any clearing operation.

1.5.2. Camouflage. If cover and concealment of the clearing operation is a concern, you should not remove standing trees and brush outside the designated cleared area unless necessary. When uprooting trees with bulldozers, take care to control their fall and avoid breaking surrounding trees.

1.5.3. Timber Salvage. For optimal use of resources, consider trimming all useful timber for logs, piles, and lumber; stockpile it for future use in bridge, culvert, protective overhead cover and other expedient construction applications. Push or skid this timber into a salvage area where it can be relocated to a sawmill or other processing location with little difficulty.

1.5.4. Disposal of Cleared Material. The choice of waste disposal method depends on the type of construction, environmental concerns, the location, the threat, and the time available. Generally, to speed disposal and keep the area clear for equipment operation, push cleared material off the construction site and into the surrounding timber. To dispose of material as rapidly as possible, assign personnel and equipment to accomplish this concurrently with the clearing and grubbing. The disposal method should be consistent with the methods of camouflage, salvage, and drainage used for clearing.

1.5.5. Safety Measures. Before starting land clearing operations, carefully consider the safety of all personnel and equipment. Ensure personnel have required personal protective equipment (PPE) during clearing operations. Department of the Air Force Manual (DAFMAN) 91-203, *Air Force Occupational Safety, Fire and Health Standards*, lists PPE for selected CE activities. Proper supervision and planning can help prevent vehicle and equipment mishaps, and accidents caused by falling trees, uprooted stumps, stump holes, and rough or broken terrain. Never compromise safety when employing expedient methods described in this publication. General safety precautions include:

1.5.5.1. Inspect equipment before use, and periodically on a regular basis.

1.5.5.2. Ensure that mechanized equipment is operated by qualified and authorized personnel only.

1.5.5.3. Use seat belts when they are available.

1.5.5.4. Provide barriers to prevent personnel from walking under loading equipment that has a hoist or lift capability.

1.5.5.5. Operate equipment in a manner that will not endanger people or property.

1.5.5.6. Observe safe load capacities and safe operating speeds.

1.5.5.7. Shut down and turn off the engine when equipment is unattended.

1.5.5.8. Stop the equipment completely (apply the parking brake if available) before mounting or dismounting.

1.5.5.9. Observe any required work/rest intervals.

1.5.5.10. Do not make any modifications to the protection system.

1.6. References. **Table 1.1** list references in this chapter relating to the design of expedient roads and drainage, and general safety standards and requirements.

Table 1.1. Chapter 1 References.

Introduction	
UFC 3-250-01	*Pavement Design for Roads and Parking Areas*
UFC 3-201-01	*Civil Engineering*
UFC 3-260-01	*Airfield and Heliport Planning and Design*
UFC 3-250-03	*Standard Practice Manual for Flexible Pavements*
UFC 3-250-04	*Standard Practice for Concrete Pavement Construction*
DAFI 90-802	*Risk Management*
DAFMAN 91-203	*Air Force Occupational Safety, Fire and Health Standards*

Chapter 2

EXPEDIENT ROAD SURFACING OVER UNSTABLE SOILS

2.1. General Information. Establishing usable roadways during beddown at contingency locations or after an attack or disaster could be critical to sustain the mission or recover the base. Procedures might be as simple as clearing or patching existing pavements or as arduous as developing expedient roads or surfaces with limited materials and equipment. Conducting and incorporating a Risk Assessment according to DAFI 90-802, should enable engineers to evaluate the best and safest option that conditions allow. Difficult terrain and poor soil conditions, especially wet sand, wet clay, and wet silt make expedient road construction and repair more problematic. Oftentimes, engineer work crews need to use ingenuity, available resources, and a lot of strenuous labor to provide temporary, workable traffic surfaces in a timely fashion. The expedient road techniques and procedures presented here is intended to provide temporary options during emergencies or in remote and austere environments when time and resources are limited. For additional ideas, options, and techniques for expedient surfacing methods over sections of unstable soils, see Army Technical Manual (TM) 3-34.63, *Paving and Surfacing Operations*.

2.2. Expedient Materials. While it can be a rare occurrence, civil engineers may need to improvise an expedient roadway surface for vehicles to traverse relatively short sections of sandy or muddy soil when traditional construction resources are unavailable. The type of materials used in expedient road construction are often divided into three groups: natural, manufactured, and non-standard materials:

2.2.1. Manufactured Materials. Manufactured materials are produced in a commercial factory. Expedient matting are rigid or semirigid portable surfaces that interlock to form the surface of an expedient pavement. These structures contribute to the soil's bearing capacity, but they do not protect the soil from infiltration (water that works its way into the soil through the surface). Membranes do not contribute to the soil's bearing capacity, but they protect the soil from losing strength due to infiltrating water. These surfaces are extremely flexible and controls dust. Under certain conditions, some lower forms of bituminous surfaces are considered expedient surfaces.

2.2.2. Natural and Non-standard Materials. Natural expedient materials are materials found in nature at the site, such as rock, sand, timber, brush, and soil. Non-standard expedient materials are produced for some purpose other than road construction, such as precut lumber, tar paper, sandbags, and bricks. Both natural and non-standard materials can be used to construct pavements and surfaces.

2.2.3. The following expedient road methods/materials are addressed here:

2.2.3.1. Corduroys (logs).

2.2.3.2. Wire mesh.

2.2.3.3. Wood planks.

2.2.3.4. Chespaling mats (flexible roll wood fencing).

2.2.3.5. Bamboo mats.

2.2.3.6. Plastic pipe mats/roads.

2.2.3.7. Roadway/airfield mats.

2.3. Techniques and Procedures. Expedient road development is usually a temporary alternative until surfaces that are more durable can be provided. Some of these techniques and procedures are rather uncommon, but in the right situation, may be the only workable option available at the time.

2.3.1. Corduroys (Logs). Corduroy roads have been around for a long time and are still used today. They are constructed of logs (split or round), laid side-by-side and crosswise to the centerline. These surfaces may be ideal expedient roadways over soft ground at remote locations where timber is readily available. The basic rule for determining the type of corduroy road needed is "the softer the ground, the heavier the corduroy required." The types of corduroy roads are standard corduroy, corduroy with log stringers, and heavy corduroy.

2.3.1.1. Standard corduroy. A standard corduroy road consists of crossed logs laid directly on the ground (**Figure 2.1**.). This type of corduroy road is the most frequently used and consists of one layer. Below are procedures to construct a standard corduroy road:

2.3.1.1.1. Place 6-8 inch-diameter, 12-foot logs next to each other (butt to tip).

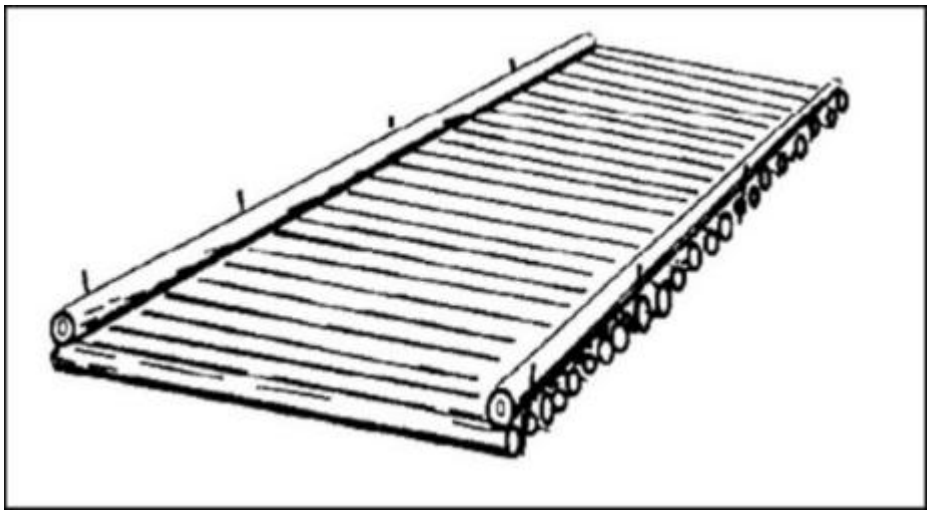
2.3.1.1.2. Place 6-inch-diameter logs along the edges of the roadway, and wire or drift-pin them in place.

2.3.1.1.3. Drive 4-foot-long pickets into the ground at regular intervals along the outside edges of the road to hold it in place.

2.3.1.1.4. Fill the chinks between logs with brush, rubble, or twigs; cover the surface with a layer of gravel or dirt to make the surface smoother.

2.3.1.1.5. Construct side ditches and culverts the same as for normal roads.

Figure 2.1. Standard Corduroy Road Section.



2.3.1.2. Corduroy with log stringers. This is a standard corduroy road that is more substantial and has two layers. Construct it by placing log stringers parallel to the centerline on 3-foot centers and lay the corduroy over the stringers as described above for standard corduroy. Securely pin the corduroy decking to the stringers, and prepare the surface as shown in **Figure 2.2**.

2.3.1.3. Heavy corduroy. A heavy corduroy road has three layers. It consists of crossed logs and stringers laid on sleepers. Place sleepers (heavy logs 10 to 12 inches in diameter and long enough to carry the entire road) at right angles to the centerline on 4-foot centers. Build a heavy corduroy road with log stringers on top of the sleepers as shown in **Figure 2.3**. Stringers and sleepers do not increase the bearing capacity of the decking. They serve as a crib to keep the road surface above the level of the soft ground. Stringers and sleepers sink into the ground until they reach a stratum that can support the load. In other words, they provide depth for a stable structure.

Figure 2.2. Corduroy Road with Log Stringers.

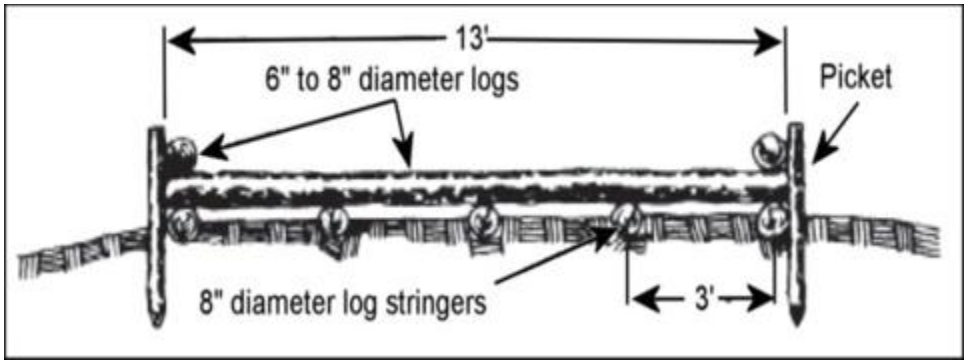
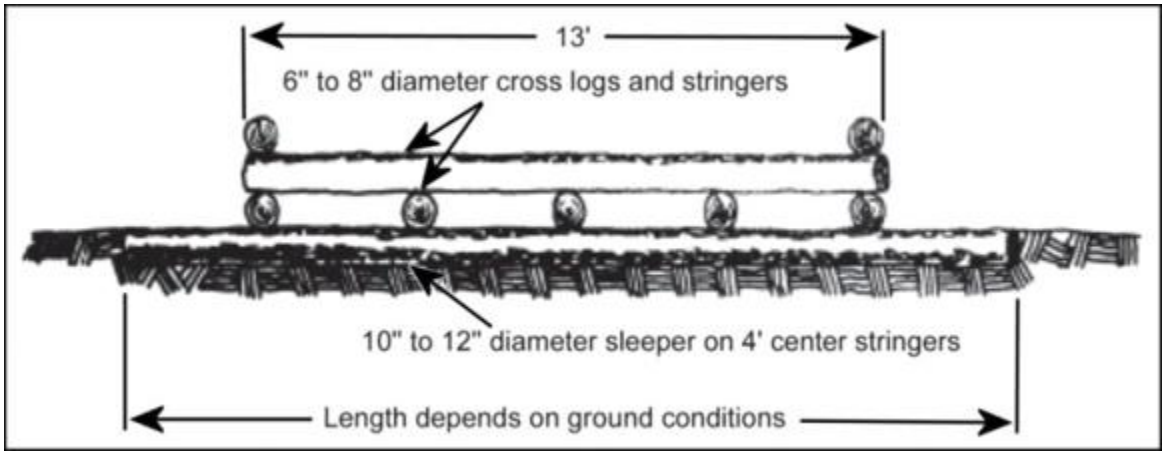


Figure 2.3. Heavy Corduroy Road.



2.3.1.4. Portable corduroy mats (made by wiring together 4-inch-diameter logs) is a hasty expedient option that personnel can prefabricate and put down quickly when needed. Corduroy mats are very similar to Chespaling mats addressed in **paragraph 2.3.5**, but corduroy mats are heavier.

2.3.1.5. The greatest objection to corduroy roads is the roughness of the surface. To make the surface smoother, consider filling the gaps between logs with brush, straw, twigs, weeds, etc., then cover with gravel or dirt. If the foundation is stable and suitable materials are available to construct a thin covering over the corduroy, a corduroy road is a good temporary road. **Note:** Provide a blanket of earth or gravel to protect the surface if tracked vehicles also use the road.

2.3.2. Wire Mesh. Various types of wire mesh, including chain-link fencing, chicken wire, expanded metal lath (used for stucco and plastering), may be used for expedient surfaces over sandy portions of roadways or during bottom preparation of shallow-water crossings. For sand surfaces, add a layer of burlap or similar material underneath the wire mesh to help confine the sand. Longer life can be obtained by proper subgrade preparation, multilayer or sandwich construction, and staking the edges of the wire-mesh at 3-foot intervals.

2.3.2.1. One of the most common types of wire mesh is chain-link fencing, commonly referred to as cyclone fencing. Since the width of chain-link fencing is normally 6 to 8 feet, lay at least two sections to obtain a width that is suitable for traffic. Ensure that sections overlap at least 1 foot. As illustrated in **Figure 2.4**, thread a 3- to 10- gauge strand of wire through the edges of the mesh and secure the mesh with pickets at 3- to 4-foot intervals.

2.3.2.2. As might be expected, the more layers used the more durable the pad will become. Other roads should never cross wire mesh road sections unless planking or some such material is placed over the mesh to protect it. In addition, do not use mesh surfaces on muddy (unstable, saturated soil) roads because they prevent grading and reshaping of the surface when ruts appear. **Note:** Commercial wire/mesh barriers (**Figure 2.5**) laid flat provide similar confinement and separation and may be more available in the local area than wire mesh and burlap.

2.3.2.3. Chicken wire and expanded metal lath are laid practically the same as the chain-link wire mesh. If 3-foot-wide chicken wire is used, lay the roadway in 5-foot widths with 1-foot overlaps and fasten the sections together with plain wire. One layer of chicken wire may be sufficient for lightweight traffic; three layers may be required to sustain trucks. Expedient road surfacing made from chicken wire usually requires constant maintenance.

Figure 2.4. Chain-Link Wire-Mesh Surface Details.

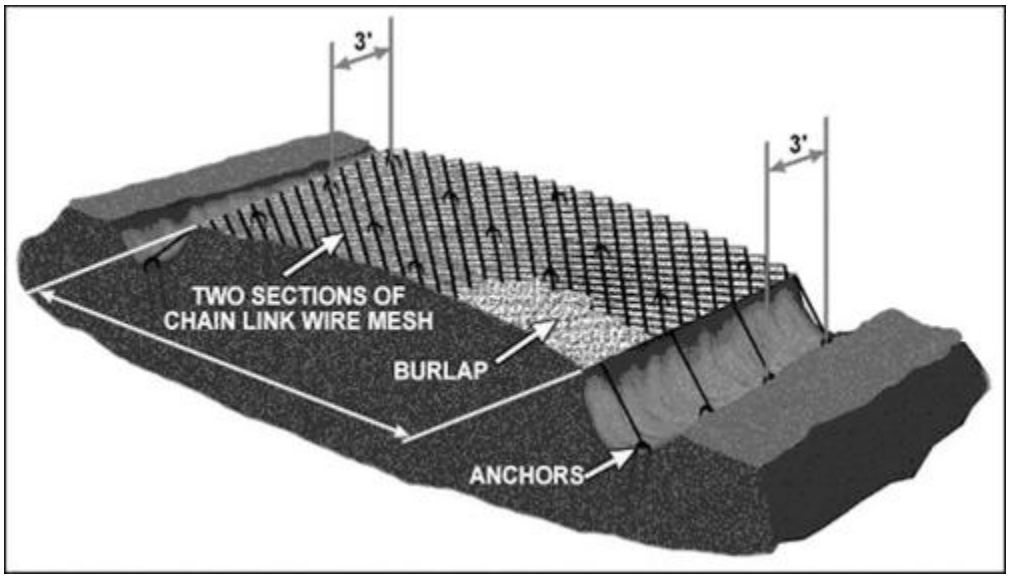


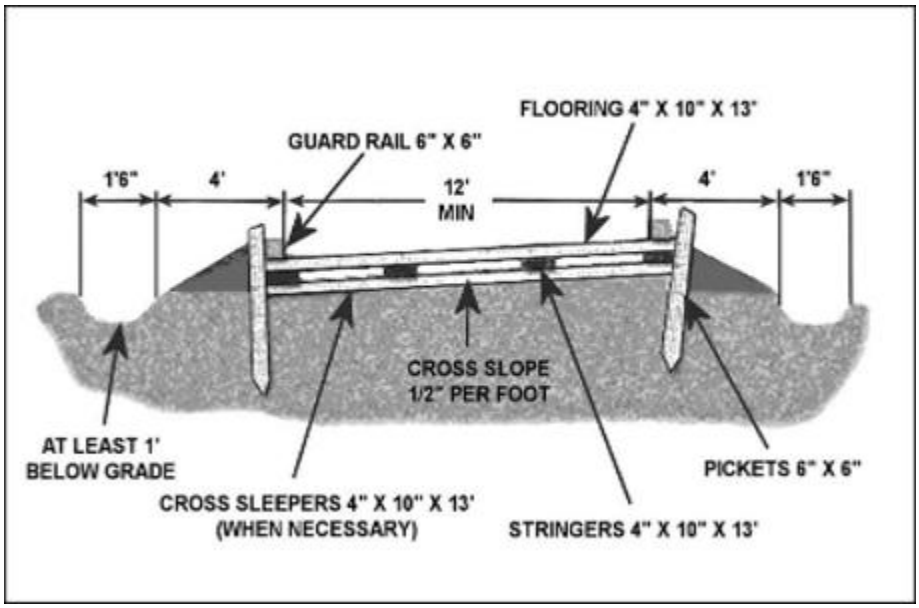
Figure 2.5. Commercial Wire-Mesh Barriers.



2.3.2.4. When roadways traverse shallow-water crossings, it may be necessary to use bottom-fill materials such as gravel and rock to fill short, deep gaps and to raise the bottom of the crossing. If so, consider using wire mesh to keep the bottom-fill material in place when exposed to traffic.

2.3.3. Plank Roads. For centuries, plank roads have been a solution to traverse sandy, rocky, and soggy bottom terrain. Engineers have constructed plank roads using planks laid across timber stringers running lengthwise. Today, these wooden plank roads are sometimes used as an expedient measure to cross short sections of loose sand or wet, soft ground. This type of road may last several months if built well and has an adequate base. Planks used for flooring, stringers, or sleepers should be 3 to 4 inches thick, 8 to 12 inches wide, and at least 13 feet long. When desired, rough, unfinished 3- by 10-inch planks can replace the 4- by 10-inch timbers shown in **Figure 2.6**.

Figure 2.6. Plank Road Construction Diagram.

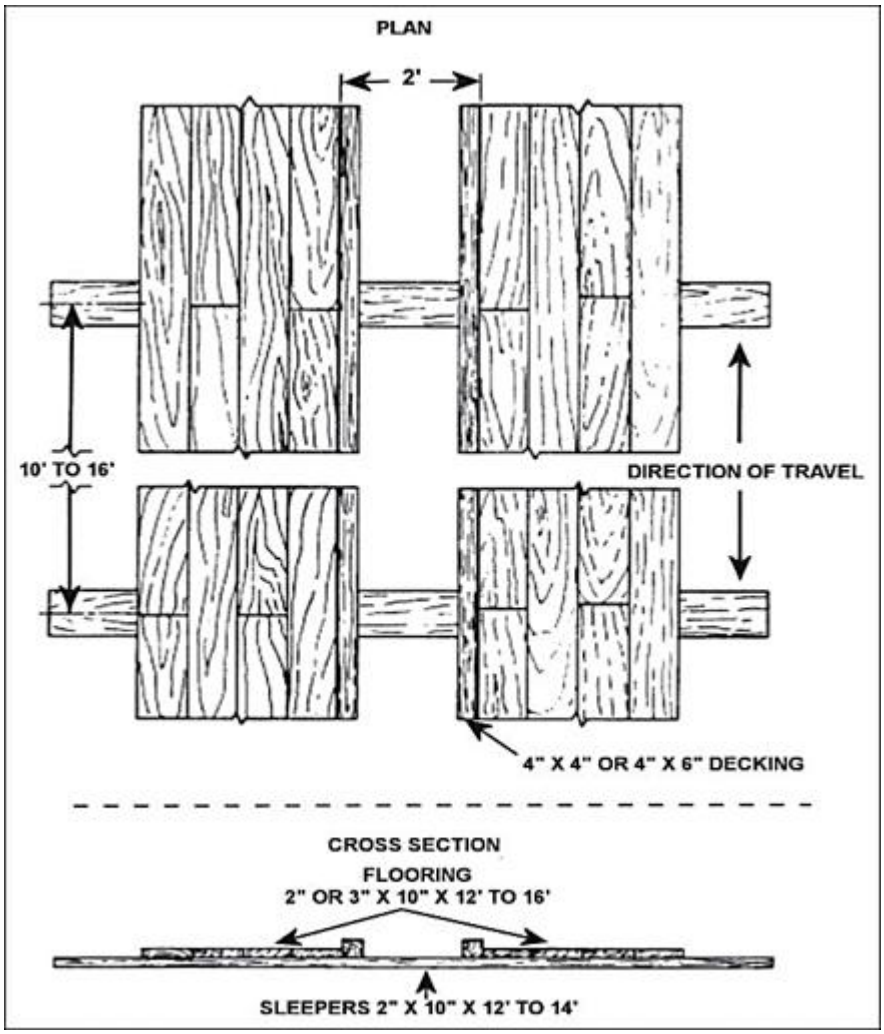


2.3.3.1. Lay stringers in regular rows parallel to the centerline on 3-foot centers. Stagger the joints, and lay floor planks across the stringers. When seasoned lumber is used, leave 1-inch gaps between the stringers to allow for swelling when the lumber absorbs moisture. Spike the planks to every stringer. Place 6-inch-deep guardrails on each side, leaving a 12-inch gap between successive lengths of the guardrail for drainage of surface water. Place pickets along each side at approximately 15-foot intervals to hold the roadway in line. If necessary, place corduroy, fascine, or other expedient cross sleepers on 3- to 5-foot centers to hold stringers in the correct position or to gain depth for the structure.

2.3.3.2. To allow for drainage, construct the base with a transverse slope versus a center crown. Place treads over the floor planks parallel to the line of traffic to produce a smoother riding surface.

2.3.4. Plank-Tread Road. Work crews can usually build plank-tread roads easily and rapidly, and with less material than ordinary plank roads. **Figure 2.7** illustrates the construction details for the plank tread road.

Figure 2.7. Plank-Tread Road Plan.



2.3.4.1. Lay the sleepers, which are made of 2- by 10-inch material that is 12 to 16 feet long, across the centerline of the road. Embed the sleepers in the ground on 10- to 16-foot centers. The spacing of the sleepers depends on the length of plank available for tread. If the sleepers are more than 2 feet thick or hold the flooring too high off the ground, reduce the amount of spacing so that the flooring can carry the load without bending excessively.

2.3.4.2. Lay flooring planks, 2 to 3 inches thick by 10 to 12 inches wide, on top of the crossties. A 2-inch plank will suffice as road surfacing in most cases; however, use a 3-inch plank for extremely severe conditions. Tread sections can be fabricated and laid directly from a truck bed to facilitate placement. Optionally, place curbs on the inside of each tread to ease passage of the narrowest vehicle using the road.

2.3.4.3. In lieu of lumber, use logs to construct log-plank or log-tread roads. To prevent the logs from rolling sideways under traffic, securely spike them to the crossties. The spacing of the crossties depends on subgrade condition, but it is normally about 3 feet. For log tread roads, use cur bs as described above.

2.3.5. Chespaling Mats. Chespaling mats are a hasty expedient surface used to cross sandy terrain or to get out of mud. Create chespaling mats using green saplings, preferably about 1.5 inches in diameter and 6.5 feet long. Wire the saplings together to form a 12-foot-long mat as shown in **Figure 2.8**.

Figure 2.8. Assembling Saplings for Chespaling Mats.

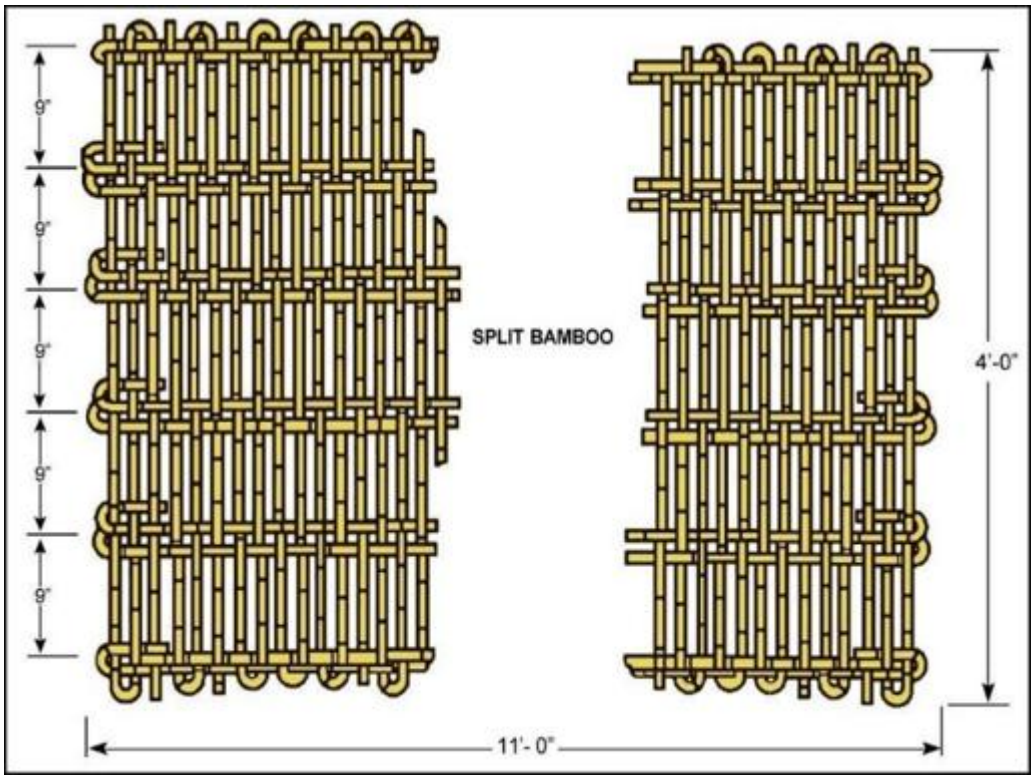


2.3.5.1. To build a chespaling road, lay a double row of mats, each mat having its long axis parallel to the centerline with a 1-foot overlay at the center. Wire the mats together. Keeping the road wet prevents saplings from becoming brittle and breaking.

2.3.5.2. Some mats are made from dimensioned timbers wired together to resemble a picket fence. Construct a variation slightly more effective for crossing sand by attaching chicken-wire netting to the bottom of the mats. **Note:** To reuse the mats, troops often roll them into bundles and carry them on wheeled vehicles to the next location.

2.3.6. Bamboo Mats. Bamboo mats are an excellent chespaling-type expedient for beach roadways. These mats are light and comparatively strong. Create these mats by splitting 2-inch bamboo rods and weaving them into a mat in a manner similar to rug weaving (**Figure 2.9**.). Soak the rods before weaving and keep the mats moist while they are in use. Place the mats with the long dimension parallel to the centerline. The mats remain serviceable for three or four months on firm ground or sand. Bamboo mats are also usable over mud.

Figure 2.9. Bamboo Mat Construction Diagram.

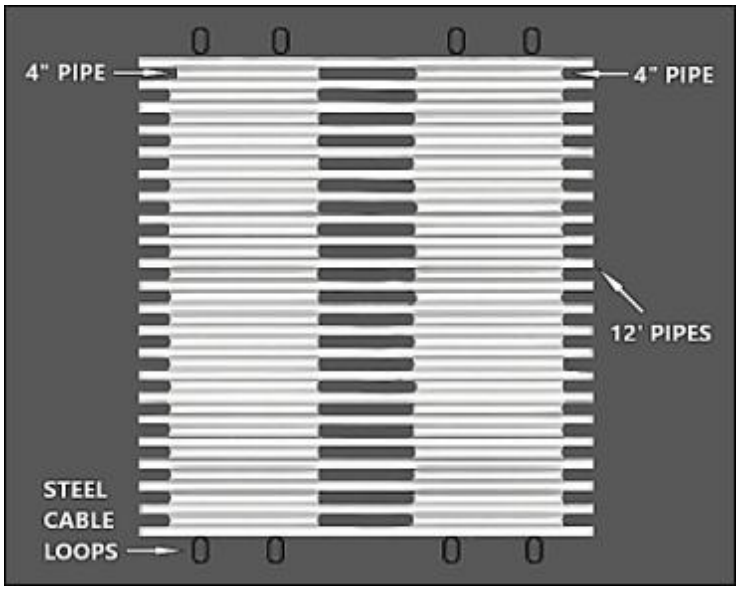


2.3.7. Plastic Pipe Mats. Portable, reusable, lightweight corduroy-type mats and roads can be built from Polyvinyl Chloride Pipe (PVC) or High-Density Polyethylene (HDPE) pipe for transitioning over bogs and wetlands (**Figure 2.10**.). Plastic pipe mats are easy to transport, install, remove, and repair. The only notable disadvantage is that the surface of this type of matting can be slippery when wet, resulting in poor traction on grades. In addition, although PVC materials can usually be bought locally, HDPE piping may need to be obtained from a national distributor. Pipe mats are usually constructed using 4-inch diameter Schedule 40 PVC or Standard Dimension Ratio 11 HDPE pipes that are tightly connected using 3/16-inch diameter galvanized steel cables to form panels (**Figure 2.11**.). Prepare and assemble materials as addressed below:

Figure 2.10. Plastic Pipe Mat Undergoing Field Tests.



Figure 2.11. Typical Pipe Mat Assembly.



2.3.7.1. Using 20-foot lengths of material, saw pipes into 12-foot lengths. Next, cut the remaining 8-foot sections in half. Drill four 1/4-inch holes completely through the 12-foot-long pipes at locations 2 feet and 4 feet from either end. Drill two holes completely through each 4-foot section 1 foot from each end.

2.3.7.2. Next, alternate one 12-foot-long section with one row made of two 4-foot-wide sections placed 2 feet from each other.

2.3.7.3. String 3/16-inch galvanized steel cable through all sections. Make loops at the end of each cable, extending beyond the last pipe. Secure the loops with 3/16-inch cable clamps.

2.3.7.4. Install mats as follows:

2.3.7.4.1. When installing roads of this type, slopes should be less than 4 percent and exclude large protrusions such as boulders and stumps. Smooth out high spots, fill ruts, and leave the root mat of any vegetation in place.

2.3.7.4.2. For stabilization purposes, cover the entire crossing area with nonwoven geotextile. Geotextile is a fabric mat that allows water drainage and provides structural support. Place the mats on the geotextile. For a pipe mat crossing, build soil ramps up to and down from the crossing. Alternatively, make ramps by interconnecting 2-, 3-, and 4-inch diameter pipes. For more information on use of geotextiles, geosynthetics, geomembranes, and geogrids in expedient road construction, see **Chapter 4** and UFC 3-220-08FA (Inactive), *Engineering Use of Geotextiles*.

2.3.7.4.3. Since sunlight can be detrimental to standard Polyvinyl Chloride material, cover exposed PVC pipe with geotextile to prevent premature failure. HDPE is ultraviolet-resistant and doesn't need to be protected from ultraviolet exposure. In addition, it tolerates cold much better than PVC. High Density Polyethylene pipe can withstand cold temperature extremes of -40° F without becoming brittle or losing shock resistance and will return to its original shape after being deformed. If traction on inclines becomes a problem, cover the surface with materials such as expanded metal grating.

2.3.7.4.4. Size the mat to meet anticipated loads, soil strength, and installation equipment. Use larger mats on very weak soils that have a low load bearing strength, such as muck or peat.

2.3.7.5. Maintain mats by simply inspecting mats and road sections during and between use for broken pieces and repairing failures by sliding off and replacing failed sections.

2.3.8. Plastic Road. The plastic road design below was conceived by the US Department of Agriculture (USDA) Forest Service as a lightweight, portable, inexpensive, and easily constructed means of crossing sensitive sites and soils of low bearing capacity (**Figure 2.12**.). The plastic road spreads wheel loads over an increased subgrade area. The geotextile underlayment helps spread the load to increase soil bearing capacity and facilitates removal of the crossing after use.

Figure 2.12. Forest Service Plastic Road.



2.3.8.1. The process for making a plastic road is very similar to that for constructing pipe mats. An exception is that pipe transition mats/panels are built into the design to ease the transition of tires ramping up and then back down again on the approach between the firm soil and the mat (**Figure 2.13**.). Furthermore, the various pipes in a plastic road are connected using 1-in. diameter Schedule 80 PVC. It is important to drill round holes to avoid creating potential stress points that could facilitate pipe shattering (**Figure 2.14**.**)**.

Figure 2.13. Plastic Road with Transition Mat.

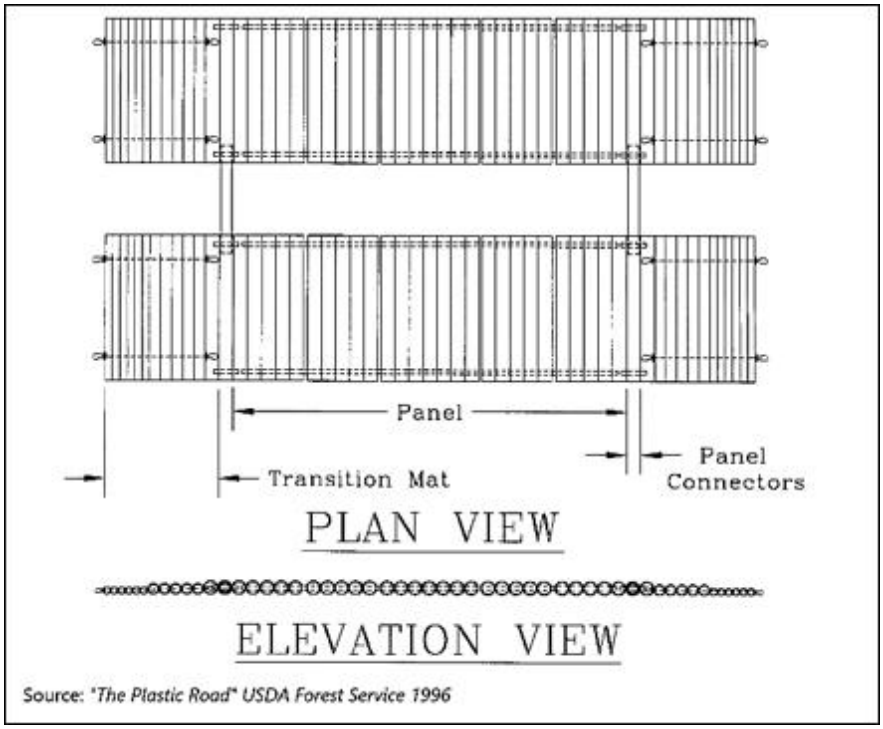


Figure 2.14. Plastic Road Connections.



2.3.8.2. USDA Forest Service tests have shown that PVC pipe road panels and transition mats for an 8-ft wide x 40-ft long crossing can be transported in a 3/4-ton pickup truck and assembled by two people in about 1 hr. The various panels in either the pipe mat or the plastic road crossing can be quickly hand-placed by two people. A plastic road can also be moved from site-to- site by attaching a chain to one end of the transition mats and then towing it with a 3/4-ton pickup. However, the distance and surface over which the plastic road is dragged should be evaluated to avoid excessive wear and breakage. Based upon Forest Service tests, a prototype plastic road installation supported over 400 loaded 18-wheel log truck passes at two sites. Detailed directions, including drawings, for constructing the plastic road are found in the following free publication: Moll, J.E., and R. Hiramoto 1996, "The Plastic Road", USDA Forest Service, Technology and Development Program, Publication 9624 1206-SDTDC available at **https://www.fs.usda.gov/t-d/pubs/pdf/96241206.pdf**.

2.3.9. Manufactured Roadway Matting Materials. If available, manufactured roadway matting materials may be an expedient option for traffic over sandy or loose soils. Expedient roadway mats are made of aluminum, fiberglass, plastic, composites, or various other materials. Below are assembly details for three types of expedient roadway matting materials:

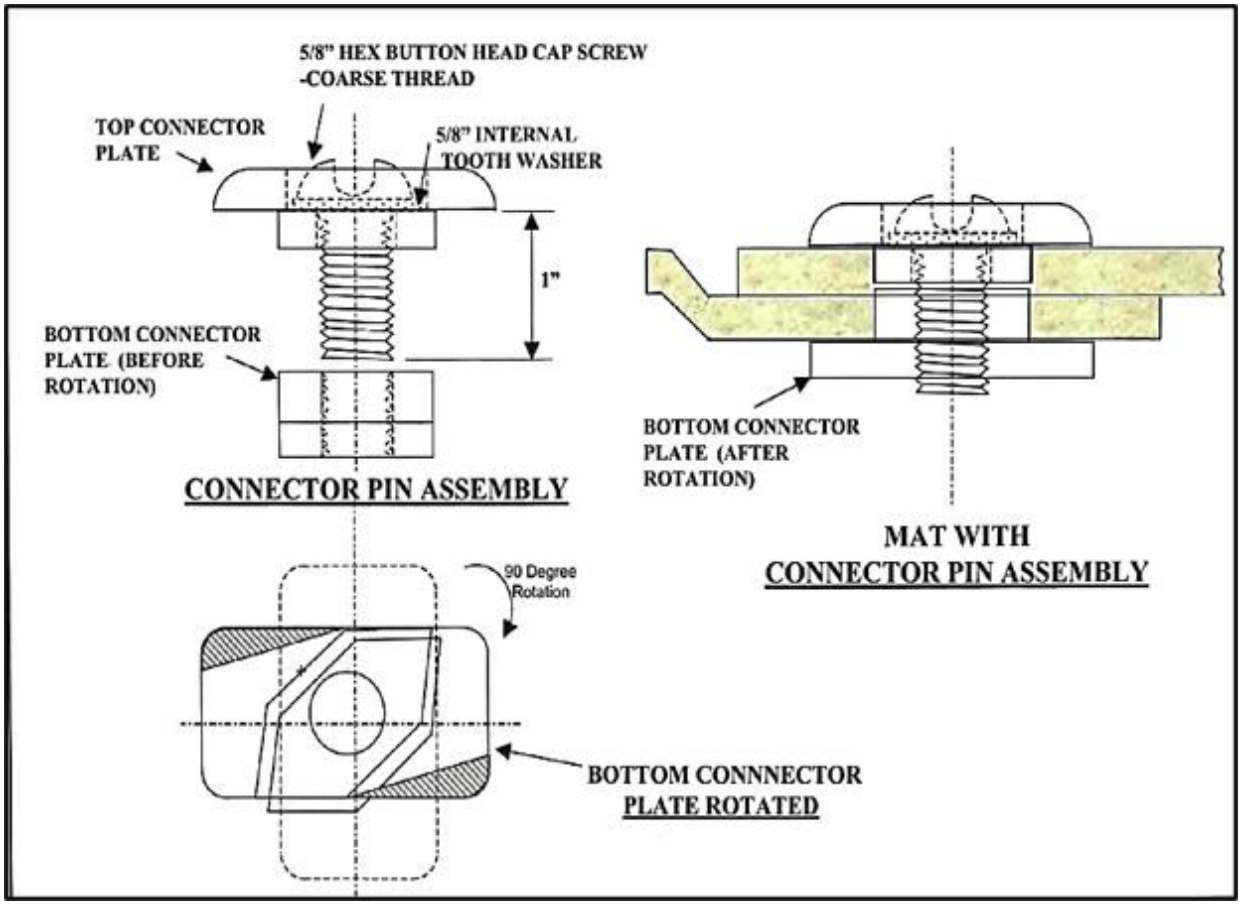
2.3.9.1. Army Corps of Engineers (ACE) Matting: Originally developed by the U.S. Army Engineer Research and Development Center (ERDC) in 1999 and patented in the United States in 2004, this fiberglass-reinforced matting system is a versatile terrain-surfacing solution that can be used over a wide range of soil and ground conditions. If available, it can provide a rapid, temporary option to improve trafficability over sandy soil. The mat is made of polyester resin reinforced with four plies of woven chopped fiberglass (**Figure 2.15**.). The weight of a 6-foot by 6-foot panel is approximately 115 pounds and the unit weight is 2.59 pounds per square foot. Each panel has an 8-inch under lap with a downward fold for connecting panels together. Ensure the matting follows the same longitudinal grade and transverse cross-slope as the adjacent roadway to maintain positive drainage. A crew of seven can install the fiberglass panels at a rate of 185 square feet per man-hour.

Figure 2.15. Fiberglass-Reinforced Mat.



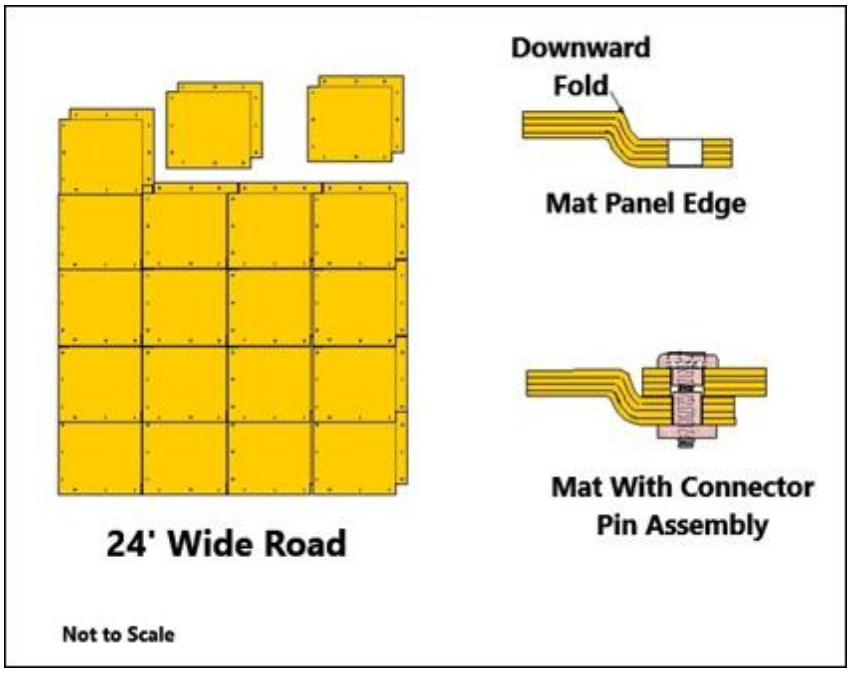
2.3.9.1.1. A quick-install connector pin bolts the panels together. The pin connectors are made of HDPE material. **Figure 2.16** shows the connector pin assembly components. This pin design can tolerate rut depths up to 3-in. The connector pin is placed in lapping rectangular holes and tightened. As the pin over-connector is tightened, the bottom plate rotates 90 degrees, locks in place, and pulls the two panels tightly together.

Figure 2.16. Fiberglass-Reinforced Mat Connector Pin Assembly.



2.3.9.1.2. According to ACE matting field tests conducted by ERDC, when installed over sand, the mat will support over 5,000 truck passes with very little rutting. Although the mat may flex significantly under heavy wheel loads during these tests, it springs back to a flat position and usually suffers no damage. An example of a 24-foot placement pattern is shown in **Figure 2.17**. For sandy soil, it is recommended that fiberglass-reinforced matting be used for small roadway sections, less than a 1/2 mile in length. When very soft soils are involved, the base must be bolstered. **Note:** Performance enhancements are being made to this system through continuous design modifications and testing.

Figure 2.17. Fiberglass-Reinforced Mat Placement Pattern.



2.3.9.2. Hexagonal Mats. There are several variations of this product on the market, including composite plastic hexagonal mats and aluminum hexagonal mats addressed below. Although the information below was adapted from testing and evaluation performed on both models at ERDC facilities, if approved for use, refer to manufacturer's instructions for detailed assembly procedures.

2.3.9.2.1. Plastic hexagonal mats. The two types of plastic hexagonal mats addressed here include the Allround and Special models (**Figure 2.18**.). These lightweight interlocking mat panels were designed for quick installation to create access roads and parking areas. The installation rate ranges from 600 to 900 square feet per man-hour. The panels are made from recycled, high-density polyethylene that will not deteriorate from exposure to ultraviolet light. Each panel has handles that ease its installation; skid resistance texture to improve tire traction; and weighs approximately 7 pounds with a surface area of approximately 3 square feet. The factory-recommended maximum wheel load is 13,000 pounds per panel when installed over a gravel base. The panel's hexagonal form permits road angles of 30, 60, and 90 degrees. When used as a roadway, the recommended vehicle speed is no more than 6 mph. As an added note, the Special model works well for making rapid parking areas, sidewalks and tent floors in bivouac areas where loose sandy soil is a particular problem.

Figure 2.18. Plastic Hexagonal Mats.



2.3.9.2.2. Aluminum hexagonal mats. The aluminum version (**Figure 2.19**) is heavier (approximately 21 pounds) and much more costly than its plastic composite counterparts. The aluminum version can be joined ahead of time and then placed as large mat assemblies. However, special equipment is required for this type of operation. Field tests conducted by the Army involving 5-ton trafficking confirm the functionality of this product for road construction over sandy soil. If drainage is an issue, stabilize the base with a geotextile. Although this system is very good when applied over sandy soils, it is, however, not effective when deployed on very soft soil subgrades—specifically those that have a CBR value of less than 1.0. The construction of roads over very soft soil is very difficult and is generally not conducive to expediency. However, information regarding common soil stabilization methods is addressed later in this chapter.

Figure 2.19. Aluminum Hexagonal Mats.



2.3.9.2.3. Installation details. When using this material, the minimum roadway width should be no less than 10.5 feet (7 panels). It is recommended that in order to achieve an effective road installation, the following procedures be used. Different installation approaches may be used, as long as one handle and two hooks remain free for connecting subsequent panels. Refer to **Figure 2.20** as the following steps are explained:

2.3.9.2.3.1. The first panel is to be positioned at the starting left corner (panel 1) of the road, as indicated in the drawing. The handle of this panel must be installed in the direction of the traffic.

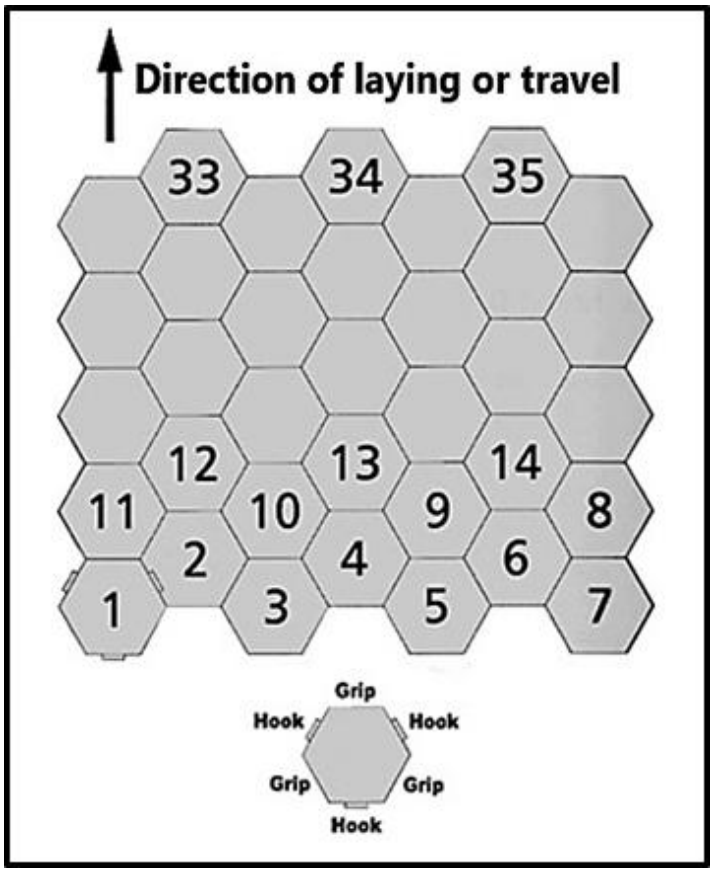
2.3.9.2.3.2. The first row of panels (panel 1 through 7) will then be hooked to the starting panel (panels 2, 4, and 6) as shown.

2.3.9.2.3.3. Next, connect rows 8-11 and 12-14 by hanging the connecting hooks into the available handle voids and joining each panel to the remaining two free hooks.

2.3.9.2.3.4. Continue this simple process until the desired length is attained. By following these steps, a team of workers can install the system without interfering with each other.

2.3.9.2.3.5. Loose panels at the beginning and end of the mat surface (panels 1, 7, 33, 34, and 35) may be fastened in any of three ways: by using SPAX® screws at the handle/hook to fix panels to each other; insertion of half panels or, by means of herrings/tent pegs.

Figure 2.20. Hexagonal Mat Installation Layout.



2.3.9.3. Submersible Matting System (SUBMAT). The SUBMAT is a new lightweight matting system developed by ERDC. The system is made from common geotextile materials and is easy to install by hand, requires no heavy equipment, and gains its strength by confining sand from a beach in a lightweight structure.

2.3.9.3.1. The SUBMAT uses a simple system of small, man-portable trash pumps to move indigenous sand into the mat's structure. SUBMAT is also scalable, and individual sections can be joined together to create the desired length of a required roadbed. A single SUBMAT is about 20 feet wide and 100 feet long.

2.3.9.3.2. Although designed to cross the gap between low- and high-tide at beach/water interface, the SUBMAT is deployable in riverine, dry beach, and inland use cases. The system is collapsible and can be transported easily on lighter vessels and vehicle convoys.

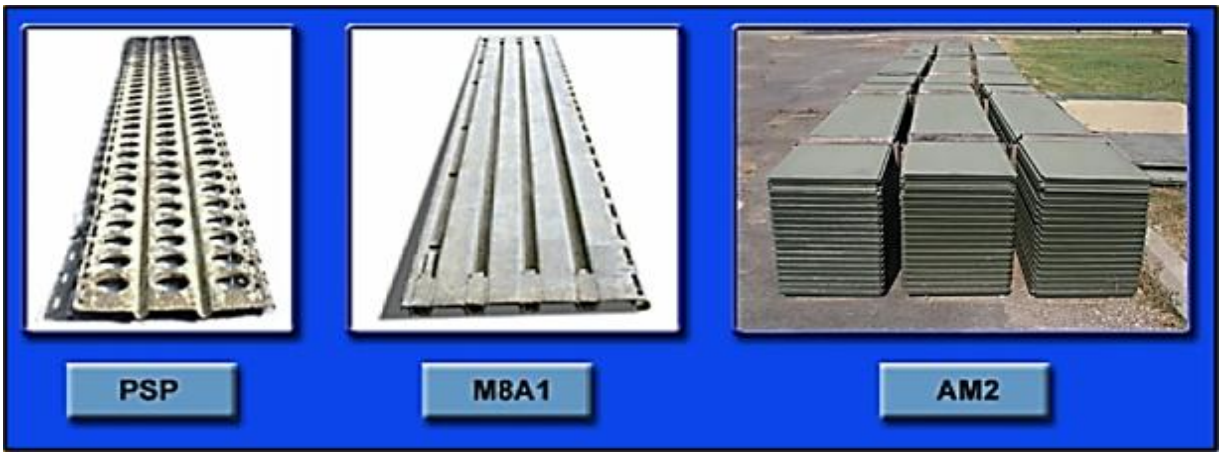
2.3.9.3.3. The ERDC test results showed that the SUBMAT system (**Figure 2.21**) was able to sustain an accumulation of 1,000 Medium Tactical Vehicle Replacement, 350 Heavy Expanded Mobility Tactical Truck, and over 150 M1A1 main battle tank passes without experiencing any significant damage. The ease of deployment, relatively low cost, and trafficability may make the SUBMAT a suitable candidate for expedient low-volume roads in austere environments such as stream beds, low-water crossings, recently flooded or flood prone areas, and areas with weak soil. **Note:** For more information on the SUBMAT system, contact the Geotechnical Structures Laboratory or Coastal and Hydraulics Laboratory at ERDC.

Figure 2.21. Submersible Matting (SUBMAT) Provides Stable Roadway for Vehicle Traffic.



2.3.10. Historical/Legacy Airfield Matting. Prior to World War II, the demand for rapidly constructed airfields led to the development of several portable, metal landing mats. As metal airfield landing mats became a standard supply item, they were quickly put to use on beaches as well as airfields. Mats such as M8 pierce steel planking (PSP) and M8A1 steel mats, or MX18, MX19, and legacy AM2 “aluminum matting system” were used throughout wartime theater of operations. If available, these mats can still be effective expedients for crossing unstable soils. Landing mats fabricated from aluminum alloys can support heavy loads and have a lower weight per square foot as compared to timbers. They also provide smoother surfaces so that the smallest-wheeled vehicle using the road can obtain traction. Of the older matting used for expedient road construction, the three most common were PSP, M8A1, and legacy AM2 (**Figure 2.22**). **Note:** Newer fiberglass and fiber reinforced plastic mats may also be an expedient option in some situations.

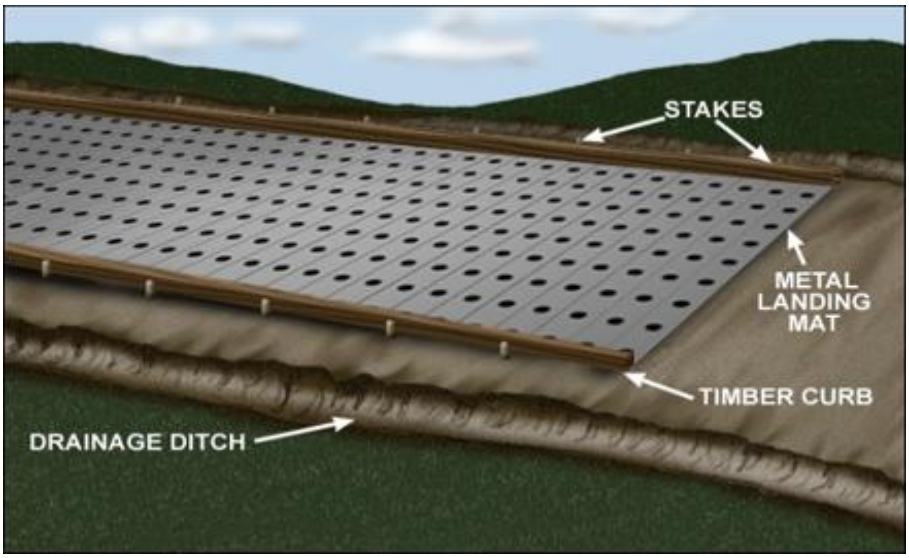
Figure 2.22. Older Mats Used in Expedient Road Construction.



2.3.10.1. When used on sand, place the metal landing mats directly on the sand to the length and width desired. Install the mat so that its long axis is perpendicular to the flow of traffic. If a width greater than the effective length of one plank is required, use half sections to stagger the joints. A second layer of the steel mat, laid as a treadway over the initial layer, increases its effectiveness. If you manage to find and plan to use PSP or similar material, place an impervious membrane under the mat to smooth and firm the subgrade, thus improving the road stability.

2.3.10.2. After extended use, these mats tend to curl at the edges. This problem can be overcome by anchoring the edges properly. Screw-type earth anchors provide the best means of anchoring. Another method of securing the edges is to use a curb of timber on the outside edge of the road and either wire it tightly to buried logs laid parallel to the road or stake it as shown in **Figure 2.23**. In addition, the road surface must be void of high spots and ruts for matting to be effective. Such irregularities cause difficulties in assembling the matting pieces.

Figure 2.23. Landing Mat Expedient Anchoring Technique.



2.4. Other Expedient Techniques. Sandy and muddy soil is a difficult undertaking due to stabilization problems. When the in-place soil is not strong enough for the road design, it can either be mechanically or chemically stabilized, covered with a bituminous surface treatment, have the unstable soil replaced, or as addressed in this chapter, covered with expedient surface materials. Muddy sections can also be improved by providing adequate drainage (e.g., ditch, drain tile or perforated pipe) to remove or divert water away from the problem area before covering the section with expedient surfacing material or backfilling with rock, gravel, or other suitable material.

2.4.1. See **Chapter 5** for information on expedient road drainage and **Chapter 4** for soil stabilization methods. Listed below are additional expedient options and techniques for sandy and muddy road situations:

2.4.1.1. Use lighter traffic loads and reduce tire pressure. A general rule of thumb is to lower tire pressure so that the tire deflection is increased by 50 percent.

2.4.1.2. Stabilize the sand by using Portland cement for constructing pavement base layers.

2.4.1.3. Mix available local ground or soil with the sand to improve traffic ability.

2.4.1.4. If firmer material lies in shallow depths, continue blading loose sand off roadway.

2.4.1.5. Place building rubble in ruts or use water-based emulsion to stabilize the sand.

2.4.2. The option of removing and replacing unstable soil can be difficult and time-consuming, but it may be an acceptable method if the site has good, stable soil underneath the poor soil and suitable fill material is available nearby.

2.4.3. The option of placing a thick layer of base course material directly over weak soil can provide a temporary solution; however, over time the base course material will sink into the weak soil and also become weak. The expedient remedy may be to place more material on the site.

2.5. References. **Table 2.1** list references in this chapter relating to construction of expedient road surfacing over sandy or muddy soils.

Table 2.1. Chapter 2 References.

Expedient Road Surfacing Over Unstable Soils	
UFC 3-220-08FA (Inactive)	*Engineering Use of Geotextiles*
TM 3-34.63	*Paving and Surfacing Operations*
ERDC TR-23-18	*Full-Scale Trafficability Testing of Prototype Submersible Matting Systems*
USDA Publication 9624 1206-SDTDC	*The Plastic Road*
DAFI 90-802	*Risk Management*

Chapter 3

FIELD-EXPEDIENT SOIL CLASSIFICATIONS AND IDENTIFICATION

3.1. General Information. Understanding on-site soil characteristics and properties is essential to determining the soil's usefulness for expedient roadway development. While characteristics such as color, odor, texture, particle size and shape aid in soil description, other properties such as compactness, moisture, plasticity, and strength provide significant insight into the engineering behavior of soils. More information on soil characteristics, testing, and classifications beyond that addressed here, is available in Air Force Handbook (AFH) 32-1034, *Materials Testing*, UFC 3-220-10, *Soil Mechanics (DM 7.1)*, and Tri-Service Pavements Working Group (TSPWG) manual 3-260-03.02-19, *Airfield Pavement Evaluation Standards and Procedures*, Appendix A, located on the Whole Building Design Guide website at **https://www.wbdg.org/dod/supp-tech-documents/tspwg-m-3-260-03-02-19**.

3.2. Overview. Generally, soil classification systems divide soils into groups and subgroups based on their engineering properties. This chapter presents techniques and procedures for determining on-site soil conditions, soil classifications, and field-expedient soil identification in preparation for expedient road construction.

3.3. Soil Categories. Information on the soil types in a region may provide some insight into the general engineering properties and trafficability characteristics of the soils. For specific soil classification, engineers mostly use those methods and systems that base classification on the grain size and consistency of the component materials.

3.3.1. Several general soil categories can be identified. Soils may be considered to belong to one of five broad categories listed below, based on the grain size, consistency, and chemical characteristics of the component particles.

3.3.1.1. Gravels are coarse materials ranging down to the size of peas.

3.3.1.2. Sand grains are also coarse, ranging in size just below peas and decrease until the individual grains are just distinguishable by the naked eye.

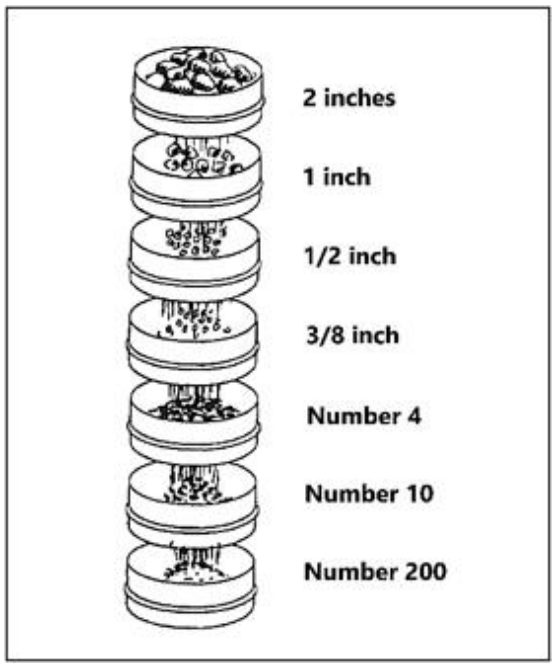
3.3.1.3. Silt is a relatively fine-grained material that lacks plasticity and is composed of individual particles indistinguishable by the naked eye.

3.3.1.4. Clay is a cohesive, highly plastic material made up of microscopic particles.

3.3.1.5. Organic matter consists of relatively fine-grained, partially decomposed animal or vegetable material.

3.3.2. Grain Size and Component Consistency. To more clearly define the specific parameters for each of the soil types, it is necessary to examine soil classification based on grain size and consistency of the component materials. Soils may be classified as gravels, sands, silts, or clays based solely on the size of the individual component grains. Soils are seldom made up of a single component; that is, pure gravel, sand, silt, or clay. More commonly, they contain particles of various sizes mixed in varying proportions. The component grain sizes and their respective proportions may be determined using dry sieve analyses (**Figure 3.1**.).

Figure 3.1. Stacked Sieves (Large to Small).



3.3.3. Soil Gradation. Gradation describes the distribution of the different size groups within a soil sample. The soil may be well graded or poorly graded.

3.3.4. Soil Plasticity. Plasticity is a property of the fine-grained portion of a soil that allows it to be deformed beyond the point of recovery without cracking or appreciable volume change. This property permits the rolling of clay into thin threads at some moisture contents without crumbling. Only clay minerals possess this property; thus, the degree of plasticity is a general index to the clay content of a soil. Engineers sometimes use the terms “fat” and “lean” to distinguish between highly plastic and moderately plastic soils. Soil plasticity is determined by observing the different physical states that a plastic soil passes through as the moisture content changes. The boundaries between the different states as described by the moisture content at the time of changes is the “consistency.”

3.3.5. Soil Strength. A soil’s strength or its ability to withstand a load relates to its shearing resistance. Shearing resistance is especially important to the bearing capacity of soils used as a base or subgrade. CBR is a measure of the shear strength (or shearing resistance) of a soil under carefully controlled conditions of density and moisture. Since testing for CBR values in a laboratory is time-consuming and impractical for expedient or contingency evaluations, several alternative methods are available for use in the field—shown here are two common methods:

3.3.5.1. One alternative involves using existing Unified Soil Classification System (USCS) empirical testing data to estimate CBR values.

3.3.5.2. Another alternative is to use penetration shear test instruments for on-site soil testing. The penetration shear tests results translate to CBR values to quantify soil strength.

3.3.6. USCS Soil Characteristics. **Table 3.1**., **Table 3.2** and **Table 3.3** lists USCS soil characteristics including soil divisions and groups, engineering properties, suitability for road construction, and range of CBR values. The group symbols generally use a two-letter value to define the soil type, see **Figure 3.2**. Using the range of values in the USCS data is the quickest and least accurate method to estimate CBR values in the field. The CBR values range from as low as 3 to as high as 80, depending on the type of soils. The fine-grained soils vary from 3 for organic clays to 15 for micaceous or diatomaceous silts and sands. The sand-silt-clay coarse-grained combinations range from 10 for the clayey mixtures to 40 for the gravelly and silty sands. Gravelly soils range from 20 for the clayey group to 80 for the well-graded gravels and gravel-sand mixtures. While these approximate values provide some useful information, when possible, use actual test values to determine the bearing capacity of soils. See American Society for Testing and Materials (ASTM) D2487-17, *Standard Practice for Classification of Soils for Engineering Purposes (USCS)*, for more information on engineering properties of various soils and grouping symbols.

Figure 3.2. Soil Group Letter Definitions.

First and/or second letters

Letter	Definition
G	Gravel
S	Sand
M	Silt
C	Clay
O	Organic
PT	Peat

Second letter

Letter	Definition
P	Poorly graded (many particles about the same size)
W	Well-graded (many different particle sizes)
H	High liquid limit
L	Low liquid limit

Table 3.1. Soil Characteristics (Part 1).

Soil Divisions		Group Symbol		Value as Subbase or Subgrade	Value as Base Course
Coarse Grained Soils (> 50% retained on No. 200 Sieve)	Gravel and Gravelly Soils (≥ 50% of coarse fraction retained on No.4 Sieve)	GW		Excellent	Good
		GP		Good to Excellent	Poor to Fair
		GM	d	Good to Excellent	Fair to Poor
			u	Good	Poor
		GC		Fair to Good	Poor
	Sands and Sandy Soils (≥ 50% of coarse fraction passing No. 4 Sieve)	SW		Good	Poor
		SP		Fair to Good	Poor to Not Suitable
		SM	d	Good	Poor
			u	Fair to Good	Not Suitable
		SC		Fair to Good	Not Suitable
Fine Grained Soils (≥ 50% passing No. 200 Sieve)	Silts and Clays LL<50	ML		Fair to Poor	Not Suitable
		CL		Fair to Poor	Not Suitable
		OL		Poor	Not Suitable
	Silts and Clays LL>50	MH		Poor	Not Suitable
		CH		Poor to Very Poor	Not Suitable
		OH		Poor to Very Poor	Not Suitable
Highly Organic Soils		Pt		Not Suitable	Not Suitable

Note: Division of GM and SM groups into subdivisions of d and u are for roads and airfields. Suffix d is used when the Liquid Limit (LL) is 28 or less and the Plasticity Index (PI) is 6 or more. Suffix u is used when the LL is greater than 28.

Table 3.2. Soil Characteristics (Part 2).

Soil Divisions		Group Symbol		Potential Frost Action	Compressibility and Expansion
Coarse Grained Soils	Gravel and Gravelly Soils	GW		None to Very Slight	Almost None
		GP		None to Very Slight	Almost None
		GM	d	Slight to Medium	Very Slight
			u	Slight to Medium	Slight
		GC		Slight to Medium	Slight
	Sands and Sandy Soils	SW		None to Very Slight	Almost None
		SP		None to Very Slight	Almost None
		SM	d	Slight to High	Very Slight
			u	Slight to High	Slight to Medium
		SC		Slight to High	Slight to Medium
Fine Grained Soils	Silts and Clays LL<50	ML		Medium to Very High	Slight to Medium
		CL		Medium to Very High	Medium
		OL		Medium to Very High	Medium to High
	Silts and Clays LL>50	MH		Medium to Very High	High
		CH		Medium	High
		OH		Medium	High
Highly Organic Soils		Pt		Slight	Very High

Note: Division of GM and SM groups into subdivisions of d and u are for roads and airfields. Suffix d is used when the LL is 28 or less and the PI is 6 or more. Suffix u is used when the LL is greater than 28.

Table 3.3. Soil Characteristics (Part 3).

Soil Divisions		Group Symbol		Drainage Characteristics	Unit Dry Weight	Field CBR
Coarse Grained Soils	Gravel and Gravelly Soils	GW		Excellent	125 – 140	60 – 80
		GP		Excellent	110 – 130	25 – 60
		GM	d	Fair to Poor	130 – 145	40 – 80
			u	Poor to Impervious	120 – 140	20 – 40
		GC		Poor to Impervious	120 – 140	20 – 40
	Sands and Sandy Soils	SW		Excellent	110 – 130	20 – 40
		SP		Excellent	100 – 120	10 – 25
		SM	d	Fair to Poor	120 – 135	20 – 40
			u	Poor to Impervious	105 – 130	10 – 20
		SC		Poor to Impervious	105 – 130	10 – 20
Fine Grained Soils	Silts and Clays LL<50	ML		Fair to Poor	100 – 125	5 – 15
		CL		Impervious	100 – 125	5 – 15
		OL		Poor	90 – 105	4 – 8
	Silts and Clays LL>50	MH		Fair to Poor	80 – 100	4 – 8
		CH		Impervious	90 – 110	3 – 5
		OH		Impervious	80 – 105	3 – 5
Highly Organic Soils		Pt		Fair to Poor	--	--

Note: Division of GM and SM groups into subdivisions of d and u are for roads and airfields. Suffix d is used when the LL is 28 or less and the PI is 6 or more. Suffix u is used when the LL is greater than 28. CBR is estimated.

3.3.7. Soil Penetration Shear Test. On-site soil testing is an accurate method to determine soil CBR values. These tests are routinely accomplished using man-portable (**Figure 3.3**) or truck-mounted penetration shear test instruments. For detailed information on soil penetration shear testing, see TSPWG 3-260-03.02-19. In addition, review AFH 32-1034 for specific soil testing and data recording procedures. The Airfield Cone Penetrometer and Dynamic Cone Penetrometer are two of the man-portable instruments used in the field to estimate CBR.

Figure 3.3. Man-Portable Soil Penetration Test Instrument.



3.3.7.1. The Airfield Cone Penetrometer (ACP) is a probe-type instrument with a 0.375-inch-diameter rod with a cone attached to one end and a handle/load indicator on the other (**Figure 3.4**.). When pushed down through the soil the ACP gives an airfield index of soil strength; these airfield index's correlate to the CBR values shown in **Figure 3.5**. This instrument was once commonly used by Special Tactics Teams for expedient evaluations because of its portability and simple operation. Its range is limited to 0 to 18 CBR and it will not penetrate many crusts, thin base courses, or gravel materials that may overlie soft soil layers. Consistency of test results is also difficult due to variability of soil strengths that impact the rate of penetration. The ACP is primarily now used in covert operations where use of a Dynamic Cone Penetrator (DCP) draws unwanted attention to personnel.

Figure 3.4. Airfield Cone Penetrometer (ACP).

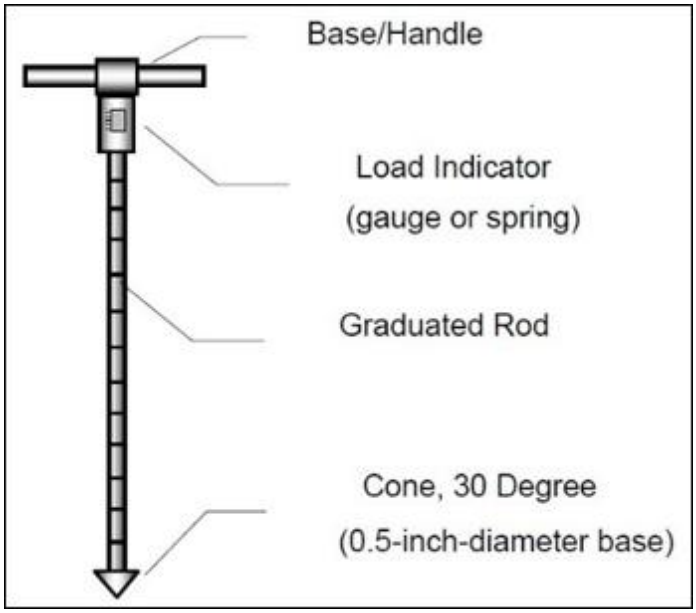
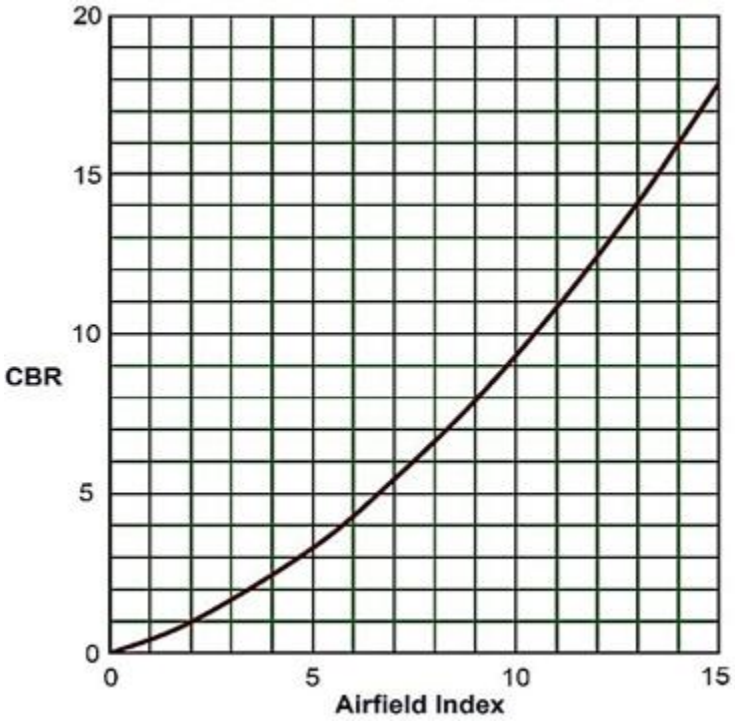


Figure 3.5. Correlation of Airfield Index and California Bearing Ratio.



3.3.7.2. The Dynamic Cone Penetrometer is an alternative method for estimating CBR. It will measure soil strengths ranging from 1 to 100 CBR. The DCP consists of a stainless-steel rod, 16 millimeters in diameter, with a cone attached to one end (**Figure 3.6**.). The DCP operator drives the cone into the soil by an 8-kilogram (17.6-pound) sliding hammer dropped from a height of 575 millimeters. The angle of the cone is 60 degrees and the diameter of the base of the cone is 20 millimeters. The rod may be either a scored version or smooth, requiring the use of an adjacent measuring scale. Operators use disposable cones that mount on an adapter in situations where the cone is difficult to remove from the soil. This disposable cone remains in the soil. Use of disposable cones can increase the number of tests accomplished per day. **Note:** The sliding hammer on dual-mass DCPs have two separate weights equaling 17.6 pounds; the hammer's outer sleeve or doughnut (7.5 pounds) can be removed, allowing use of the inner 10.1 pound weight alone when testing weaker (CBR < 10) soils. However, tests using the inner weight alone are not generally recommended for Air Force evaluations. For more information, see TSPWG 3-260-03.02-19 and ASTM D6951M-18, *Standard Test Method for Use of the Dynamic Cone Penetrometer in Shallow Pavement Applications*.

Figure 3.6. Dynamic Cone Penetrometer (DCP).



3.3.7.2.1. At least two people are required to operate the DCP. One person, the operator, holds the device by its handle in a vertical position and taps the device using the slide hammer until the base of the cone is flush with the surface of the soil. The second person, or recorder, then measures the distance between the cone and the surface, ensures the device remains in a vertical position, measures the cone penetration, counts the number of hammer drops between measurements, and records the data. The DCP operator should:

3.3.7.2.1.1. Adjust the number of blows between measurement increments in an attempt to record a measurement approximately every 25mm of penetration.

3.3.7.2.1.2. Increase the number of blows between readings if the measurement penetration is less than 20mm.

3.3.7.2.1.3. Decrease the number of blows if the measurement is greater than 30mm, for greater sensitivity of the measurement.

3.3.7.2.2. Both the operator and recorder should be alert to any sudden increases in cone penetration rates, which indicate weaker soil layers. Consult AFH 32-1034 and TSPWG 3-260-03.02-19 for detailed soil testing procedures using the DCP. See **Table 3.4** for DCP kit contents.

Table 3.4. Dynamic Cone Penetrometer Kit.

1	case assembly
1	bottom rod threaded and welded to the anvil
1	dual-mass hammer
1	vertical scale in millimeters and inches
1	go/no go gauge
6	hardened, 60-degree, fixed cones
3	hardened cone adapters and 200 disposable cones
2	channel lock pliers
1	can of light, lubricating oil
1	thread-locking compound
1	hexagonal wrench set (5/64 to 1/4 inch)

3.4. Field Expedient Identification. Although soil testing in a laboratory is usually the preferred method, during the initial stages of a contingency operation, a lack of time or facilities may make laboratory testing difficult. Consider field-testing to assess soil properties and classify soil material. The calculations can distinguish between different soil types and properties and estimate potential engineering behavior of the soil during and after construction. These expedient classifications are approximations and should not be used for designing permanent or semi-permanent road construction projects.

3.4.1. Equipment. Practically, all field-expedient soil testing is possible with no equipment or accessories other than a small amount of water. However, improved accuracy and uniformity of results increase by using certain equipment. The equipment listed below in **Table 3.5** is easily transportable and should be available in most engineer units. If necessary, simply improvise items using available resources.

Table 3.5. field-expedient soil testing equipment.

No. 40 Sieve (identifies the fine-grained portions of any soil)
No. 4 Sieve (defines the limit between gravels and sands)
No. 200 Sieve (defines the limit between sands and fines)
Entrenching tools
Pan and oven or other heating device
Mixing bowl and pestle
Scales or balances
Knife or small spatula

3.4.1.1. Sieves. A Number 40 US standard sieve. Any screen with about 40 openings per lineal inch is usable, or an approximate separation by sorting the materials by hand is an option. Use a Number 4 sieve to define the limit between gravels and sands and use the Number 200 sieve for separating sands and fines.

3.4.1.2. Excavation hand tools. A pick and shovel or a set of entrenching tools for obtaining soil samples. A hand auger is useful if the desire is to obtain samples from depths more than a few feet below the surface.

3.4.1.3. Pan and heating element (oven, stove, etc.), for drying soil samples.

3.4.1.4. Mixing bowl and pestle (rubber/wooden) for pulverizing the fine-grained portion of the soil; may be improvised using a canteen cup and wooden dowel.

3.4.1.5. Scales or balance for weighing soil samples.

3.4.1.6. Knife or small spatula for obtaining and trimming soil samples to the desired size.

3.4.1.7. Brushes or improvised cleaning tools (e.g., paintbrush, toothbrush, etc.) for removing soil particles from sieves after analysis, ensuring accurate results and preventing cross-contamination.

3.4.2. Data Collected. Soil data collected should provide a basic description of the soil and its properties. Soil properties forming the basis for the USCS are percentage of gravels, sands, or fines; shape of the grain-size distribution curve; and plasticity of the soil. **Table 3.6** lists soil properties and example data usually included in a soil description. Complete descriptions with classification symbol(s) are much more useful than a symbol or other isolated portion of a description alone. See **paragraph 3.4.3** for test methods that aid is soil identification.

Table 3.6. Soil Properties and Data Examples (Representing Poorly Graded Gravel).

Soil Property	Data
Color	Dark brown to white
Grain size (including estimated maximum grain size and estimated percent by weight of fines (material passing No. 200 sieve))	Coarse-grained soil, maximum particle size 2-3/4 inches, estimating 60 percent gravel, 36 percent sand, and 4 percent passing the No. 200 sieve
Gradation	Poorly graded (insufficient fine gravel, gap-graded)
Grain shape	Gravel particles sub-rounded to rounded
Water content (i.e., dry, moist, wet)	Dry
Plasticity	Non-plastic
Predominant type	Predominately gravel
Secondary components	Considerable sand and a small amount of non-plastic fines (silt)
Classification symbol	(GP)
Remarks (i.e., content; compactness; cohesiveness near PL; dry strength; and geologic name/source	Slightly calcareous, no dry strength, dense in the undisturbed state

3.4.3. Test Methods. Soil testing methods that aid in the field identification of soils include visual examinations, sedimentation tests, and plasticity tests. Each of these tests provide insight into the potential engineering behavior of the soil tested, both during and after construction. For additional instructions, review AFH 32-1034 and TSPWG 3-260-03.02-19, Appendix A.

3.4.3.1. Visual examination. This test should establish the color, grain sizes, grain shapes of the coarse-grained portion, approximate gradation, and some properties of the undisturbed soil.

3.4.3.1.1. Color. Color helps to identify and distinguish between various soil types. It may also indicate the presence of certain chemicals, minerals, or impurities. Describe the color of the sample when initially taken in the field at the as-sampled water content, because the color may change with changes in water content. Include the apparent moisture content at the time of identification (**Table 3.7**.). Colors generally become darker as the moisture content increases and lighter as the soil dries. Some fine-grained soil (OH, OL) with dark, drab shades of brown or gray, including almost black, contain organic material. In contrast, clean, bright shades of gray, olive green, brown, red, yellow, and white are associated with inorganic soils. Gray-blue or gray-and yellow

mottled colors frequently result from poor drainage. Red, yellow, and yellowish-brown result from the presence of iron oxides. White to pink may indicate considerable silica, calcium carbonate, or aluminum compounds.

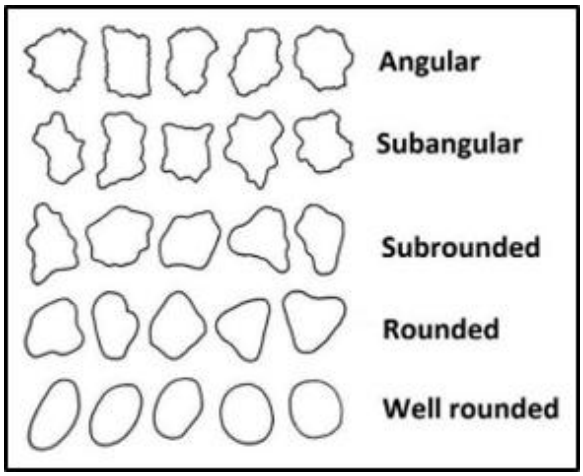
Table 3.7. Water Content of Soil.

Description	Conditions
Dry	No sign of water and soil is dry to touch
Moist	Signs of water and soil are relatively dry to touch
Wet	Signs of water and soil are definitely wet to touch; granular soil exhibits some free water when densified

3.4.3.1.2. Grain size. Establish the maximum particle size of each soil sample to determine the upper limit of the gradation curve. Gravels range down to the size of peas; sands start just below this size and decrease until the individual grains are just distinguishable by the naked eye. Silt and clay particles are indistinguishable as individual particles. See **paragraph 3.4.3.1.4** and UFC 3-220-10 for more information on grain sizes.

3.4.3.1.3. Grain shape. The shape of individual grains in a soil mass plays an important role in the engineering characteristics (strength and stability) of the soil. Illustrated in **Figure 3.7** are the five subdivisions in descending order of desirability for construction. See UFC 3-220-10, Section 1-3.2.1.3 for more detailed information.

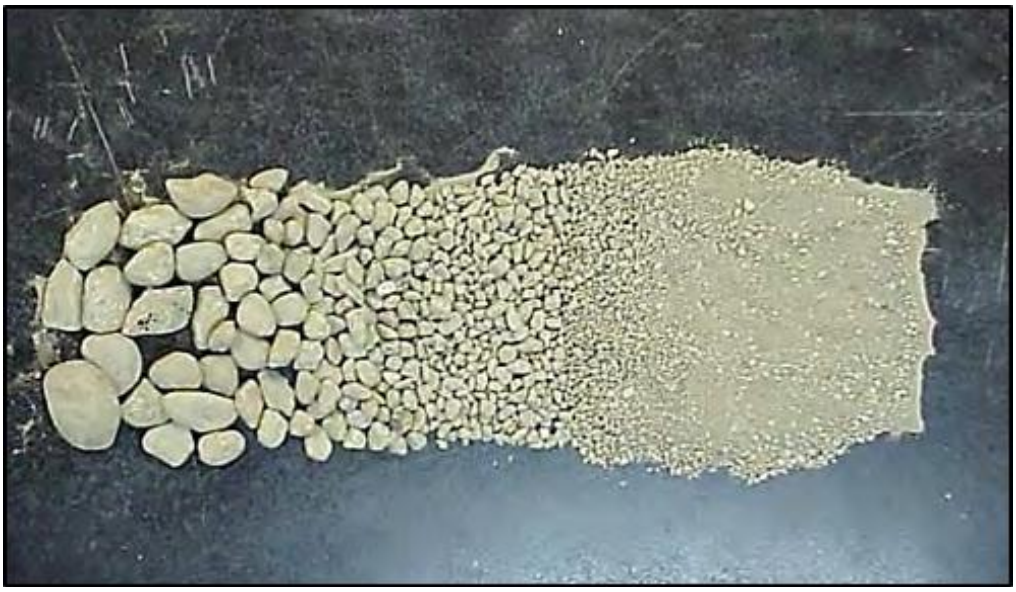
Figure 3.7. Bulky Grain-Shaped Particles.



3.4.3.1.4. Grain size distribution. Examining a dry sample spread on a flat surface can make an approximate identification. Pulverize all lumps until individual grains are exposed but not broken. The recommended method is using a rubber-faced or wooden pestle and a mixing bowl but mashing the sample underfoot on a smooth surface will suffice for an approximate identification. Separate the larger grains (gravels and some sands) by picking them out individually. Examine the remainder of the soil and estimate the proportions of visible individual particles and fines (**Figure 3.8**.). Convert these estimates into percentages by weight of the total sample. If the fines exceed 50 percent,

consider the soil fine-grained (M, C, or O). If the coarse material exceeds 50 percent, the soil is coarse-grained (G or S). Examine coarse-grained soil for gradation of the particle sizes from the largest to the smallest. A good distribution of all sizes means the soil is well graded (W). Overabundance or lack of any size means the material is poorly graded (P). Estimate the percentage of the fine-grained portion of the coarse-grained soil for further classification. Fine-grained soils and fine-grained portions of coarse-grained soils require other tests for identification.

Figure 3.8. Grain Size Distribution.



3.4.3.1.5. Undisturbed soil properties. Characteristics of the soil in the undisturbed state may be helpful in identification. The compactness of gravels or sands may be loose, medium, or dense. Clays may be hard, stiff, or soft. Record the ease or difficulty of sample removal. The moisture content of the soil influences the in-place characteristics. It is helpful to know the weather just prior to and during the field evaluation to determine how the soil has reacted or will react to weather changes. The presence of decayed roots, leaves, grasses, and other vegetable matter in organic soils produces soil which is usually dark when moist, having a soft spongy feel and a distinctive odor of rotting organic matter. The odor may be musky and slightly offensive. The odor is especially apparent in undisturbed conditions or in fresh samples. The odor becomes less obvious as the sample is exposed to air and more intense when the wet sample is heated.

3.4.3.2. Sedimentation test. From visual examination, it is relatively easy to approximate the proportions of gravels and sands in a soil sample. Determining the proportion of fine-grained particles is more difficult but just as important. Although you can separate the fines from the sample using a No. 200 sieve, the sedimentation test is an alternate field method to separate fines from the sand particles in a soil sample.

3.4.3.2.1. Placing a small amount of the fine fraction of a soil (such as a heaping teaspoon) in a transparent cup or jar, covering it with 5 inches of water, and agitating it by stirring or shaking will completely suspend the soil in water. With cohesive soils,

it will be necessary to break up all lumps of soil before adding the water. After soil particles have been suspended in the water, they will start to settle to the bottom, beginning with the larger sized particles, in time-periods indicated in **Table 3.8**. Smaller particles will settle through water at a slower rate than large particles. Since all of the particles of soil larger than the No. 200 sieve will have settled to the bottom of the cup or jar 30 seconds after the mixture has been agitated, it follows that the particles remaining in suspension are fines.

Table 3.8. Sedimentation Test.

Approximate Time of Settlement in 5 Inches of Water	Grain Diameter	Differentiates
2 seconds	0.4 mm	Coarse sand – fine sand
30 seconds	0.072 mm (No. 200 sieve)	Sand – fines
10 minutes	0.03 mm	Coarse silt – fine silt
1 hour	0.01 mm	Silt – clay

3.4.3.2.2. Carefully pour the water containing the suspended fines into another container 30 seconds after agitation, add more water to the cup or jar containing the coarse fraction, and repeat the procedure until the water-soil mixture becomes clear 30 seconds after mixing. The cup or jar will contain the coarse fraction of the soil and the other container will hold the fines.

3.4.3.2.3. After absorbing or evaporating the water off, the relative amounts of fines and sands can be accurately determined. In clay soils, the clay particles will often form small lumps (flocculate) that will not break up in water. If after several repetitions of the test substantial amounts of clay are still present in the coarse material, the sand will feel slippery. Further mixing and grinding with a stick will be necessary to help break up these lumps.

3.4.3.3. Plasticity tests. Fine-grained soil particles are generally not classified using gradation criteria but primarily by characteristics related to plasticity. Expedient field tests can help determine the cohesive and plastic characteristics of fine-grained soil. Except where noted, these tests are performed on material passing the No. 40 sieve, and include the breaking or dry strength test, roll or thread test, ribbon test, wet shaking test, cast test, wash test, bite or grit test, shine test, feel test, and sedimentation test. For more detailed information beyond that here, see plasticity tests in TSPWG 3-260-03.02-19, Appendix A.

3.4.3.3.1. Breaking or dry strength test. Prepare a pat of soil about 2 inches in diameter and 0.5 inch thick by molding it in a wet, plastic state. Let the pat dry completely (in the sun, in an oven, or inside the engine compartment), then grasp the pat between the thumbs and forefingers of both hands and attempt to break it. If the pat breaks, try to powder it by rubbing it between the thumb and forefinger of one hand. **Note:** Dry pats of highly plastic clays often display shrinkage cracks. Breaking the pat along such a crack may not give a true indication of strength. It is important to distinguish between a break along such a crack and a clean, fresh break that indicates the true dry strength of the soil. If the:

3.4.3.3.1.1. Pat cannot be broken nor powdered by finger pressure; it is a very

highly plastic soil (CH).

3.4.3.3.1.2. Pat can be broken with great effort, but cannot be powdered, it is a highly plastic soil (CL).

3.4.3.3.1.3. Pat can be broken and powdered, but with some effort, it is a medium plastic soil (CL).

3.4.3.3.1.4. Pat breaks easily and powders readily, it is a slightly plastic soil (ML, MH, or CL).

3.4.3.3.1.5. Pat has little or no dry strength and crumbles or powders when picked up, it is a non-plastic soil (ML or MH) or (OL or OH).

3.4.3.3.2. Roll or thread test. Mix a representative portion of the sample with water until you can mold or shape it without it sticking to your fingers. Describe this moisture content as being just below the sticky limit. Prepare a nonabsorbent rolling surface by placing a sheet of glass or heavy wax paper on a flat or level support, then shape the sample into an elongated cylinder and rapidly roll the prepared soil cylinder on the surface into a thread approximately 0.125 inch in diameter. If the moist soil rolls into a thread, it has some plasticity. The number of times you can roll it into a thread without crumbling is a measure of the degree of plasticity. The higher the soil is on the plasticity chart, the stiffer the threads are as they dry out and the tougher the lumps are if the soil is remolded after rolling. Soils that cannot be rolled are non-plastic. **Note:** Micaceous silts and sands can be rolled due to the flaky nature of the mica. The wet shaking test is the only way to distinguish this property. If the:

3.4.3.3.2.1. Soil can be molded into a ball or cylinder and deformed under firm finger pressure without crumbling or cracking, it is high plasticity (CH).

3.4.3.3.2.2. Soil can be molded, but it cracks or crumbles under finger pressure, it is medium plasticity (CL).

3.4.3.3.2.3. Soil cannot be lumped into a ball or cylinder without breaking up, it is low plasticity (CL, ML, or MH).

3.4.3.3.2.4. Soil forms a soft, spongy ball or thread when molded, it is an organic material (OL or OH).

3.4.3.3.2.5. Soil cannot be rolled into a thread at any moisture content; it is a non-plastic soil (ML or MH).

3.4.3.3.3. Ribbon test. Prepare a soil sample as in the roll or thread test. Form a roll of soil about 0.5 to 0.75 inch in diameter and 3 to 5 inches long. Lay the roll across the palm of one hand (palm up) and starting at one end, squeeze the roll between the thumb and forefinger over the edge of the hand to form a flat unbroken ribbon about 0.125 to 0.25 inch thick. Allow the ribbon as formed to hang free and unsupported. Continue squeezing and handling the roll carefully to form the maximum length of ribbon that can be supported only by the cohesive properties of the soil. If the:

3.4.3.3.3.1. Sample holds together for a length of 8 to 10 inches without breaking; it is highly plastic and highly compressive (CH).

3.4.3.3.3.2. Soil can be ribboned only with difficulty to 3- to 8-inch lengths; it is low plasticity (CL).

3.4.3.3.4. Wet shaking test. Form a ball of soil about 0.75 inch in diameter, moistened with water to just below the sticky limit. Smooth the soil pat in the palm of the hand with a knife blade or small spatula, shake it horizontally, and strike the back of the hand vigorously against the other hand. When shaking, water comes to the surface of the sample producing a smooth, shiny, or livery appearance. Squeeze the sample between the thumb and forefinger of the other hand. The surface water will disappear. The surface will become dull and the sample will become firm, resisting deformation. Cracks will occur as pressure is continued and the sample will crumble. If the water content is still adequate, shaking the broken pieces will cause them to liquefy again and flow together. This process can only occur when the soil grains are bulky and non-cohesive. It is easy to identify very fine sands and silts by this test. Even small amounts of clay will tend to retard the reaction to this test. A rapid reaction is typical of non-plastic, fine sands and silts. A sluggish reaction indicates slight plasticity, indicating the silt has small amounts of clay or organic silts. **Note:** No reaction at all does not indicate a complete absence of silt or fine sand.

3.4.3.3.5. Cast test. Compress a handful of damp (not sticky) soil into a cylinder and observe its ability to be formed and handled. If the:

3.4.3.3.5.1. Soil crumbles when touched - GP, SP, SW, GW.

3.4.3.3.5.2. Soil cast withstands careful handling - SM, SC.

3.4.3.3.5.3. Soil cast can be handled freely - ML, MH.

3.4.3.3.5.4. Soil cast withstands rough handling - CL, CH.

3.4.3.3.6. Wash test. Place a small dry sample of soil into the palm of the hand and cover with water. Note how quickly the water discolors and how long the fines are suspended. One variation is to look for mud puddles or create them, disturb the soil surface and note how the water discolors and how long the fines are suspended. If the water becomes completely discolored and hides the sand particles there is evidence of greater than 5 percent silt content.

3.4.3.3.7. Bite or grit test. Grind a small pinch of soil lightly between the teeth. For sandy soils, the sharp hard particles of even fine sands will grate very harshly between the teeth and will be highly objectionable. For silty soils, silt grains are not particularly gritty, but their presence is still quite unpleasant and easily detected. With clayey soils, the clay grains feel smooth and powdery like flour, and dry lumps will stick when lightly touched with the tongue. **Note:** Ideally, perform this test on material passing the No. 200 sieve to avoid biting into sand and providing separation between silty and clayey fines.

3.4.3.3.8. Shine test. Rub a clay sample with a fingernail or smooth metal surface such as a knife blade. Highly plastic clay will produce a definite shine; lean clays will remain dull.

3.4.3.3.9. Feel test. For consistency, squeeze a piece of undisturbed soil between the thumb and forefinger; it may be hard, stiff, brittle, friable, sticky, plastic, or soft. Remold the soil by working it between the hands. This can indicate the natural water content. Clays that become fluid on remolding are probably near their liquid limit (LL); if they remain stiff and crumble, they are probably below their LL. Check the texture by rubbing a portion of fine-grained soil between the fingers or on a more sensitive area such as the inside of the wrist. Results are similar to the bite or grit test. **Note:** Ideally, perform this test on material passing the No. 200 sieve to avoid sand particles giving an erroneous reaction.

3.4.3.4. Recording Test Results. **Table 3.9** is a convenient way to track various field identification tests. As tests are completed, mark the results on the chart. The results from the different tests may vary, but by plotting the test results, you will have a general indication of the soil type. Essentially, soil types and their characteristics can affect how the soil will behave as a construction material. Once soil testing and surveys are completed, consider the preparations and actions needed to accomplish the specific expedient roadway construction or repair project.

Table 3.9. Recording Field Identification Test Results (Notional).

Field Identification of Soils						
Test	**Material**	**Soil Types**				
		ML	MH	CL	CH	OL/OH
Dry strength	< 40 sieve (wet)	No to low	Low to medium	Medium to high	Very high	Low
Roll/thread	< 40 sieve (sticky)	Low	Low to medium	Medium	High	Spongy
Ribbon	< 40 sieve (sticky)	No cohesion	Little cohesion	3 to 8 inches	8 to 10 inches	
Wet shake	< 40 sieve (sticky)	Slow to rapid	No to slow	No to slow	No	
Cast	Damp	Handle freely		Handle roughly		
Bite/feel	< 40 (< 200) Sieve	Unpleasant		Smooth		
Shine		Dull			Shine	
Wash		Discolors quickly, > 5% silt				
Dust		> 10% silt				
Sedimentation		30 seconds		1 hour		

3.5. References. Table 3.10 list references in this chapter relating to soil characteristics, testing, and classifications.

Table 3.10. Chapter 3 References.

Field-Expedient Soil Classifications and Identification	
UFC 3-220-10	*Soil Mechanics (DM 7.1)*
AFH 32-1034	*Materials Testing*
TSPWG 3-260-03.02-19	*Airfield Pavement Evaluation Standards and Procedures*, Appendix A
ASTM D2487-17	*Standard Practice for Classification of Soils for Engineering Purposes (USCS)*
ASTM D6951M-18	*Standard Test Method for Use of the Dynamic Cone Penetrometer in Shallow Pavement Applications*

Chapter 4

SOIL STABILIZATION FOR EXPEDIENT ROADS

4.1. General Information. Soil stabilization methods are often used to strengthen expedient roads at contingency locations and many of the methods can be effective for extended periods. Mechanically stabilized soil mixtures are widely used as surfaces for military roads to carry light traffic where difficult soil conditions exist. Additive stabilization is often used to stabilize soils when mechanical stabilization methods are inadequate and replacing an undesirable soil with a desirable soil is not possible or is too costly. In addition, both processes are much less costly, material wise, than standard road development practices involving asphalt or concrete.

4.2. Overview. Soil stabilization can increase soil strength and durability, prevent erosion, and lower dust generation. The desired result is the creation of a soil material or soil system that remains in place under the design use conditions for the design life of the project. There are numerous methods to stabilize soils; however, all stabilization methods fall into two broad categories—mechanical and additive stabilization. Some stabilization techniques use a combination of the two. Both methods are addressed here; for additional information on soil stabilization methods, see UFC 3-250-11, *Soil Stabilization and Modification for Pavements*.

4.3. Mechanical Stabilization. Mechanical stabilization relies on physical processes to stabilize the soil, altering the physical composition of the soil (soil blending) or placing a barrier in or on the soil to obtain the desired effect. Mechanical stabilization through soil blending is the most economical and expedient method of altering the existing material. When soil blending is not feasible or does not produce a satisfactory soil material, consider geosynthetics (i.e., geotextiles, geomembranes, geogrids, mats, nets, and other composites) or additive stabilization methods as potential options.

4.3.1. Soil Blending and Compaction. Mechanically mix or blend two or more gradations of soil material to get a mixture that meets the required specifications. For example, the natural soil at a selected location has low load-bearing strength because of excess clay, silt, or fine sand; however, within a reasonable distance, suitable granular materials may be present, and if blended with the existing soils can markedly improve the soil at a much lower cost in labor and materials than involved with importing surfacing. Consider checking local areas for potential sites that may provide the required unconsolidated materials for soil blending. Use soil surveys, intelligence reports, and local inhabitants such as contractors and engineers as sources of additional information about the location of borrow material, sand and gravel deposits, and highly organic soils. See **Attachment 3** for more information on types and sources of potential aggregate materials.

4.3.1.1. The sand-clay road is a type of mechanically stabilized soil surface. It consists of a natural or artificial mixture of sand and clay that is graded and drained to form a road surface. While difficult to obtain, the Plasticity Index (PI) should be less than 5 and Liquid Limit (LL) less than 25 in case this layer becomes a subbase after placing additional layers above the sand clay.

4.3.1.2. Sand-clay roads will carry light traffic reasonably well and heavy traffic except under bad weather conditions. The amount of moisture these roads absorb determines their stability under traffic loads. They require dust control, blading, and dragging. Sand-clay roads withstand traffic better than ordinary earthen roads, but their use is limited to areas where a suitable mixture of sand and clay occurs naturally or where a deficiency of either soil is readily corrected. As a base course for future surfacing, sand-clay roads produce poor results, unless engineers reduce the plasticity by additive stabilization with an agent such as lime.

4.3.1.3. For wearing surfaces on unpaved roads, sand-clay-gravel mixtures (GC) are generally considered the most satisfactory. The addition of fine gravel (slightly larger than the No. 4 sieve) usually adds stability. However, the percentage of fines should be kept at a reasonable level (below approximately 15%).

4.3.1.4. Gravel roads consist of a compacted layer of gravelly soil that meets the plasticity requirements for mechanically stabilized soil mixtures. The gravel is graded from coarse to fine, with a maximum allowable size of 1 inch. A natural pit- or bank-run gravel may meet these requirements without further processing other than screening. Some pit- or bank-run gravels may require both screening and washing to meet the requirements. River-run gravels normally require the addition of binder to the soil, as do mechanically stabilized soil mixtures. River-run gravels may also require crushing to provide a rough, angular surface rather than the natural, smooth surface characteristic of river-run materials. The ability to carry heavy, sustained traffic depends on the:

4.3.1.4.1. Strength and hardness of the gravel.

4.3.1.4.2. Cohesiveness of the clay binder.

4.3.1.4.3. Thickness of the layer.

4.3.1.4.4. Stability of the subgrade.

4.3.1.5. Listed in **Table 4.1** are typical gradation requirements for aggregate surface courses. Personnel can build these roads rapidly, even in cold weather. Like other untreated surfaces, these roads require considerable maintenance such as blading and dust control in dry weather. During wet weather, proper maintenance is difficult, especially under heavy traffic. **Note:** Gravel road surfaces with low plasticity make excellent base courses for later-stage pavements.

Table 4.1. Gradation for Aggregate Surface Courses.

Gradation for Aggregate Surface Courses (% passing by weight)				
Sieve Size	**No. 1**	**No. 2**	**No. 3**	**No. 4**
1 in. (25 mm)	100	100	100	100
3/8 in. (9.5 mm)	50-85	60-100	---	---
No. 4 (4.75 mm)	35-65	50-85	55-100	70-100
No. 10 (2.00 mm)	25-50	40-70	40-100	55-100
No. 40 (0.425 mm)	15-30	24-45	20-50	30-70
No. 200 (0.075 mm)	8-15	8-15	8-15	8-15
Note: The percent by weight finer than 0.04 in. (0.02 mm) should not exceed 3 percent.				

4.3.1.6. On-Site Blending Steps. Mechanical mixing of soil-aggregate materials can be accomplished at the construction site, a central plant, or a borrow area, and then spread and compacted to the required densities by conventional means. If the mixture does not meet the specifications after blending, other stabilization methods may be an option. See **Table 4.2** for the steps involved in on-site blending.

Table 4.2. On-Site Soil Blending Steps.

Soil Blending (On-Site)
Preparation: - Shape the area to crown and grade - Scarify, pulverize, and adjust the moisture content of the soil, if needed - Reshape the area to crown and grade
Add imported soil materials using one of the following methods: - Distribute evenly by means of an improved stone spreader - Use spreader boxes behind dump trucks - Tailgate each measured truck, loading to cover a certain length - Dump in equally spaced piles; form into windrows with a motor grader before spreading
Mixing: - Add water to obtain moisture content of about 2 percent above optimum and mix with a rotary mixer, pulvimixer, blade, scarifier, or disk - Continue mixing until the soil and aggregate particles are in a uniform, well-graded mass - Blade to crown and grade, if needed
Compaction: - Compact to specifications determined by the results of compaction tests performed on the blended soil material - Select the appropriate types(s) of compaction equipment, based on the gradation characteristics of the blended soil

4.3.2. Geotextiles. Another method available for improving the condition of soils besides blending and chemical stabilization involves the use of geotextiles (**Figure 4.1**.). The term geotextile refers to any permeable textile used with foundation, soil, rock, earth, or any other geotechnical engineering-related material as an integral part of a human-made project, structure, or system. Geotextiles are also commonly referred to as geofabrics, engineering fabrics, or just fabrics. The terms geotextiles and geofabrics are used interchangeably here.

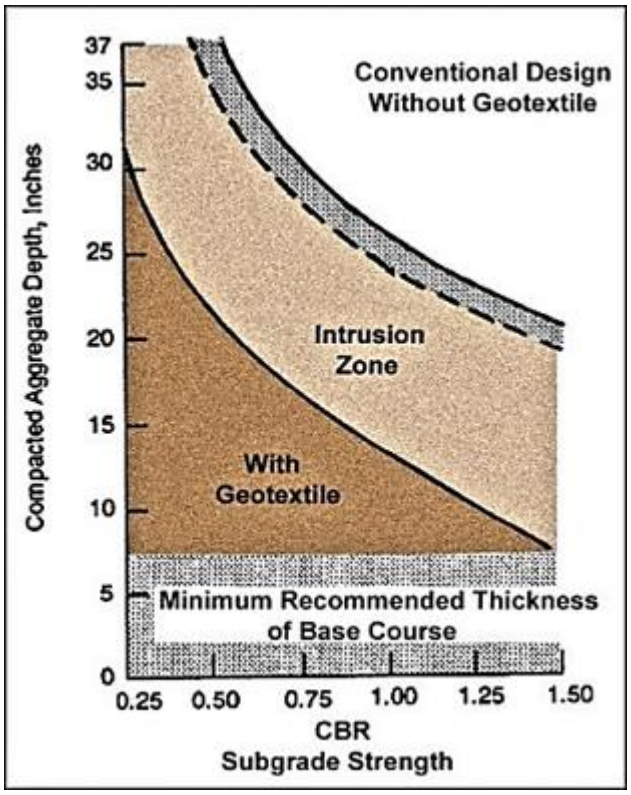
Figure 4.1. Installation of Geotextile Fabric on Expedient Roadway.



4.3.2.1. Geotextiles serve four primary functions: reinforcement, separation, drainage, and filtration. In relation to expedient road construction geotextile fabrics can be used to reinforce weak subgrades and separate soil layers. In theater-of-operations horizontal construction, the primary concern is with separating and reinforcing low load-bearing soils to reduce construction time. These geotextiles were also used to prevent the pumping of subgrade fines into the structural layers because of the soft soil conditions and heavy truck traffic. In many situations, using geofabrics can reduce soil requirements, which saves time, materials, and equipment costs.

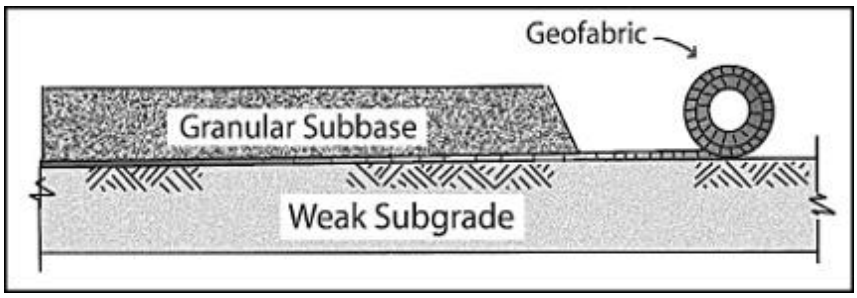
4.3.2.2. The design engineer attempts to reduce the thickness of a pavement structure whenever possible. Tests show that for low load-bearing soils, generally anything less than five CBR, the use of geofabrics can often decrease the amount of subbase and base course materials required. The fabric lends its tensile strength to the soil to increase the overall design strength. **Figure 4.2** illustrates this concept.

Figure 4.2. Comparison of Aggregate Depth Requirements.



4.3.2.3. Engineers also use geosynthetics fabrics for rapid stabilization of swamps, peat bogs, and beach sands. The normal solution is to remove the poor base material by a process commonly referred to as "mucking." However, this is not an option during most contingencies for a number of reasons. Specifically, the removal process is often difficult and time-consuming. Furthermore, when using mucking, the area should have a sound, stable soil underneath the soft soil and suitable fill material nearby. Consequently, during contingencies engineers usually place a bridge of base course construction material directly over the weak soil, separated by a layer of geosynthetics material, as illustrated in **Figure 4.3**. This method of separating and confining weak subgrades has historically performed well as an expedient road construction option in many situations, however different types of subgrades can result in different performance.

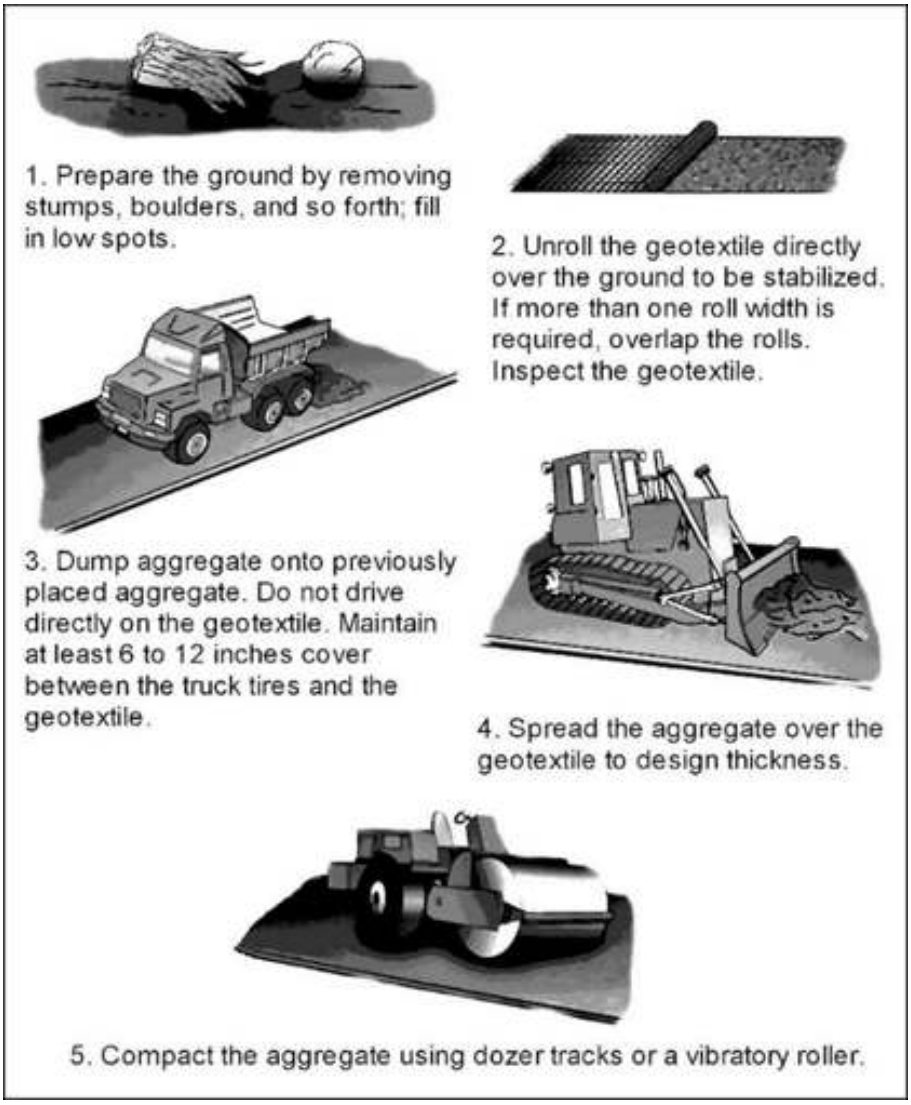
Figure 4.3. Geosynthetic Fabrics Separate and Confine Weak Subgrades.



4.3.2.4. A wide range of geofabrics are available to meet various soil conditions. Selection of the proper geosynthetics will depend on the actual soil and hydraulic conditions. **Note:** See UFC 3-250-01 and UFC 3-220-08FA (Inactive) for additional information on road design and the types and application of geosynthetics.

4.3.2.5. Begin construction by clearing, grubbing, and grading along the planned route. Illustrated in **Figure 4.4** is the basic geotextile construction sequence for the direct fill process.

Figure 4.4. Basic Geotextile Construction Sequence.



4.3.3. Geocells. Using Geocells (often referred to as sand grid) is another method to mechanically stabilize loose sands and produce a load-distributing base layer for expedient road construction or repair. Sand-grid road construction greatly improves trafficability of wheeled vehicles over sand and sandy soils. The sand grid stabilizes the soil by confining the sand or sandy materials in interconnected cellular elements improving load-bearing capability. It has application in areas where sandy materials are abundant and quality construction aggregates are not available. The plastic honeycomb grids are typically manufactured and shipped in unexpanded sections of various sizes, which are easily expanded for field use (**Figure 4.5**.). Grid sections are relatively lightweight and expand to form a honeycomb arrangement; for example, an 11 feet by 5 inch by 8 inch section weighs approximately 100 pounds and covers an area 8 by 20 feet and 8 inches thick. For more information on using sand grids beyond what is addressed here, see Army Technical Manual (TM) 3-34.64, *Military Soils Engineering*.

Figure 4.5. Expanding Sand-Grid Material.



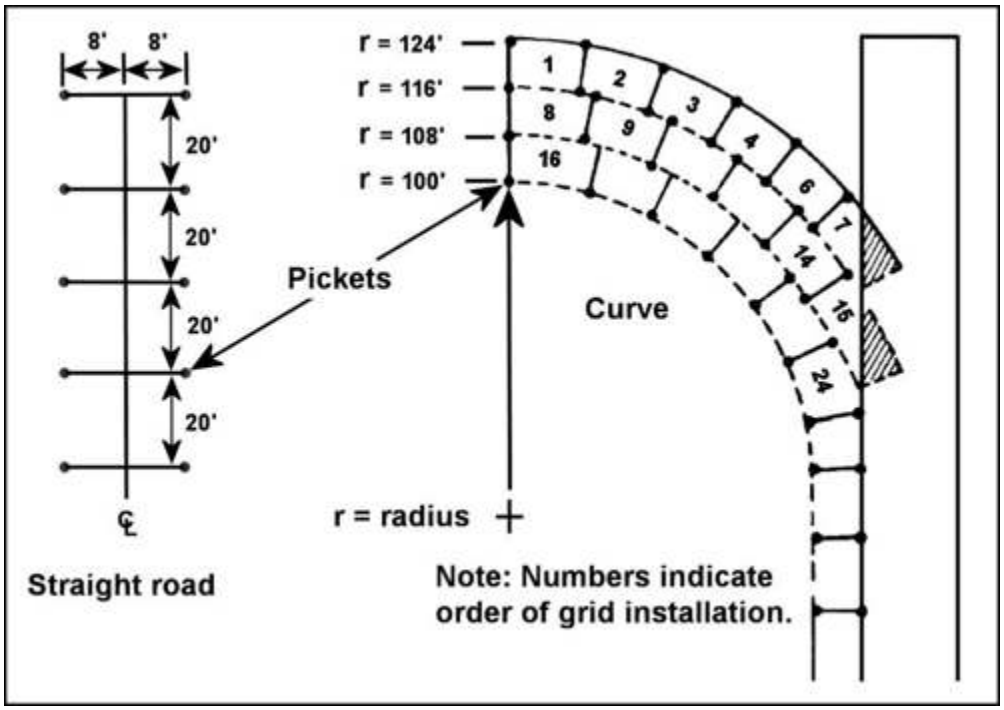
4.3.3.1. Work crews can quickly construct a sand-grid road using bucket loaders, graders, and vibratory compaction rollers for site preparation. The use of rough-terrain forklifts, water distributors, asphalt distributors, shovels, and plywood sheets can increase the effectiveness and speed of construction. Once the equipment is on hand, accomplish site preparations, i.e., perform normal cut and fill operations to reach the desired grade; back blade the surface for smoothness; compact the sand subgrade at a moisture content approaching saturation using the vibratory roller. Afterwards, layout the sand-grid pattern and begin installation. Installation rate is approximately 65 square feet per man-hour. **Table 4.3** below is the recommended equipment for building sand-grid roads using geocell example above where each section covers an area 8 by 20 feet.

Table 4.3. Recommended Equipment for Sand Grid Geocells.

Graders
Bucket loaders (smooth-bucket, no teeth)
Rough-terrain forklifts
Vibratory rollers
Water distributors
Bituminous distributors
Long-handled round-pointed shovels
Plywood sheets (3/8-inch by 8-foot by 4-foot)

4.3.3.2. Sand-grid road designs are simple, require minimum skilled labor, and use local materials whenever possible. Install the grid by setting the stakes and string lines in 8-foot by 20-foot boxes in the desired pattern. See **Figure 4.6** for layout patterns.

Figure 4.6. Example Sand-Grid Layout Pattern.



4.3.3.3. Expand each grid section by pulling the ends outward to slightly over 20 feet, then shaking the section in the air to obtain uniform cell openings. Place the section according to your layout pattern. Use pickets or place sand in the corner cells and approximately every fifth cell on the sides to anchor the grid in place.

4.3.3.4. To construct joints between grids, use small, 3/8-inch plywood sheets to allow workers to stand on the unfilled grid to have access to the joints. For end joints, the rounded end cells from different sections should touch each other. For longitudinal (side-by-side) joints, interlock the “welded” cell portions of each section, as if fitting a puzzle. Fill the jointed cells with sand. If needed, level the joints by placing plywood over the joints and have crews walk on the plywood.

4.3.3.5. Fill each grid section using a bucket loader. Drop the sand vertically into the cells from a height of at least 2 feet. Do not push the sand forward or cells will be displaced.

Overfill the grids by 2 to 4 inches so bucket loaders can operate on the sand-filled grid layer without damaging any cells. Use shovels and rakes to spread overfill across the cells to form a uniform layer. Have bucket loaders vary wheel paths to achieve uniform compaction.

4.3.3.6. If water is readily available, wet the sand using a water distributor. This will significantly aid in the compaction process. Compact the road surface with one or two passes of the vibratory roller. Remove excess sand from the grid surface with a grader or by back-blading with a bucket loader. Do not use a bucket loader with teeth because cell damage can result. Compact the road surface again with one pass of the vibratory roller. **Note:** Do not over compact the surface because it could damage the sand grid and result in premature failure.

4.3.3.7. To stabilize and seal the road surface, a sand-asphalt surfacing is formed by spraying a suitable liquid-asphalt cement, emulsion, or cutback (rapid curing (RC) 250 at 165^{+0} F is preferred) on the surface of the sand-grid layer. Spray the asphalt product on the road surface and allow enough time for the asphalt product to soak into the grid structure (usually about 10 hours). The asphalt should penetrate into the top 1/2 to 1 inch of sand in the cells. Apply a very light coat (1/4 inch) of blotter sand using shovels. Compact the road using one pass of a non-vibrating roller. Vibrating the road at this point will break the asphalt bonds.

4.3.3.8. To further increase road life, apply and compact a 3-inch surface coat of 1-inch (maximum) gravel, if available. This layer will add a protective cover over the sand-grid road, significantly increasing the road life. Over a sand subgrade, this sand-grid road is capable of supporting over 10,000 passes of heavy truck traffic including tandem axle loads of up to 53,000 pounds. However, avoid tracked vehicles traveling over this road, as their tracks will easily damage the grid cells. **Figure 4.7** shows an expedient sand-grid road under construction over unstable sandy terrain.

Figure 4.7. Using Sand-Grid in Road Construction.

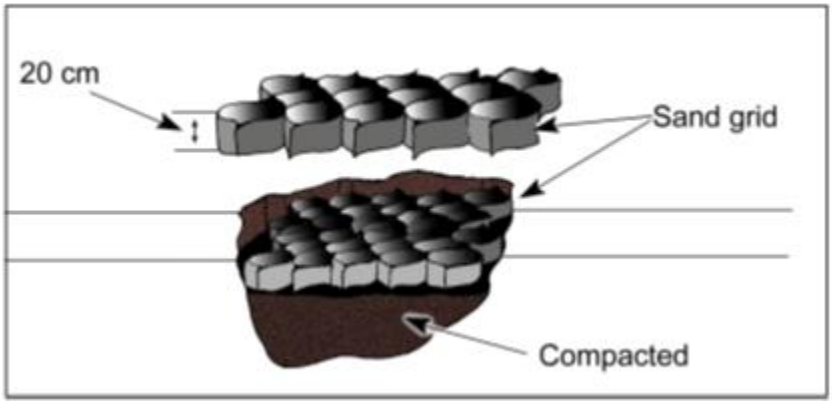


4.3.3.9. When making road repairs with this material, place the initial 4-inch layer of sand-grid in the crater, parallel to the centerline of the roadway. Expand the sand-grid, using

shovels to fill the grid-edge sections to hold it in place. Cover the entire bottom of the cavity with grid material in this manner so it conforms to the shape of the repair, curving or cutting as necessary (**Figure 4.8**.). **Note:** Sections of the grid that do not fully expand only serve to add strength to the base.

4.3.3.10. After positioning the first layer, fill the grid, working from one edge of the repair towards the center. Compact the cells level with the pavement using a rubber-tire, steel-wheel roller, or vibratory-plate tampers and remove any excess material after compaction. If more than one layer is needed, follow the previous instructions for road construction.

Figure 4.8. Sand-Grid Roadway Repair.



4.3.4. Geosynthetics Fibers. Sometimes referred to as geofibers or sand fibers, these products may be an option in rapid road construction to overcome sandy and soft soil deficiencies. Sandy soil stabilization using this method involves mixing small amounts of hair-like, polypropylene fibers into moist sand with a field or rotary mixer (**Figure 4.9**.). The sand-fiber mix is then compacted with a smooth-drum vibratory roller.

Figure 4.9. Blending Fibers into Soil for Stabilization.



4.3.4.1. Next, add a wearing surface by spraying a resin-modified emulsion or emulsified asphalt onto the surface to bond the sand grains with the fiber filaments and protect the sand-fiber surface. Using this soil stabilization and road construction method, engineer

work crews can quickly provide expedient roadway surfaces at remote sites, over beaches, or across desert sands with fewer resources than required for other road-building methods. Testing indicates that roads constructed with this technology can carry over 10,000 passes of heavy wheeled vehicle traffic with little or no maintenance required.

4.3.4.2. Sand-fiber stabilization uses existing military construction equipment and requires no special construction skills. In addition, sand-fiber soil stabilization is an option for a wide variety of sands and silty soils found around the world.

4.3.4.3. Although many types of synthetic fibers are available, polypropylene fibers are low cost, have different lengths and diameters, and have an extensive test history. Primarily, they are available as monofilament, fibrillated, and tape fibers. Testing has shown that monofilament and tape fibers are more efficient for rapid stabilization.

4.3.4.4. Fibrillated fibers can require extensive mixing for maximum efficiency. Polypropylene fibers mixed into soil without adhesives (cement) rely exclusive on mechanical interlock, friction, and fiber entanglement between soil particles to anchor the fiber in place. Thus, a high surface area fiber (such as a tape fiber) may be more efficient in a fine-grained clay soil than in a coarse-grained soil (such as gravel) where fewer soil grains are in contact with the fiber.

4.3.4.5. The most important fiber property for soil stabilization is fiber length. Longer fibers generally result in improved performance but hamper construction. The longer fibers are more difficult to mix and considerable balling may occur, diminishing efficiency. Testing of different fiber lengths indicate fiber lengths from 3/4 inch to 1-1/2 inch results in good soil improvements with minimal balling. Furthermore, adding too many fibers can cause poor performance by increasing fiber-to-fiber contact resulting in a more porous mix with lower density. Tests indicate 1 percent fiber is the optimum amount for wheeled traffic loads.

4.4. Additive Soil Stabilization. Additive soil stabilization involves adding a manufactured commercial product to the soil in the proper quantities to improve the quality of the soil layer. Additives may stabilize the soil chemically, physically or by a combination of the two. Below is basic information on methods, application, and equivalent thickness. See UFC 3-250-11 and UFC 3-250-01 for more information on soil stabilization techniques.

4.4.1. Methods. Examples of pure chemical stabilizers are lime, acids, and enzymes that cause a chemical change in the soil. Listed in **Table 4.4** are soil stabilization applications and methods. While some of the methods appear to be labor intensive, they are quite the opposite when compared against standard road construction methods.

4.4.1.1. Examples of pure physical stabilizers are bitumen's that bind the soil by adhesion and fibers that bind by mechanical friction and interpenetration. Other stabilizers such as lime-cement-fly ash (LCF) may bind soil particles together but also induce chemical changes in the soil. Most additive stabilization projects involve the use of cement, lime, or fly ash.

4.4.1.2. Emulsion polymers (sometimes referred to as latex) used for soil stabilization are generally vinyl acetate or acrylic-based polymers suspended in water by surfactants. They typically consist of 45 to 50 percent polymer content by weight. Analogous to latex paint, the emulsion polymers cure by water loss. The polymers used for soil stabilization typically

have excellent tensile and flexural strength, adhesion to soil particles, and resistance to water. However, limits to these materials often include a short shelf life (less than 2 years) due to bacterial growth and settling. **Note:** Do not mix emulsion polymers with gray water or salt water for dilution.

Table 4.4. Stabilization Methods for Specific Applications.

Application	Soil Type	Treatment Method
Subgrade Stabilization Improves load-carrying and stress-distribution characteristics	Fine-grained	SA, SC, MB, C
	Coarse-grained	SA, SC, MB, C
	Clays of low PI	C, SC, CMS, LMS, SL
	Clays of low PI	SL, LMS
Reduces frost susceptibility	Fine-grained	CMS, SA, SC, LF
	Clays of low PI	CMS, SC, SL, LMS
Improves waterproofing and runoff	Clays of low PI	CMS, SA, LMS, SL
Controls shrinkage and swell	Clays of low PI	CMS, SC, C, LMS, SL
	Clays of low PI	SL
Reduces resiliency	Clays of low PI	SL, LMS
	Elastic silts or clays	SC, CMS
Base Course Stabilization Improves substandard materials	Fine-grained Clays of low PI	SC, SA, LF, MB, SC, SL
Improves load-carrying and stress-distribution characteristics	Coarse-grained	SA, SC, MB, LF
	Fine-grained	SC, SA, LF, MB
Reduces pumping	Fine-grained	SC, SA, LF, MB, membranes
Dust Palliative	Fine-grained	CMS, SA, oil or bituminous surface spray, PSB
	Plastic soils	CMS, SL, LMS, APSB, DCA 70
Legend: APSB = Asphalt penetration surface binder; LMS=Lime-modified soil; C=Compaction; MB=Mechanical blending; CMS=Cement-modified soil; DCA 70=Polyvinyl acetate emulsion; LF=Lime-fly ash; SA=Soil-asphalt SC=Soil-cement; SL=Soil-lime; PI=Plasticity Index		

4.4.2. Application. When there is not enough time to perform thorough mix design tests, the suggested stabilizer amounts in **Table 4.5** may be acceptable for listed soil types and classes.

4.4.2.1. Avoid adding excessive amounts of cement because it will likely lead to an increase in shrinkage cracking, which will lead to deterioration of the pavement surface. The use of cement, lime, or fly ash to stabilize a soil with a high content of sulfates can lead to heave from ettringite formation. Engineers may need to have soil tested for sulfate content before selecting a stabilization agent. Selecting the type and quantity of additives depend on the soil classification and the degree of improvement in the soil quality desired.

4.4.2.2. Although effective, there are hazards associated with soil stabilizers. While the hazard of airborne particles is reduced significantly in the open air on the construction site, when working with these materials, refer to detailed information listed in the Safety Data Sheet prepared by the product manufacturer.

Table 4.5. Soil Stabilizers for Different Soil Types.

Soil Type	Class	Fine Content	Plasticity Index	Stabilizer Type (a)	Percent Additives (d)
Sand (1A)	SW or SP	Low	All	Portland Cement	3-5
Sand (1B)	SW-SM, SP-SW, SW-SC, or SP-SC			Emulsion Polymer	3-5
				Fiber (b)	0.2-0.5
Sand (1C)	SM, SC, or SM-SC	High	High	Lime and Portland Cement	3-5 and 3-5
			Low	Portland Cement	5
			Intermediate	Fiber (b)	As above 0.2-0.5
Gravel (2A)	GW or GP	Low	All	Portland Cement	3-5
				Emulsion Polymer	3-5
				Fiber (b)	0.2-0.5
Gravel (2B)	GW-GM, GP-GM, GW-GC, or GP-GC				
Gravel (2C)	GM, GC, or GM-GC	High	High	Lime and Portland Cement	3-5 and 3-5
			Low	Portland cement	5
			Intermediate	Fiber (b)	As above 0.2-0.5
Clays, Silts (3)	CH, CL, MH, ML, OH, OL, or ML-CL	High	High (>20)	Lime and Portland Cement	5-7 and 5-7 c)
		High	Low (<12)	Portland Cement	7 (c)
		High	Intermediate (12-20)	Lime or Portland Cement Fiber (b)	As above 0.2-0.5

(a) All of these stabilizers require a polymer emulsion surface coating of 0.25–1 gallon/square yard (1:2 dilution with water) for weatherproofing. Reapplication of surface coating may be necessary depending on traffic and conditions.

(b) Monofilament polypropylene fiber used in combination with stated stabilizers—length and denier will vary depending on soil type. In general, the finer the soil, the more fiber surface area (length and shape) is needed to interact with the soil-stabilizer matrix.

(c) In some cases, intermediate and high plasticity clays may require stabilizer amounts significantly higher than suggested.

(d) The suggested stabilizer amounts are determined by dry weight of a soil.

4.4.3. Thickness. The design thickness of a base or subbase course can be reduced if the stabilized material meets the specified gradation, strength, stability, and durability requirements indicated in UFC 3-250-11 for the particular type of material. To design a pavement containing stabilized soil layers requires applying equivalency factors to a layer or layers of a conventionally designed pavement. An equivalency factor represents the number of inches of a conventional base or subbase that can be replaced by 1 inch of stabilized material. Equivalency factors for stabilized materials are determined as shown in **Table 4.6**. Limit the cement content to 4 percent by weight or less to prevent excessive reflective cracking. Selecting an equivalency factor from the table is dependent upon the classification of the soil to be stabilized.

Table 4.6. Equivalency Factors for Stabilized Material.

Material	Equivalency Factors	
	Base	Subbase
Asphalt-Stabilized		
All-bituminous concrete	1.15	2.30
GW, GP, GM, GC	1.00	2.00
SW, SP, SM, SC	(*)	1.50
Cement-Stabilized		
GW, GP, SW, SP	1.15	2.30
GM, GC	1.00	2.00
ML, MH, CL, CH	(*)	1.70
SC, SM	(*)	1.50
Lime-Stabilized		
ML, MH, CL, CH	(*)	1.00
SC, SM, GM, GC	(*)	1.10
Lime, Cement, Fly Ash-Stabilized		
ML, MH, CL, CH	(*)	1.30
SC, SM, GM, GC	(*)	1.40
Unbound crushed stone	1.00	2.00
Unbound aggregate	(*)	1.00
*Not used for base course material.		

4.5. References. **Table 4.7** list references in this chapter having more information relating to soil stabilization methods for expedient roads.

Table 4.7. Chapter 4 References.

Soil Stabilization for Expedient Roads	
UFC 3-250-11	*Soil Stabilization and Modification for Pavements*
UFC 3-250-01	*Pavement Design for Roads and Parking Areas*
UFC 3-220-08FA (Inactive)	*Engineering Use of Geotextiles*
USACE ETL 1110-1-189	*Use of Geogrids in Pavement Construction*

Chapter 5

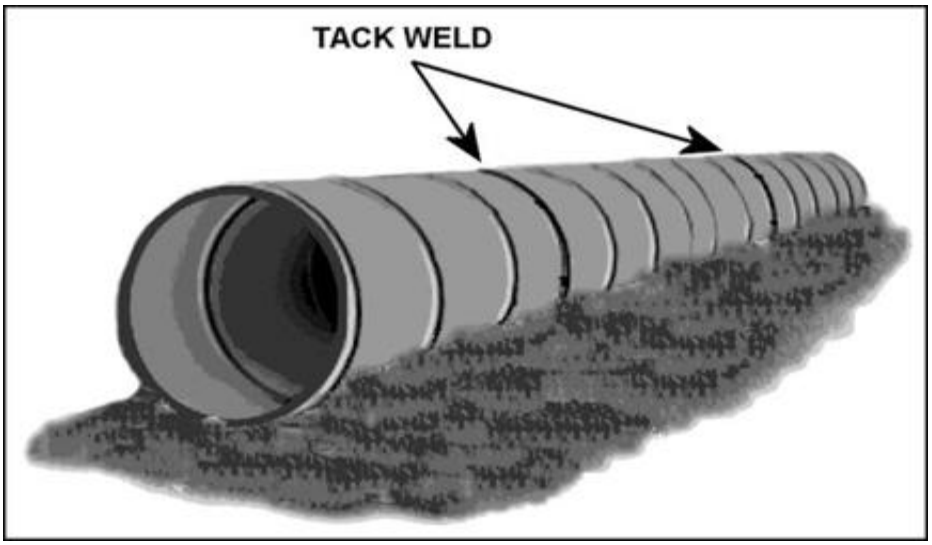
EXPEDIENT DRAINAGE METHODS

5.1. General Information. Quickly removing surface water and water runoff away from critical infrastructure can be vital for military operations. This may be especially true at contingency locations vulnerable to heavy rains and flooding. Civil engineer tasks related to this endeavor include conducting a risk assessment and developing expedient drainage methods for conveying stormwater runoff and floodwaters away from critical roadways, work facilities, temporary housing, flightline areas, etc. Surface drainage should convey flow to a point of discharge capable of handling the anticipated runoff. The expedient drainage methods presented here are intended to provide temporary options during emergencies or in remote and austere environments when time and resources are limited. The focus of expedient drainage methods in this chapter are culverts; deflectors; open channels and ditches; drainage system erosion control and maintenance; French drains; soakaways; vertical drains, and flood control dikes.

5.2. Expedient Culverts. Construct field-expedient culverts to convey water from one area to another, typically when building a roadway across a stream, diverting water to a reservoir, or managing stormwater runoff. Be prepared to improvise culverts from available material. Below are common examples of expedient culvert materials and designs, including oil and gas drums, logs and timber, sand-filled bags and containers, and metal planks and mats. **Note:** If available, recommend 24” (600mm) storm drainpipe (or equivalent cubic diameter if pipe is not used) minimum with an emphasis on the proper support and backfill in loose sandy soil.

5.2.1. Metal Drum Culverts. When empty metal drums are used (**Figure 5.1**), the capacity of the culverts is equivalent to the same diameter of corrugated metal pipe set at a comparable slope. **Note:** If using drums as culverts, take extreme care when using a cutting torch to remove the lids. Ensure the drum does not contain any flammable liquids, vapors, and materials or toxic gases before using the cutting torch. Joints between drums should be welded solid; however, if that is not possible, tack welds may be used if provisions are made to seal the joints to prevent the culvert water from saturating the surrounding fill material.

Figure 5.1. Expedient Metal Drum Culvert.



5.2.2. Log or Timber Box Culverts. Culverts constructed of logs or timber are usable in the field as temporary measures until structures that are more permanent can be employed (**Figure 5.2** and **Figure 5.3**.). In both instances, an area equal to the end area of an appropriately sized corrugated metal pipe or concrete pipe outlet, on the same slope, can be used for hydraulic capacity.

5.2.2.1. When constructing a log or a timber culvert, the joints should be as tight as practicable and, if plausible, covered with some type of sealing material (such as plastic sheeting, asphalt, or tar) to prevent water seepage. If such material is not available, caulk with leaves and brush.

5.2.2.2. To prevent the movement of soil through log openings, double-layer the top logs with burlap or brush. This step also applies to culverts made of sandbags and steel planks (**Figure 5.4**), if a paved surface is not used. **Note:** A minimum cover of three feet over a culvert of this type is required.

Figure 5.2. Log Box Culvert Example.

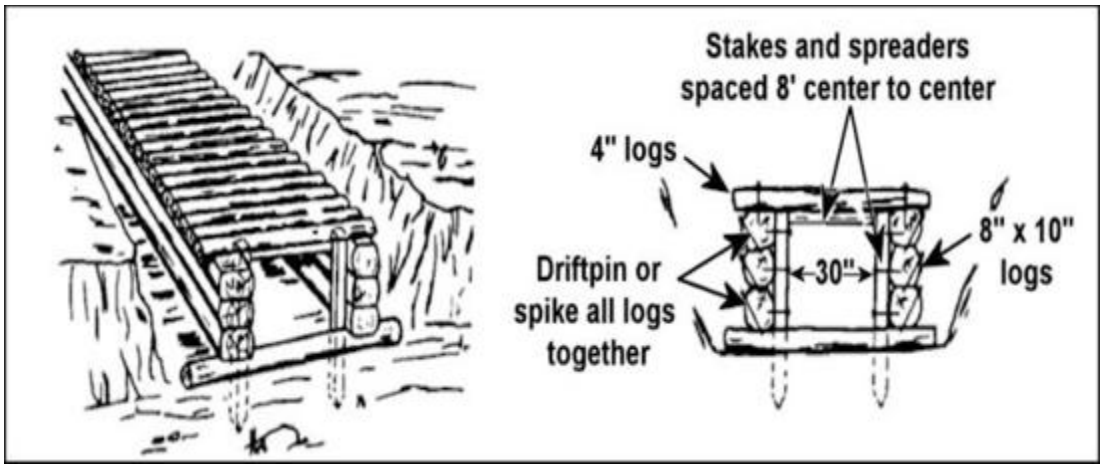


Figure 5.3. Timber Box Culvert Example.

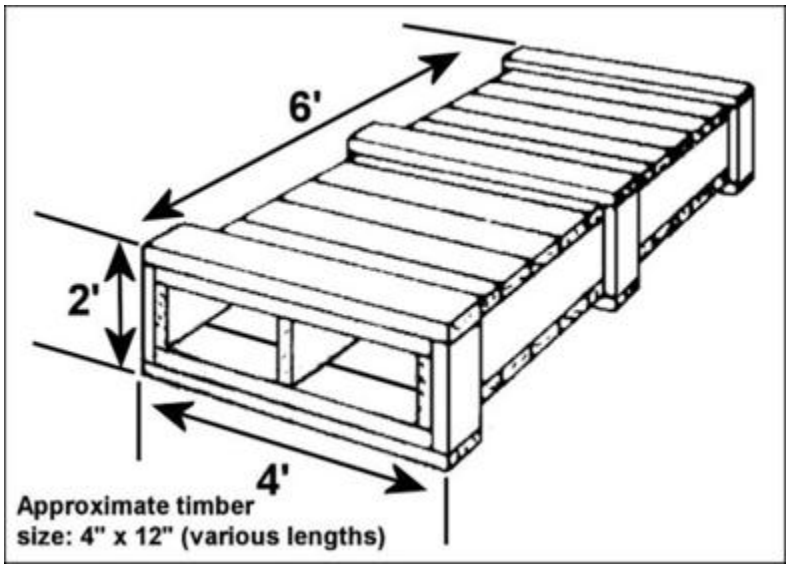
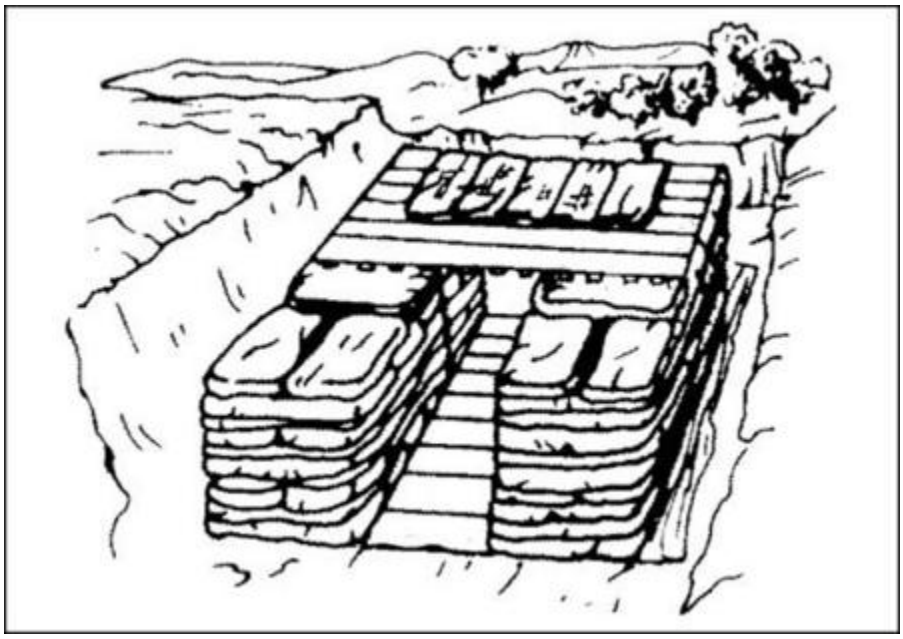


Figure 5.4. Expedient Sandbag and Metal Planks Culvert.

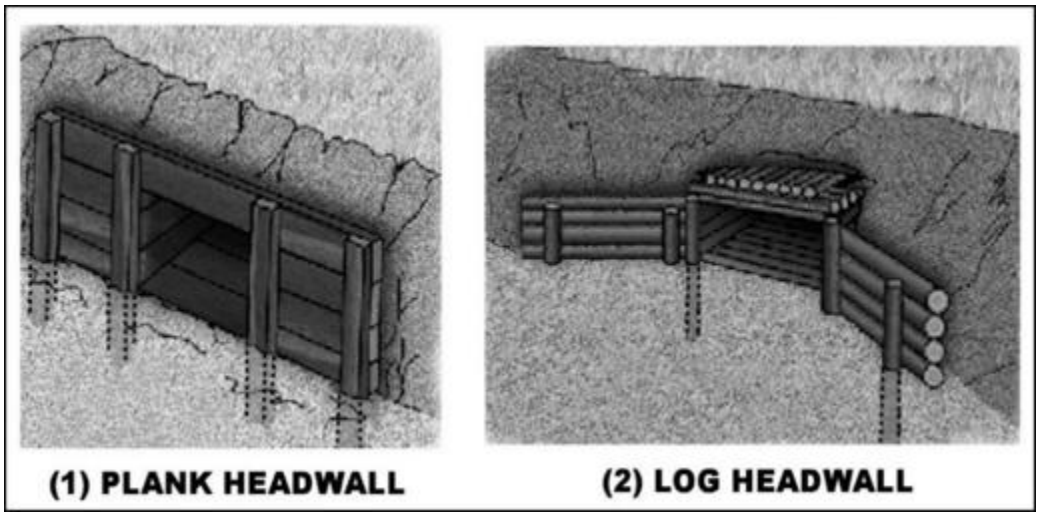


5.2.3. Culvert Headwalls and Wingwalls. Construct headwalls and wingwalls to prevent and control erosion, guide water into a culvert, reduce seepage, and hold the end of a culvert in place.

5.2.3.1. Headwalls. Always use headwalls (also known as retaining walls) at the upstream end of a culvert. As an option, use headwalls downstream where steep grades are involved to support and protect the soil mass at the end of the culvert and to hold the culvert sections in place. Construct headwalls using materials as durable as the culvert. If necessary, use sandbags or rubble as expedient measures. Extend the inlet end of the headwall pipe so that a minimum-sized headwall is required. Headwalls should not protrude above the shoulder grade and should be located at least two feet outside the shoulder to avoid becoming traffic hazards. **Figure 5.5** illustrates expedient headwalls made of wooden planks and logs.

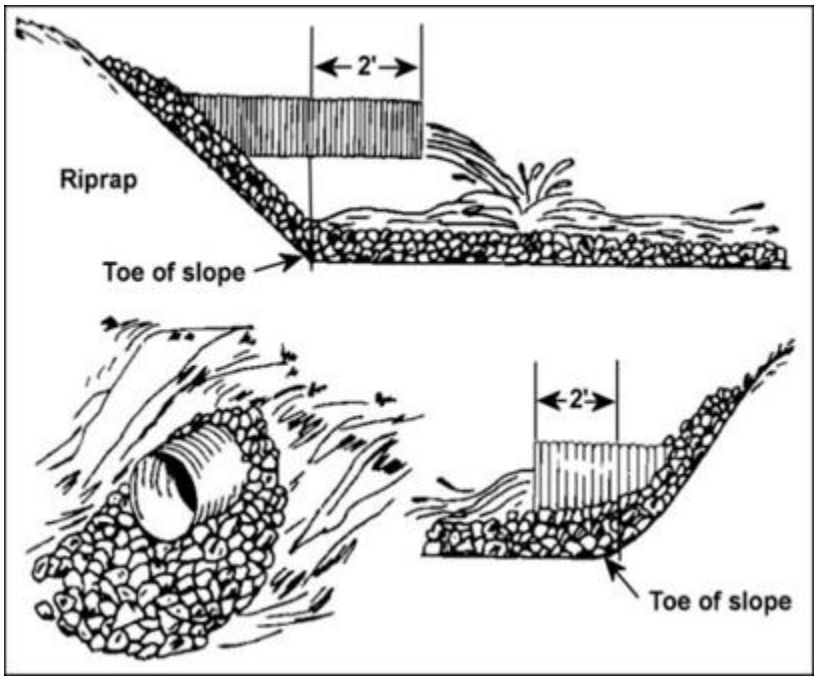
5.2.3.2. Wingwalls. Wingwalls channelizes water and prevents washing out of headwalls. They are set at an angle to the headwall to further support the surrounding fill and help direct the water flow. Their height should prevent embankment material from spilling into the waterway. The log headwall, shown as (2), includes wings.

Figure 5.5. Expedient Headwall Examples.



5.2.3.3. Alternatives. Headwalls and wingwalls can be costly to build in both time and materials. If possible, avoid this effort by extending the outlet a minimum distance of two feet beyond the toe of the fill (**Figure 5.6**.). In addition, protect the area directly beneath the projecting culvert using erosion-resistant material such as sandbags or rock riprap. Riprap should consist of rock not less than 6-inches in diameter. Riprap should be two layers deep on the sides of the channel. Riprap under the culvert outfall should be at least three layers deep.

Figure 5.6. Use of Riprap and Rubble with No Headwall.



5.2.4. Expedient Open Top Culverts. The open top culvert is another way to remove surface water away from the surface of roads. It is easy to construct and install using common hand tools. However, debris can easily fill the culvert or mechanized road maintenance equipment could cause damage. Typically, the open top culvert method consists of a “U” shaped wooden trough made of treated lumber (**Figure 5.7**.). Install the trough flush with the roadway surface and angled 30 to 45 degrees downgrade. Extend the outlet end of this type of culvert beyond the edge of the road, and place riprap at the outlet end to prevent erosion. This type of culvert generally requires regular maintenance; consequently, use it only on roadways with minimum grades. If treated lumber is not readily available, an open culvert can also be fabricated using natural material as shown in **Figure 5.8** where logs are used.

Figure 5.7. Open Top Culvert Design.

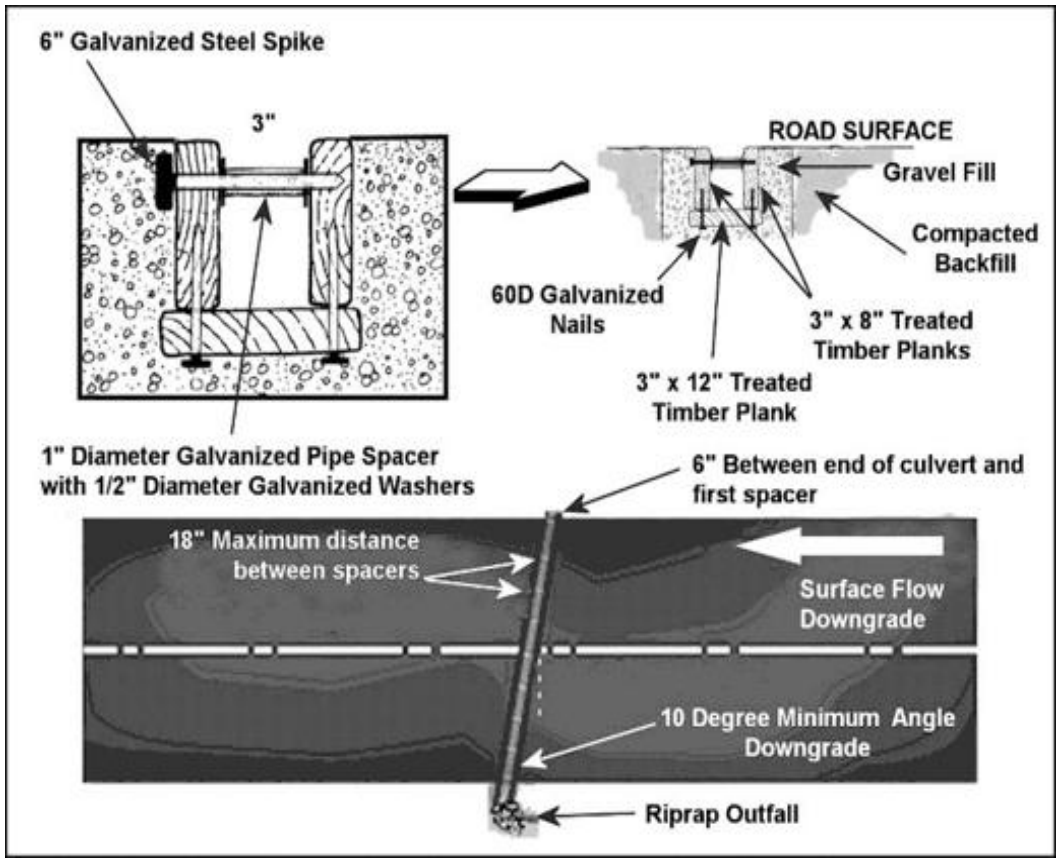
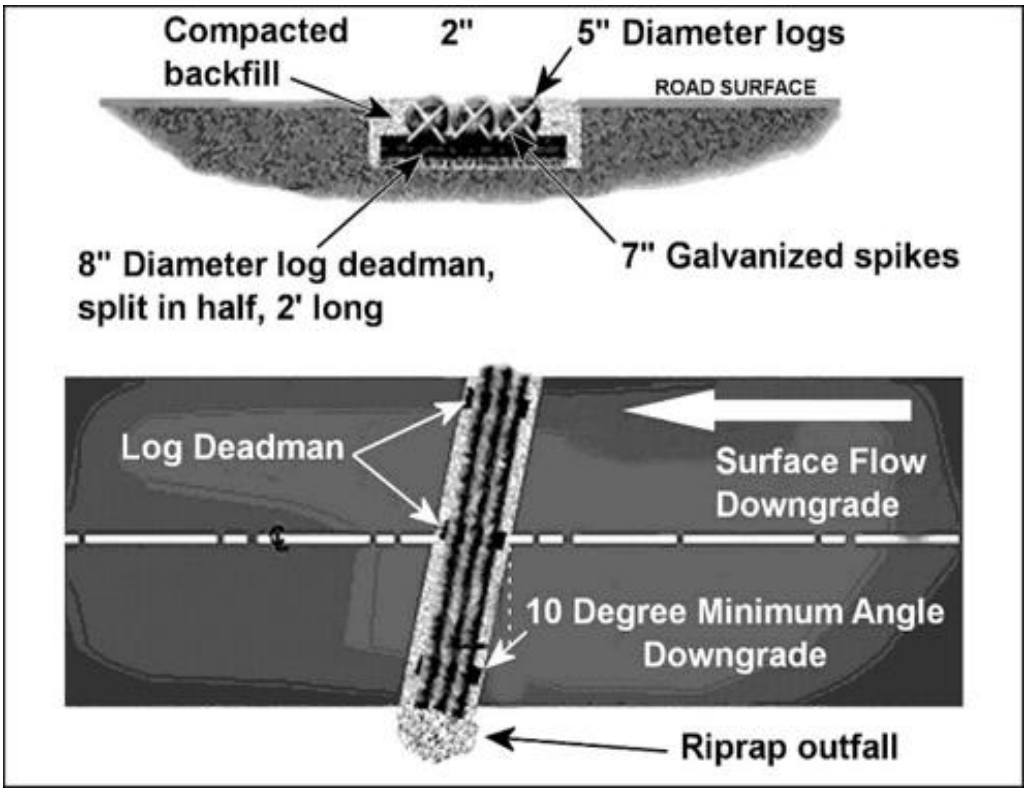
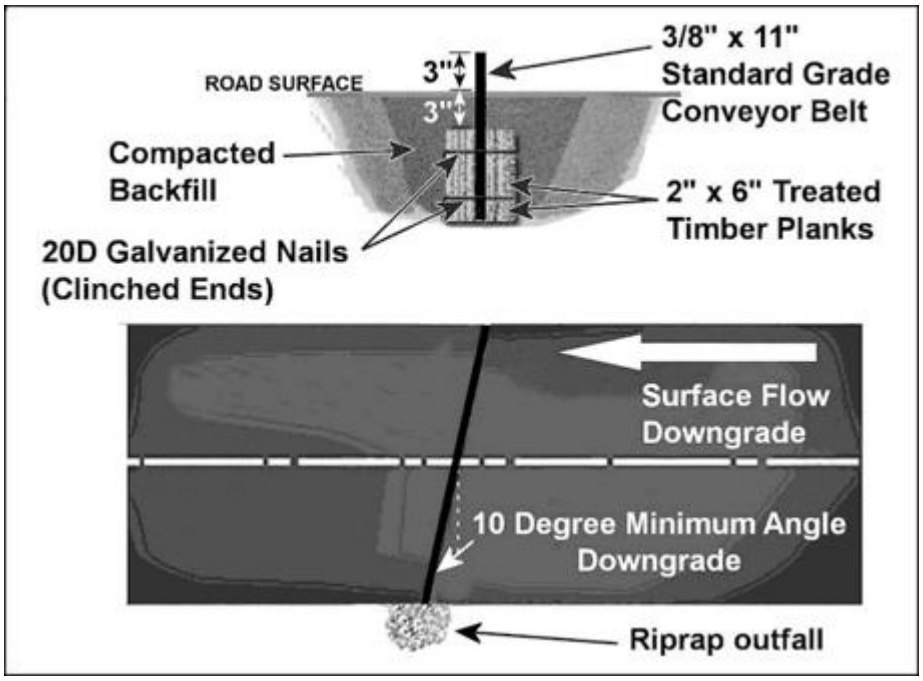


Figure 5.8. Open Top Culvert Built Using Logs.



5.3. Deflectors. Install a deflector to remove water off the surface of an expedient road. Deflectors are particularly effective on sloping roads and help to stop channeling and rutting of the road surface. A deflector is simply rubber belt material (3/8-inch to 1/2-inch thick) fastened between treated timbers. If flat rubber belt material is not available, use unserviceable arresting barrier tape. As illustrated in **Figure 5.9**, bury the timbers at a slight angle across the roadway with only about 3 inches of the belt exposed. As water runs down the surface of the roadway, the belt intercepts the water and disperses it off to the side. It is common practice to place riprap at the dispersal end of the deflector to prevent erosion of the shoulder and side slopes of the road.

Figure 5.9. Deflector Construction Details.



5.4. Open Channels or Ditches. The ditch is the most common open-channel cut into the soil used to convey water. Often critical after a flood, ditches and channels handle unwanted surface water and controls or enhances water runoff. Generally, for expedient ditches, the topography and available engineer equipment determines ditch size, shape, location, and construction method.

5.4.1. The main types of ditches used in road and airfield construction are interceptor, side ditch, and diversion.

5.4.1.1. An interceptor ditch is generally located on a hillside above a roadway or other feature requiring protection. Its function is to intercept runoff and direct the flow to a more desirable location. It is usually located above side hill cuts to prevent erosion of the cut.

5.4.1.2. A side ditch is located along the side of a road. It collects runoff from the road and adjacent areas and transports it to a culvert or diversion ditch.

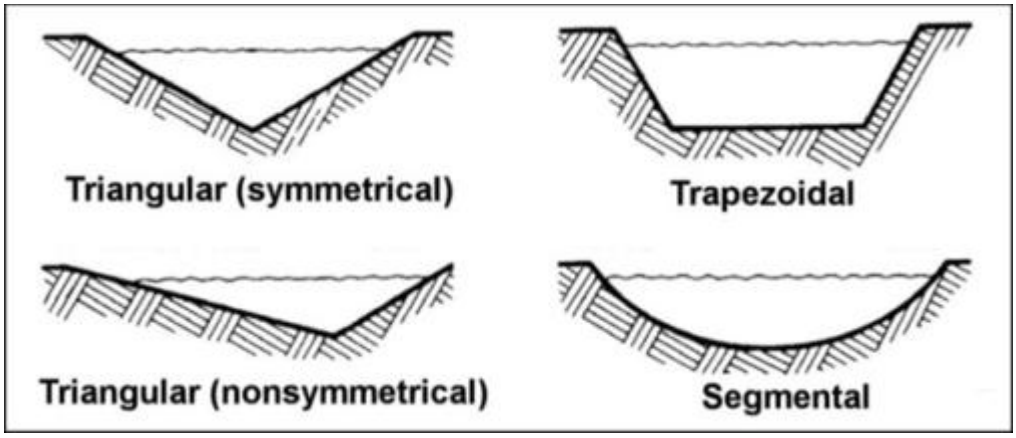
5.4.1.3. When the topography allows, build a diversion ditch to transport water away from roadways and airfields. Use it in conjunction with interceptor and side ditches to transport water between culverts or to divert an existing stream channel around a project.

5.4.2. The most common shapes of ditch cross sections are triangular, trapezoidal, and segmental (**Figure 5.10**.).

5.4.2.1. Triangular (V-shaped) ditches are commonly considered appropriate for minor channels with flows up to 60 cubic feet per second (cfs); they can be symmetrical or nonsymmetrical. Trapezoidal ditches are usually the better option if flows greater than 60 cfs are probable. Guard against being too conservative when it comes to sizing drainage ditches; an oversized ditch causes less problems than one too small. See **paragraph 5.4.3** when estimating peak runoff rates.

5.4.2.2. The segmental-ditch shape usually results when using explosives to create the ditch. Work crews often use this technique when the terrain is too soft to support excavating machinery. Ditches cut by hand will often bear this shape as well.

Figure 5.10. Common Ditch Cross-Section Shapes.



5.4.3. When developing an expedient open ditch, it may be necessary to estimate the surface water runoff from the drainage area. The Rational Method as addressed in UFC 3-201-01 and the Hydraulic Design Series No. 2 (HDS-2) from the U.S. Department of Transportation (USDOT), calculates peak design flows from small watersheds.

5.4.3.1. The assumptions in the Rational Method are listed below. Although these assumptions are not fully valid in every case, for small drainage areas, the errors will not be large.

5.4.3.1.1. The drainage area is typically 200 acres or smaller.

5.4.3.1.2. The peak flow occurs when the entire watershed is contributing runoff.

5.4.3.1.3. The rainfall intensity is uniform over a period of time equal to the time of concentration.

5.4.3.1.4. The time of concentration is the time for water to travel from the hydrologically most remote point of the basin to the outlet or point of interest.

5.4.3.1.5. The frequency of the computed peak flow is equal to the frequency of the rainfall intensity. For example, the 10-year rainfall intensity is assumed to produce the 10-year peak flow.

5.4.3.2. The most common formula for the Rational Method is shown below. To calculate the peak design flows, simply substitute the relevant values into the equation (Q = CiA): See example equation in **Table 5.1** below.

Table 5.1. Sample equation values.

Q = CiA Where: *Q = peak flow (rate of runoff) in cubic feet per second (cfs).* *C = a dimensionless coefficient that is a function of the watershed characteristics.* *i = rainfall intensity in inches per hour.* *A = watershed area in acres.*

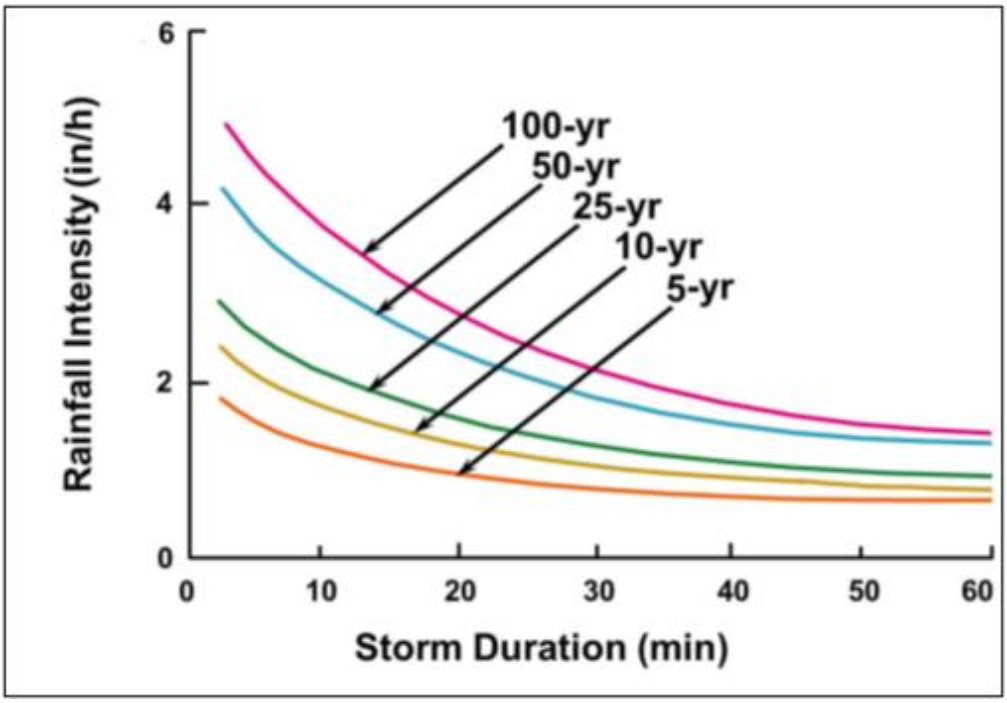
5.4.3.3. The runoff coefficient (C) in equation Q = CiA is a function of surface conditions including slope, soil type, and presence of vegetation. These conditions affect infiltration and other hydrologic abstractions and, therefore, the value of C. Since the physical interpretation of the runoff coefficient is the fraction of the rainfall on the watershed that becomes surface runoff, it's value must be between one and zero. Listed in **Table 5.2** are some typical values of C for the Rational Method. **Note:** For additional runoff coefficients relating to business and residential drainage areas, see USDOT Publication No. FHWA-HIF-24-007 (HDS-2).

Table 5.2. Runoff Coefficients for the Rational Method (ASCE 1960).

Land Use Category	Type of Drainage Area	Runoff Coefficient, C
Industrial	Light areas	0.50 - 0.80
	Heavy areas	0.60 - 0.90
Lawns	Sandy soil, flat, <2%	0.05 - 0.10
	Sandy soil, average, 2 – 7%	0.10 - 0.15
	Sandy soil, steep, >7%	0.15 - 0.20
	Heavy soil, flat, <2%	0.13 - 0.17
	Heavy soil, average, 2 - 7%	0.18 - 0.22
	Heavy soil, steep, >7%	0.25 - 0.35
Open	Parks	0.10 - 0.25
	Railroad yard areas	0.20 - 0.40
	Unimproved areas	0.10 - 0.30
Streets	Asphaltic	0.70 - 0.95
	Concrete	0.80 - 0.95
	Brick	0.70 - 0.85
Other impervious	Drives and walks	0.75 - 0.85
	Roofs	0.75 - 0.95
Source: Extracted from U.S. DoT Publication No. FHWA-HIF-24-007, Hydraulic Design Series No. 2.		

5.4.3.4. The rainfall intensity (i) in equation Q = CiA is obtained using intensity-duration-frequency (IDF) curves (**Figure 5.11**) developed by state or local regulatory agencies for specific locations. In addition, the National Oceanic and Atmospheric Administration and Natural Resources Conservation Service also created regional rainfall intensity curves that can be used to create IDF relationships for drainage design. For more details, see USDOT Publication No. FHWA-HIF-24-006, Hydraulic Engineering Circular No. 22 (HEC-22).

Figure 5.11. Family of Intensity-Duration-Frequency (IDF) Curves (Notional).



5.4.3.4.1. Obtaining accurate IDF curves for developing countries and remote areas outside the US can be challenging because historical rainfall gauge data and other precipitation records may be unavailable or of poor quality. If IDF curves are not available, particularly in the deployed area, civil engineers may need to develop them on a project-by-project basis.

5.4.3.4.2. UFC 3-201-01 specifies engineers should use a minimum 10-year storm frequency, minimum for the facility type, or minimum required by the state or local governing authority; whichever is more stringent.

5.4.3.5. The area of the watershed (A) in equation Q = CiA is the drainage area (in acres) that contributes to the runoff.

5.4.3.6. The calculation of peak stormwater runoff rate from the equation, Q = CiA, is quite straightforward if values are known for C, i, and A. Values for the runoff coefficient (C) and the watershed (A) can typically be obtained without much trouble. Determining the design rainfall intensity (i) usually requires the most effort. **Note:** See example equation in **Table 5.3** below.

Table 5.3. Rational Method Formula for Peak Water Runoff Calculations (Notional).

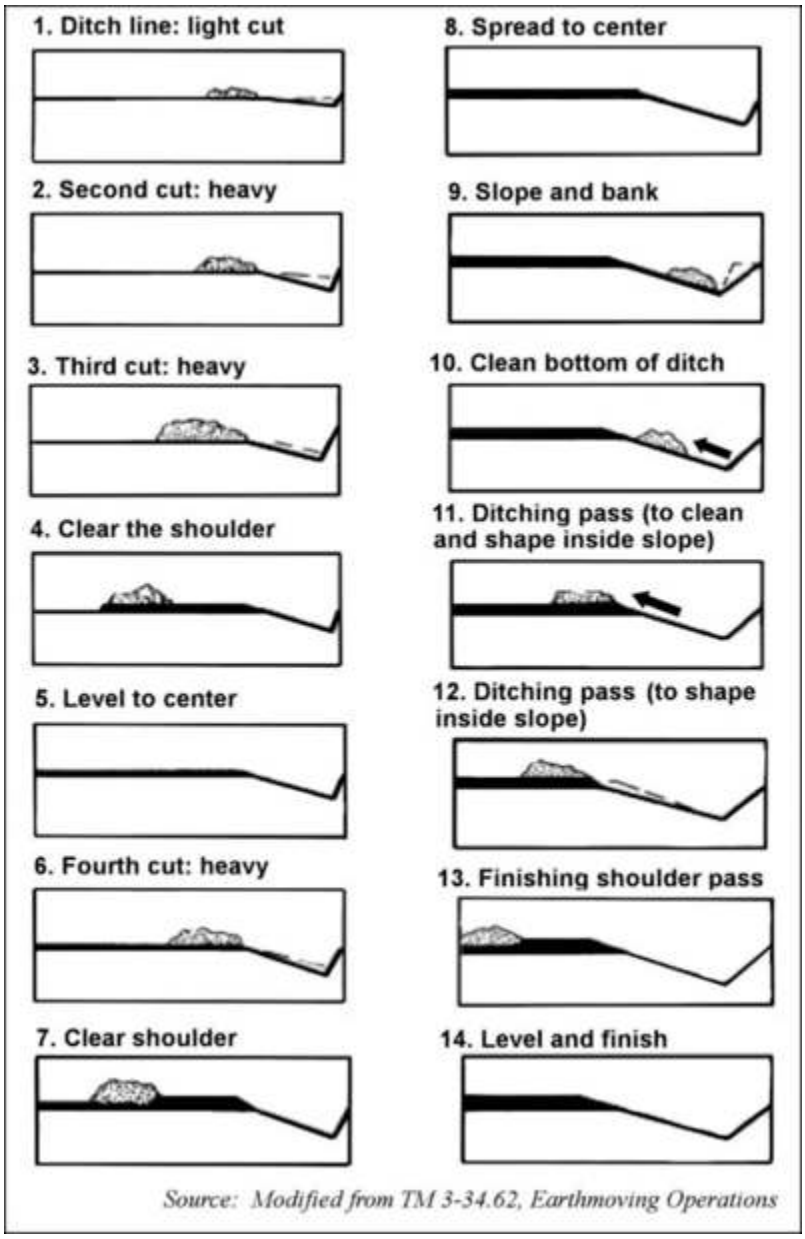
Example: Calculate the peak stormwater runoff rate from a watershed of 15 acres, with a runoff coefficient of 0.35, from a storm of intensity 2.4 in/hr.
Solution: Substitute values into the equation (Q = CiA), thus: Q = (0.35)(2.4)(15) = 12.6 cfs

5.4.4. When constructing expedient ditches/open channels; graders, dozers, or wheeled scrapers are usual choices for excavation. However, the backhoe, front-end loader, and crane are also options. A maximum side slope of 2:1 (horizontal to vertical) is typical, unless using steeper side slopes are supported by a geotechnical report. With flows greater than 100 cfs, provide 1 foot of freeboard on side slopes.

5.4.4.1. The road grader is probably the most common heavy equipment item used for expedient ditch construction. An operator can quickly excavate the necessary cross section of triangular, or V ditches handling flows up to 60 cfs. Grader efficiency drops significantly when cross sections of larger dimensions are required. When using the road grader to excavate triangular ditches, make a 3- to 4-inch marking cut on the first pass at the outer edge of the bank slope. This can enhance grader control and produce straighter ditches. Make each subsequent ditch cut as deep as possible without stalling or losing control of the grader. Normally, make ditching cuts in second gear at full throttle. Start with the blade positioned so that the toe is in line with the center of the lead wheel. Bring each succeeding cut in from the edge of the bank slope so that the toe of the blade will be in line with the ditch bottom on the final cut. **Figure 5.12** illustrates the steps of the V-ditching method. The steps shown are for a single roadside ditch. Repeat the steps on the opposite side of the road.

5.4.4.2. Use a wheeled scraper to rapidly excavate the flat bottom and midsection of trapezoidal ditches and dress back the side slopes with subsequent passes of a road grader. Excavate smaller bottom widths using the road grader (with its blade turned to a high angle) or other engineer equipment. Production rates will be much lower for the grader than those of the wheeled scraper. The 60 cfs guideline is flexible. If a scraper is not available to excavate a ditch carrying 100 cfs, use a grader to construct an oversized V ditch rather than using equipment with a lower production rate to construct a trapezoidal ditch.

Figure 5.12. Road Grader V-ditching Method.



5.5. Drainage System Erosion Control and Maintenance. Erosion presents many difficult problems in designing and maintaining open channels. Erosion tends to change the shape of channels, which can cause troubles near structures such as roads, bridges, and culverts. Additionally, erosion presents problems when the displaced material deposits in undesired locations. The primary cause of erosion is the velocity of the water in the channel. Controlling erosion helps to maintain an effective and clear drainage system with a minimum of maintenance. Most methods of control rely on dissipating the energy of the water, providing an erosion-resistant surface, or some combination of these techniques. This section addresses ways to reduce or eliminate soil erosion in drainage systems.

5.5.1. Controlling Velocity. Water velocity in a channel is dependent on the slope, shape, and depth of the channel. Control erosion in the channel by lowering the velocity below the soil-erosion flow. Achieve lower water velocity by reducing the ditch side slope; making the ditch wider and shallower; lining the ditch with erosion-resistant material; or by constructing check dams. Building the ditch wider and shallower or with a reduced slope increases the surface area. Larger surface areas cause greater friction and lowers the water velocity. Since constructing new ditches at a different slope, width, and depth is usually impractical during field operations, expedient alternatives may include lining existing channels with erosion-resistant riprap and rubble or constructing check dams. Other methods such as terracing, using geosynthetics liners or natural liners (growing grass, turf), or paving the channel sides with asphalt or concrete may be used when time, materials, and other resources are available. **Note:** Consult UFC 3-220-08FA (Inactive) for more information on use of geosynthetic liners in drainage structures and for erosion control measures.

5.5.2. Riprap. Increasing the effective roughness of a ditch decreases ditch soil erosion. One easy way of doing this is with riprap (**Figure 5.13**.). This method involves placing rocks or stones in the ditch bottom and sides to provide a protective cover over the soil to prevent soil erosion. To prevent underlying soil from moving through the riprap, consider applying filter fabric, geotextile material, or a layer of gravel before applying the riprap. To prevent riprap failure due to velocity and turbulence, resulting in movement of rock and bed material, ensure appropriate rock size is used. Normally, rocks should be larger than 6 inches to assure stabilization. Additionally, rocks should be hand-placed and compacted individually in at least two layers. The riprap not only prevents soil erosion but will also tend to decrease the velocity in the channel. This makes it useful for lowering velocities in sections of channel when making transitions to soil ditches from paved or other high-velocity ditches. Other riprap considerations include:

5.5.2.1. Gradation—The riprap should be well graded and resistant to erosion and crumbling.

5.5.2.2. Size and depth—Riprap size and depth should be based on the anticipated volume and velocity of the flow over the rocks.

5.5.2.3. Stone quality—Stone for riprap should consist of field stone or quarry stone that is angular, variably sized and resistant to cracking during freeze and thaw cycles. **Note:** For suitability and properties of igneous, sedimentary, and metamorphic rocks, see **Attachment 4**.

Figure 5.13. Riprap-Lined Drainage Ditches.



5.5.3. Check Dams. Check dams are the most common structure used to reduce velocity in ditches that have longitudinal grades not exceeding 8 percent. They work by decreasing the slope of the water surface. Eventually soil deposits on the ditch bottom, decreasing the ditch slope itself. Consider using check dams when the slope ranges between 2 and 8 percent. Channels with slopes less than 2 percent generally do not require extensive erosion controls. With more than 8 percent slopes, it is usually more economical to pave the ditch with asphalt or concrete than to build check dams.

5.5.3.1. Expedient check dams may be constructed using lumber, logs, sandbags, rocks, or other suitable and locally available materials. As illustrated in **Figure 5.14**, each dam should extend at least 2 feet into the sides of the ditch, and the ditch top should be at least 12 inches above the top of the dam. Correct spacing between check dams can be determined using the procedure in **Table 5.4**.

Figure 5.14. Typical Check Dam Design.

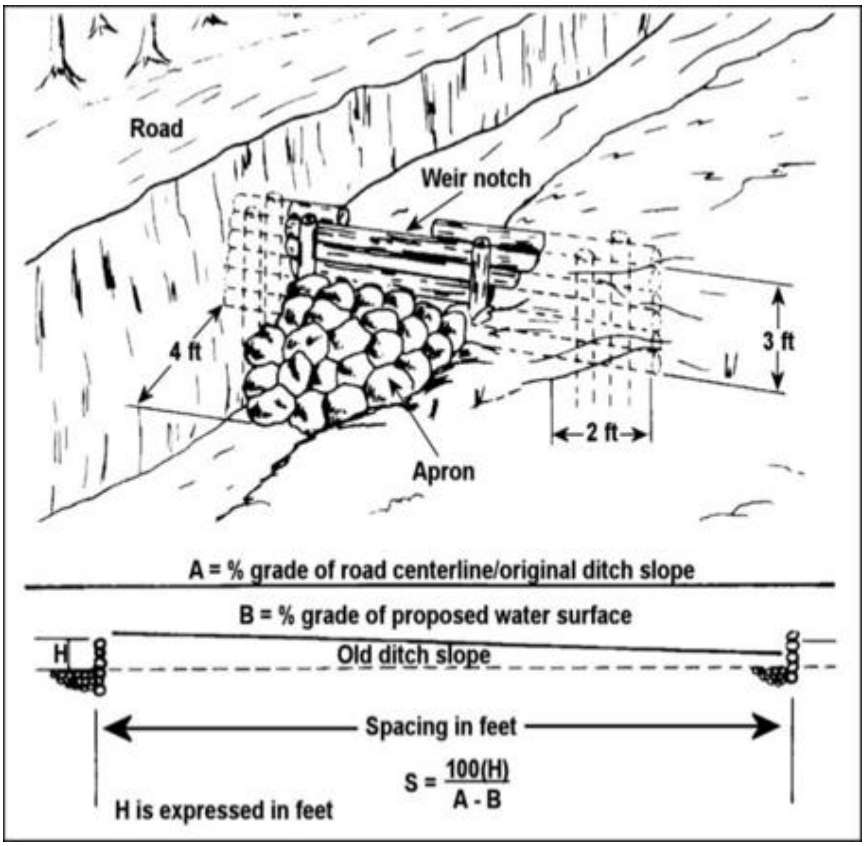


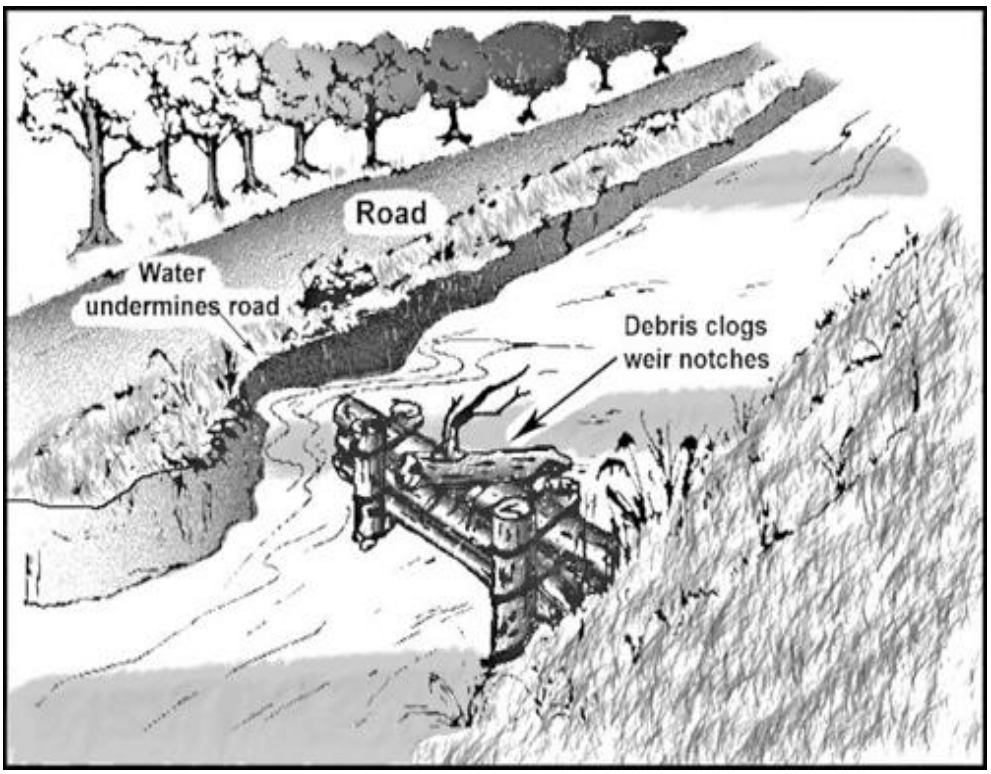
Table 5.4. Check-Dam Spacing Formula.

Check-Dam Spacing	
Formula: $S = \frac{100(H)}{A-B}$	
Where: S =	Spacing between check dams (This value should not be less than 50 feet).
H =	Height from the channel bottom to the lower edge of the weir notch
A =	Slope of the original ditch in percent.
B =	Desired slope in percent (Value should be set at two percent; the maximum slope that will not require additional erosion control).
Example Calculation:	
Original slope =	5 percent
Desired slope =	2 percent
H =	3 feet
S =	

5.5.3.2. The height between the bottom of the channel to the lower edge of the weir notch should not be greater than 3 feet (unless structurally designed) nor less than the practical limit of 1 foot. An apron of rubble or riprap should extend at least 4 feet from the face of the check dam on the discharge side to prevent erosion. Cut a weir notch in the top of the check dam with a capacity large enough to discharge the anticipated runoff to prevent water from cutting around the edges of the dam.

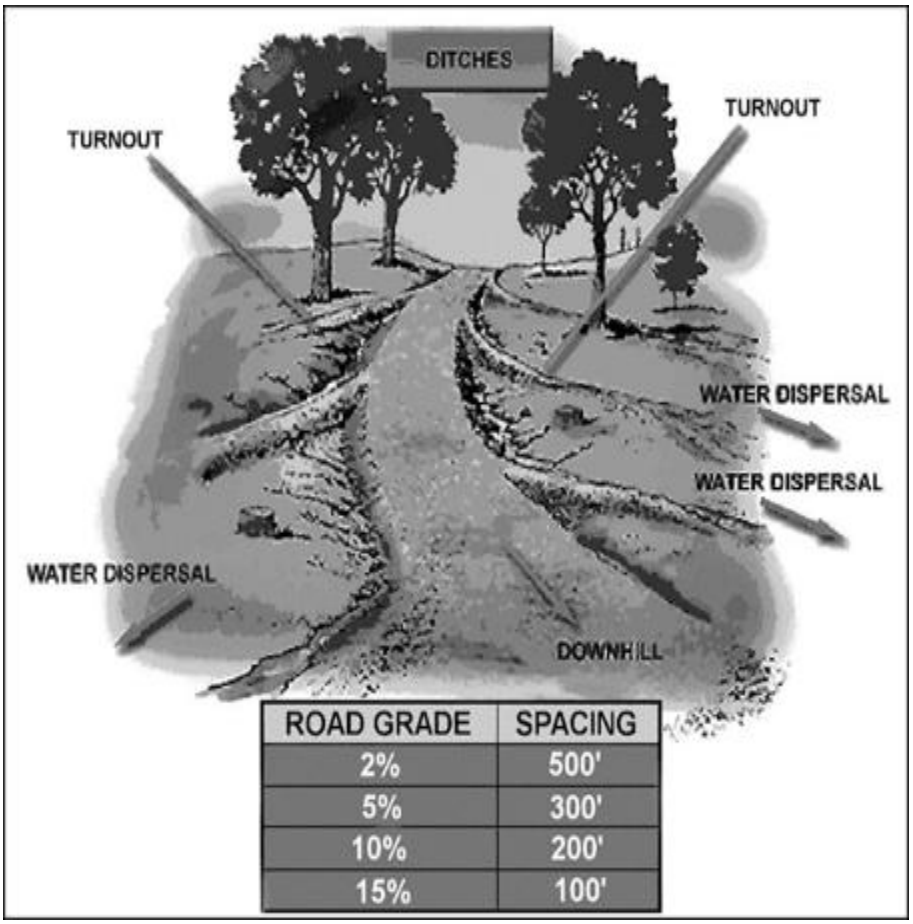
5.5.3.3. Erosion around the dam is the most common problem associated with check dams. The primary cause for this situation is that the weir notch is too small or clogged with debris causing water to flow over the top of the dam. This situation results in erosion where the dam is anchored into the ground (see **Figure 5.15**), and if left uncorrected over time, it will often result in more damage to adjacent structures than if the dam was never installed.

Figure 5.15. Dam Erosion Caused by Weir Notch Debris.



5.5.4. Roadside Turnouts/Diversion Ditch. If encountering hilly terrain during the construction of expedient roads, erosion in the supporting drainage ditches may be a problem. One method of fighting such erosion involves using riprap, but the use of turnouts might be more effective. Turnouts and diversion ditches are channels cut from the roadside ditches to adjacent areas that can handle larger runoff quantities. Attempt to have collected water spread over an open land area. Refrain from diverting the water directly into nearby streams or lakes. **Figure 5.16** shows a typical turnout arrangement and a chart that indicates the spacing of turnouts in respect to the grading of the road.

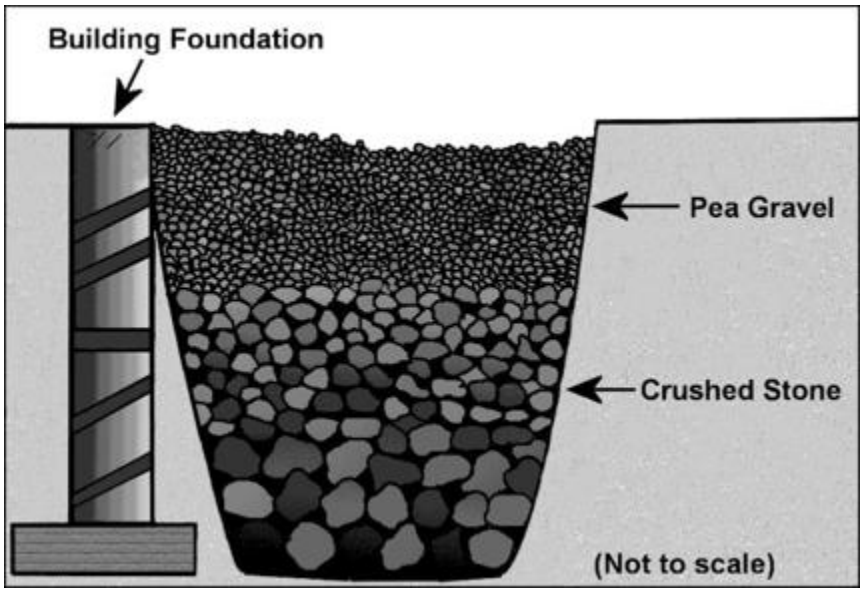
Figure 5.16. Turnout Layout and Spacing Requirements.



ROAD GRADE	SPACING
2%	500'
5%	300'
10%	200'
15%	100'

5.6. French Drains. When using French (or blind) drains near buildings and other structures, install them at the lowest point where standing water is found (see **Figure 5.17**.). The drain may terminate at any point where the water will not drain back toward the structure. Normally, French drains are shallow systems and function on the principle of gravity. As such, to be effective, the drain must slope downward with a minimum recommended slope of 1/4 inch per foot. If the adjacent ground grade runs upward along the drain path, dig deeper to maintain a downward slope. A basic French drain is constructed by simply filling a ditch or trench with broken or crushed rock. Consider leaving the top surface of the rock exposed so the trench will act as a combination drain (surface and subsurface drainage) or cover the rock with a relatively impervious soil so that no surface water can penetrate. The latter is general practice. However, consider using other methods for permanent construction because French drains tend to silt up with prolonged use.

Figure 5.17. Open French Drain without Pipe.



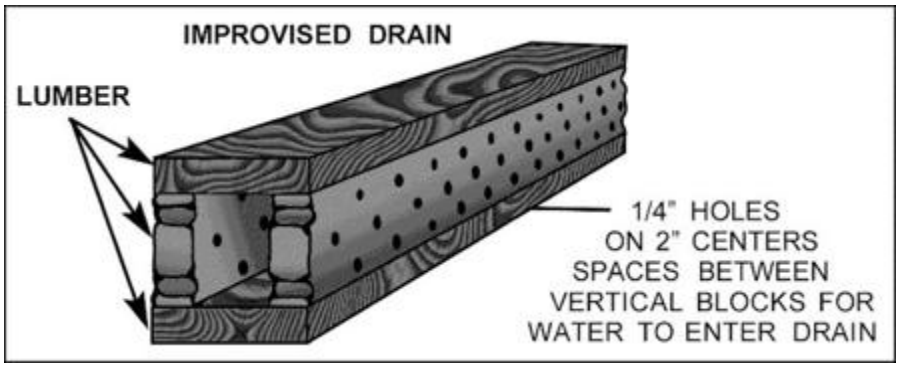
5.6.1. In field construction, consider using French drains as a substitute for perforated or open joint pipes because of logistical limitations on piping or on filter materials suitable for use with such piping. However, consider using a French drain with a perforated land drain in the bottom, which when wrapped in a layer of geotextile fabric, ensures longer life by stopping any fines from clogging the system. In areas with severe drainage problems, run multiple perforated pipes together to act as water collectors or interceptor drains. This way, the water carries away more quickly when the surrounding earth is saturated.

5.6.2. Installing a French drain with a perforated pipe is usually simple, but obstacles such as tree roots, boulders, and underground utilities, can make the project difficult and time-consuming. The basic procedure for installing a French drain that includes perforated pipe is as follows. Keep in mind, installing a system without piping and geotextile fabric is an option, but greatly reduced longevity due to silt transfer is likely.

5.6.3. From the beginning point of the drain, dig a trench about 12 inches deep and approximately 8 inches wide. If the ground slopes upward, dig deeper from the starting point to maintain a downward slope. Line the cavity with a geotextile, leaving sufficient excess to allow it to overlap the subsequent piping and fill material. Next, pour in about 2 inches of rock (pea gravel or crushed stone). Lay a 4- to 6-inch perforated pipe over the rock with the perforations facing down. If perforated pipe is not available, fabricate a substitute from lumber as shown in **Figure 5.18**.

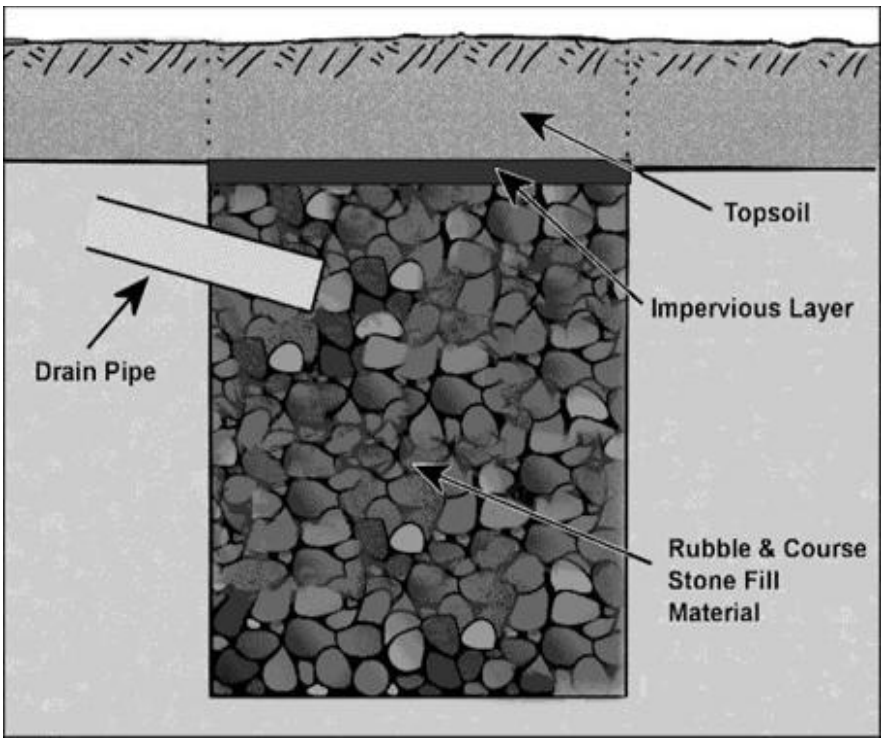
5.6.4. After pipe installation, place another 2 inches of rock in the opening to completely cover the pipe and fold the remaining geotextile over the stone to encapsulate the drain field. Lastly, back fill with soil (about 2 inches) and plant grass. **Note:** Water entering the trench will have to run to a suitable drain-off point, and that, in most cases, is a soak-away or watercourse. However, never discharge stormwater runoff into a cesspool or septic tank, and never discharge wastewater into a French drain or soakaway.

Figure 5.18. Improvised Drain Line.



5.7. Soakaways: A soakaway is simply a hole in the ground filled with rubble and coarse stone with a drainage pipe laid into it to aid removing surface (rain) water from other areas. Construct the soakaway in granular soil with good drainage properties. Do not use a soakaway in clay soils. Most local authority regulations dictate that soakaways be located at least 15 feet from any habitable building. The pipe flowing to them should be at least 3 inches in diameter; however, a 4-inch pipe diameter is the recommended size. Lay the pipe to a fall of 1 inch to every 40 inches. The size of the soakaway should be a minimum of 3 feet by 3 feet by 3 feet below the bottom of the incoming pipe. The stone fill should surround the pipe and finish approximately 4 inches above it. As indicated in **Figure 5.19**, place an impervious layer on the stone. Recommend using materials such as thick polyethylene, tarpaulin, or even a bed of concrete. Afterward, place topsoil on top of this layer to restore the garden level.

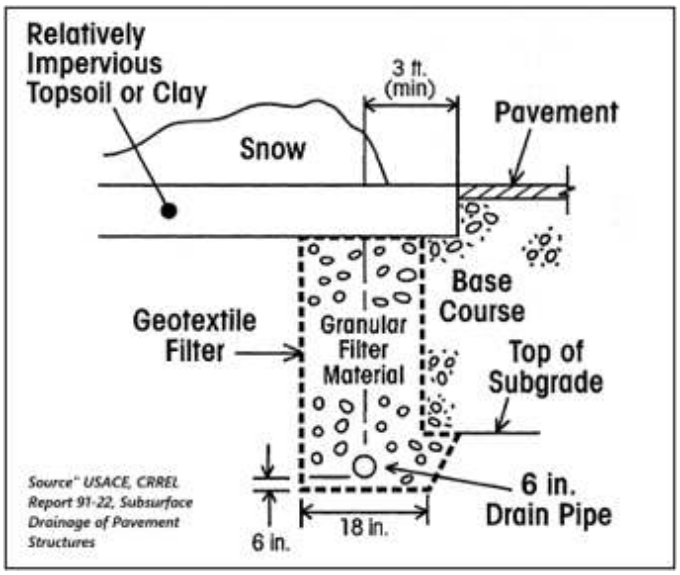
Figure 5.19. Typical Soakaway.



5.8. Vertical Drains. Vertical drains can transmit surface or trapped subsurface water through an impervious soil or rock layer to a lower, porous or freely draining layer of soil. The designs range from vertical well, pit, or pipe drains with sand and/or gravel filters to contemporary prefabricated drains made of geotextile or composite materials. If drainage is obstructed, additional wells are built, or the pocket is drained with an easily maintained lateral subdrain system. When vertical drains are constructed or inserted through compressible soil layers, they can intercept horizontal water flow and transmit the water to a drainage layer at the surface and/or to underlying coarse-grained soil, depending on the drainage conditions. **Note:** During advance planning for vertical drains, consider any potential effects on groundwater such as contamination, increased water table levels, etc. For information on drainage characteristics of various soils, see AFH 32-1034 or TSPWG 3-260-03.02-19, Appendix A.

5.8.1. Vertical wells are often used in northern latitudes where deep freezing is common. They permit fast runoff from melting snow to get through the frozen soil and reach a pervious stratum (**Figure 5.20**.). Under such conditions, the bottoms of these wells are treated with calcium chloride or a layer of hay to prevent freezing.

Figure 5.20. Subsurface Vertical Drain Along Pavement Edge.



5.8.2. Today, most vertical drains used are prefabricated vertical drains (PVD) comprised of a plastic core encased in geotextile fabric, like the examples shown in **Figure 5.21**. These PVDs are inserted into the soil by a mandril and are often used to accelerate the consolidation of compressive soils using radial drainage to dissipate excess pore water pressures faster than natural consolidation methods. For more information on the design and use of constructed and prefabricated vertical drains, see UFC 3-220-10, *Soil Mechanics (DM 7.1)*. Other helpful design and use information is available in Federal Highway Administration, FHWA-NHI-16-027, *Ground Modification Methods Reference Manual–Volume I* (2017), Geotechnical Engineering Circular 13, Washington D.C.

Figure 5.21. Prefabricated Vertical Drains (PVD).



5.9. Raised Flood Control Dikes. Expedient raised flood control dikes or barriers (sometimes called levees) have historically consisted of soil or sandbags. However, using soil-filled wire and fabric containers in lieu of sandbags is a contemporary option for building expedient dike walls because it requires less time and personnel than filling and placing sandbags. Expedient dikes provide emergency flood protection for periods of only a few days or weeks. Dikes and levees that are more enduring require additional planning, analysis, time, and materials to build and usually not considered expedient. Although there is typically little time to prepare for an unforeseen flood emergency, local emergency planners should address certain basics that engineers will need prior to dike construction. Issues such as the location of earth borrow sites; heavy equipment routes; proposed alignment for dike, equipment, and supplies, and other matters should be determined before dike construction begins.

5.9.1. General Dike Construction Design. The design of raised flood control dikes is similar to that of freestanding soil berms; however, they serve two very different purposes. Since the purpose of flood control dikes is to hold back floodwater, their failure could potentially result in devastation over a wide area and adversely affect many people. Build raised dikes to hold throughout the entire flood event. Anytime there is water against the embankment, there is a potential danger that the dike could fail. Even under emergency conditions, use proven construction methods when constructing raised flood control dikes—an emergency is usually not the time to experiment. The foundation soils and the materials available for construction generally dictate the standard design of a raised dike. Always attempt to make the dike or embankment compatible with the foundation. Information on foundation soils may be available from local officials or engineers. Use the information in **paragraph 5.9.5** as a guide when determining vertical and horizontal embankment design. Finally, actual dike construction will depend, in most cases, on time, materials, and right-of-way available.

5.9.2. Alignment. Before heavy equipment operators can begin moving earth to build a raised dike, engineer planners must establish an alignment for the proposed barrier. The alignment should be the shortest route possible, provide the maximum practical protection, and take advantage of any high ground where available. Keep the flood barrier as far landward of the body of water as possible to prevent encroachment on the floodway. Avoid sharp bends. Keep as many trees and as much brush between the dike and the water as possible to help deflect water current and debris. The earth borrow areas should have adequate material for construction of the dike and must always be accessible. Also, the borrow area should be in an area that will not become isolated from the dike project by high water.

5.9.3. Foundation Preparation. Prior to constructing a raised embankment, the foundation area along the dike alignment should be prepared.

5.9.3.1. Cut trees and remove the stumps. Remove all obstructions above the ground surface, if possible. This includes brush, structures, snags, and similar debris. In snow-covered areas, push the snow riverward to decrease ponding when the snow melts.

5.9.3.2. Strip the foundation of topsoil and surface humus. Stripping may be impossible if the ground is frozen. In this case, rip or scarify the foundation, if possible, to provide a rough surface for bonding with the embankment. Make every effort to remove all ice or soil containing ice. Frost or frozen ground can also give a false sense of security in the early stages of a flood fight. It can act as a rigid boundary and support the dike; but on thawing, soil strength can reduce sufficiently to allow cracks or slides to develop. Frozen soil forms an impervious barrier to prevent seepage. This may result in a considerable buildup in pressure under the soils landward of the dike and, upon thawing, pressure may be sufficient to cause sudden blowouts. If this condition exists, monitor and be prepared to act quickly if sliding or sand boils develop. If stripping is possible, the material should be pushed landward and riverward of the toe of the dike. **Note:** Perform clearing and grubbing, structure removal, and stripping only if time permits.

5.9.4. Materials. Earth fill materials for emergency dikes will usually come from local borrow areas. Attempt to utilize materials that are compatible with the foundation materials. However, due to time limitations, consider using local materials and follow reasonable construction procedures. The materials should be relatively clean (free of debris) and should not contain large frozen pieces of earth.

5.9.4.1. Clay. Clay is preferred because dike sections can be made smaller (steeper side slopes). In addition, clay is relatively impervious and has a relatively high resistance to erosion when it is compacted. A disadvantage in using clay is that adequate compaction is difficult to obtain without proper equipment. Additionally, the water content in impervious fill can affect the compaction needs. Efforts are typically made at the borrow site to obtain material with optimal moisture; otherwise, if that is not possible, more time may be required for compaction. Another disadvantage is that the clay may be wet, and sub-freezing temperatures may cause the material to freeze in the borrow pit and in the hauling equipment. Be sure to consider weather conditions because it could cause delays in the overall construction effort.

5.9.4.2. Sand. If sand is used, see the recommendations in **paragraph 5.9.5.1** below. Steep slopes without poly coverage on the riverside slope will result in seepage through the dike to outcrop high on the landward slope and may cause the slope to slump and potentially fail if it occurs over an extended period.

5.9.4.3. Silt. Try to avoid material that is primarily silt. If used, apply poly facing to the river slope. In addition to being very erodible, silt, upon wetting, tends to collapse if not properly compacted.

5.9.5. Dike Sections. As previously addressed, the foundation soils and available construction materials generally dictate dike or levee design. Therefore, even under emergency conditions, try to make the embankment compatible with the foundation. Consider any available information on foundation soils from local officials or engineers. The two dike sections cited below are classical and idealized, and usual field conditions depart from them to various degrees. However, the information can potentially lessen serious flood-fight problems during high water. In determining the top width of any type of section, consider whether a revised forecast can potentially require placing additional fill. A top width adequate for construction equipment will facilitate raising the levee. Dike height should provide at least 2 feet of freeboard above forecast flood crest.

5.9.5.1. Sand section. If the dike section is to be made of sand, use a minimum of 1V:3H river slopes. A 1V:4H river slope is preferable and will be less susceptible to erosion, but a 1V:3H slope is considered an adequate minimum for emergency purposes. Use 1V:5H for the landward slope and a 10-foot top width.

5.9.5.2. Clay section. If the dike section is made of clay, use 1V:2-1/2H for both slopes. While 1V:3H slopes are preferable, 1V:1-1/2H is an acceptable minimum for emergency purposes. The bottom width should comply with soil creep ratio criteria, i.e., L (across bottom) should be equal to C x H; where C is the foundation soil creep ratio (**Table 5.5**), and H is the dike height. Refer to UFC 3-220-10 for additional information related to foundation soil creep.

Table 5.5. Creep Ratios on Pervious Foundations.

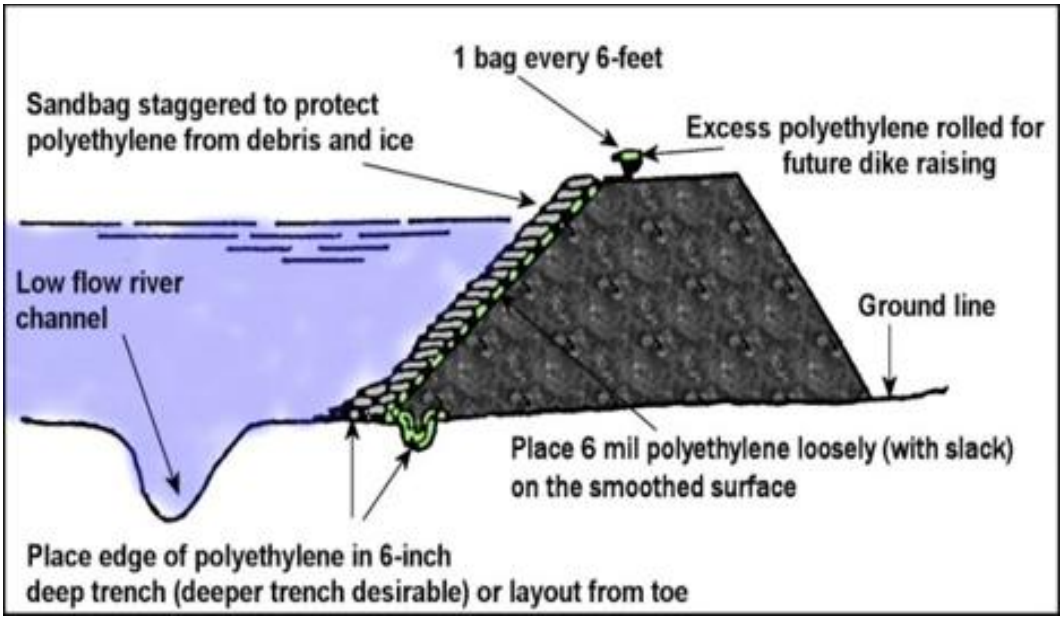
Soil Creep Ratio	
Soil Type	**Critical Creep Ratio**
Very Fine Sand or Silt	18
Fine to Medium Sand	15
Coarse Sand	12
Fine Gravel or Sand and Gravel	9
Coarse Gravel including Cobbles	3.0
Very Hard Clay or Hardpan	1.6

5.9.6. Placement and Compaction. Obtaining proper compaction equipment for a given soil type will be difficult. In most cases, the only compaction may be from hauling and spreading equipment, i.e., construction traffic routed over the fill.

5.9.7. Slope Protection. Methods of protecting embankment slopes from current scour, wave wash, seepage, and debris damage are numerous and varied. During a flood emergency, time, availability of materials, cost, and construction capability preclude the use of all accepted methods of permanent slope protection. Field personnel may expeditiously determine the type and extent of slope protection the emergency barrier requires. Several methods of protection exist which prove highly effective in an emergency. Polyethylene (poly) and sandbags and riprap are two expedient methods addressed here.

5.9.7.1. Poly and sandbags. Experience has shown that a combination of polyethylene (poly) and sandbags is an effective, expedient, and economical method of combating slope attack in an imminent flood situation. Illustrated in **Figure 5.22** is an example of slope protection using poly and sandbags. Ideally, place poly and sandbag protection in the dry. However, many cases of unexpected slope attacks occur during high water. Below are expedient techniques for placement of poly and sandbags under both (wet/dry) scenarios.

Figure 5.22. Polyethylene and Sandbags Slope Protection.



5.9.7.1.1. Placement in the dry. After completion of the embankment, begin by anchoring the poly along the waterside toe:

5.9.7.1.1.1. Excavate a trench along the toe and place poly inside the trench and backfill the trench.

5.9.7.1.1.2. Place poly flat-out away from the toe and push earth over the flap.

5.9.7.1.1.3. Place poly flat-out from the toe and place one or more rows of sandbags over the flap.

5.9.7.1.1.4. After anchoring the poly, unroll the poly up the slope and over the top, leaving enough excess to anchor with sandbags.

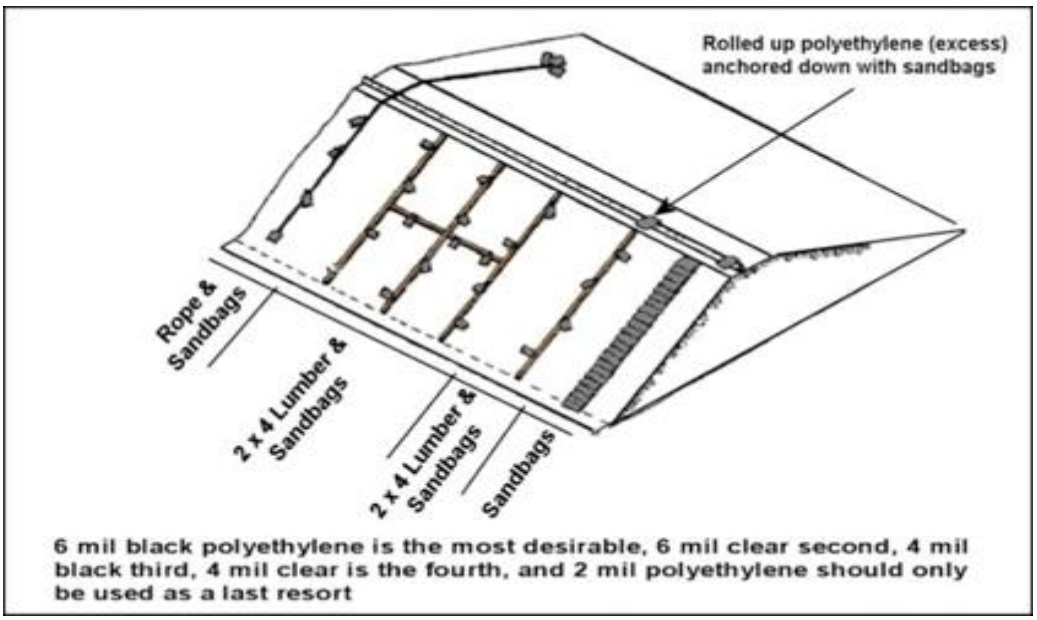
5.9.7.1.1.5. Place the poly from downstream to upstream along the slopes and overlapping at least 2 feet. The poly is now ready for the “hold down” sandbags.

5.9.7.1.1.6. Anchor or hold down poly placed on raised embankments.

5.9.7.1.2. An effective method of anchoring poly is a grid system of sandbags. Workers can construct a grid system faster and with fewer bags and much less labor than a total covering.

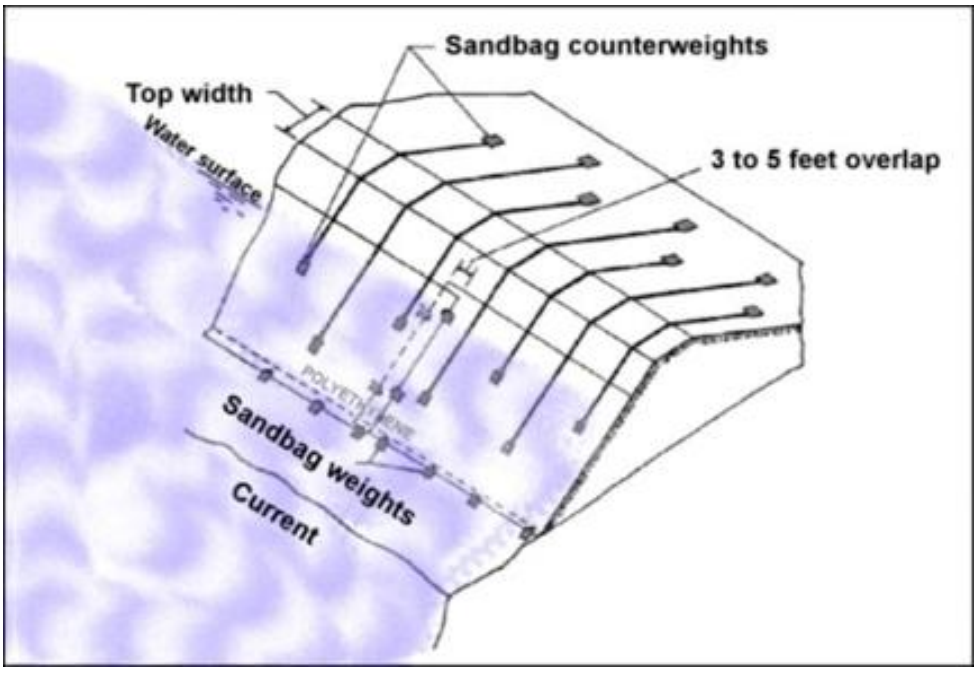
5.9.7.1.3. Various grid systems (**Figure 5.23**) include vertical rows of lapped bags, 2 by 4-inch lumber held down by attached bags, and rows of bags held by a continuous rope tied to each bag. In addition, you can hold down poly using a system of two sandbags tied with rope and the rope saddled over the embankment crown with a bag on each slope. **Note:** If anticipating extremely high velocities of water, heavy debris, or a large amount of ice, use a solid blanket of sandbags over the poly.

Figure 5.23. Anchoring Polyethylene.



5.9.7.1.4. Placement in the wet. In many situations during high water, poly and sandbags placed in the wet must provide emergency protection. Wet placement may also be required to replace or maintain damaged poly or poly displaced by current action. **Figure 5.24** shows a typical section of embankment covered in the wet.

Figure 5.24. Placement of Polyethylene Sheeting in the Wet.



5.9.7.1.4.1. Form the sandbag anchors at the bottom edge of the poly by bunching the poly around a fistful of sand or rock and tying the sandbags to this fist-sized ball.

5.9.7.1.4.2. Use counterweights consisting of two or more sandbags connected by a length of 1/4-inch rope to hold the center portion of the poly down. The number of counterweights depends on the uniformity of the dike slope and current velocity.

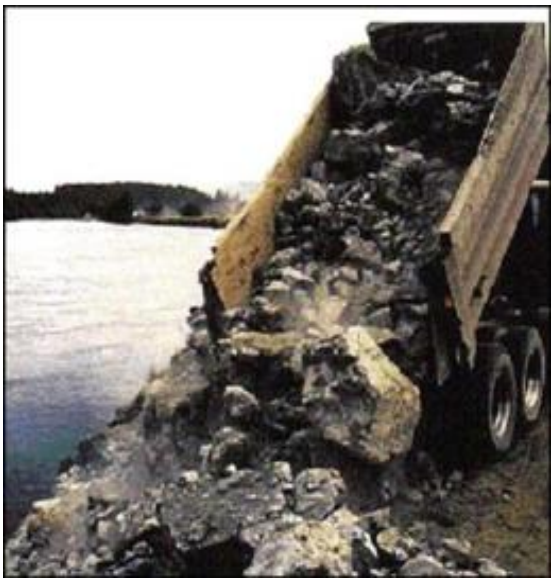
5.9.7.1.4.3. Place the poly by first casting out the poly sheet with the bottom weights and then adding counterweights to slowly sink the poly sheet into place. The poly, in most cases, continues to move down slope until the bottom edge reaches the toe of the slope.

5.9.7.1.4.4. Add sufficient counterweights to ensure that no air voids exist between the poly and the dike face and to keep the poly from flapping or being carried away by the current. Be sure to have enough counterweights prepared for this purpose prior to placement of the sheet.

5.9.7.1.5. Overuse of poly. Workers should not place poly on landward slopes to prevent seepage. It only forces seepage to another exit and may prove detrimental. However, using poly on landside slopes of dikes to prevent rainwater from entering a crack where slope movement has occurred is an option, particularly in fat clay soils. Keeping water out of the cracks caused by slope movements is desirable to prevent lubrication and additional hydrostatic pressure on the slip surface.

5.9.7.2. Riprap. Riprap is a positive means of providing slope protection (**Figure 5.25**) and its use may be an option in a few cases where erosive forces are too large to effectively control by other means. Objections to using riprap when flood fighting are: (1) rather costly; (2) large amount necessary to protect a given area; (3) availability; and (4) little control over its placement, particularly in the wet. In situations where suitable rock is not available within economical hauling distance, soil cement slope protection may be a viable option.

Figure 5.25. Placement of Riprap.



5.9.7.3. Miscellaneous measures. Several other methods of slope protection proved useful in the past. Small groins extending 10 feet or more into the channel were effective in deflecting current away from the dikes. Using log booms to protect dike slopes from debris or ice attack. Cable the logs together and anchor with a deadman in the levee. The boom will float out in the current and, depending on log size, will deflect floating objects. Straw bales pegged into the slope may be successful against wave action as well as straw spread on the slope and overlain with snow fence.

5.9.8. Using Wire/Fabric Earth Filled Barriers. Use earth filled barriers for fast construction of a water dike wall. Reinforce the walls on the opposite side (landside) of the water with earthen material filled with organic/rock mixture for strength.

5.9.9. Other considerations. Potentially, someone could request to locate irrigation and/or drainage ditches near the landside dike toe. This could lead to serious seepage and/or slope stability problems. Work crews should consult with the engineer before constructing drainage ditches, especially if raised dikes are in the immediate area.

5.10. References. Table 5.6 list references in this chapter relating to roadway and airfield drainage structures and methods, and soil drainage characteristics.

Table 5.6. Chapter 5 References.

Expedient Drainage Methods	
UFC 3-201-01	*Civil Engineering*
UFC 3-220-08FA (Inactive)	*Engineering Use of Geotextiles*
UFC 3-260-01	*Airfield and Heliport Planning and Design*
UFC 3-220-10	*Soil Mechanics (DM 7.1)*
TSPWG 3-260-03.02-19	*Airfield Pavement Evaluation Standards and Procedures, Appendix A*
AFH 32-1034	*Materials Testing*

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFI 90-802, *Risk Management*, 20 January 2026

AFI 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*,

25 October 2023

AFI 10-209, *RED HORSE Program*, 7 January 2026

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USDA Publication 9624 1206-SDTDC, *The Plastic Road*, September 1996

ASTM D2487-17, *Standard Practice for Classification of Soils for Engineering Purposes (USCS)*, 12 February 2025

ASTM D6951M-18, *Standard Test Method for Use of the Dynamic Cone Penetrometer in Shallow Pavement Applications*, 3 August 2023

Prescribed Forms

None.

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication.*

Abbreviations and Acronyms

ACE Matting—Army Corps of Engineers Matting

ACP—Airfield Cone Penetrometer

AF—Air Force

AFCEC—Air Force Civil Engineer Center

AFH—Air Force Handbook

AFI—Air Force Instruction

AFPAM—Air Force Pamphlet

AFTTP—Air Force Tactics, Techniques and Procedures

APSB—Asphalt Penetration Surface Binder

ASTM—American Society for Testing and Materials

CBR—California Bearing Ratio

CE—Civil Engineer

CFS—Cubic Feet per Second

CMS—Cement-Modified Soil

DAF—Department of the Air Force

DAFMAN—Department of the Air Force Manual

DCP—Dynamic Cone Penetrometer

DOW—Department of War

ERDC—Engineer Research and Development Center

HDPE—High Density Polyethylene

LCF—Lime-Cement-Fly Ash

LF—Lime-Fly Ash

LL—Liquid Limit

LMS—Lime-Modified Soil

MB—Mechanical Blending

OPR—Office of Primary Responsibility

PCASE—Pavement-Transportation Computer Assisted Structural Engineering

PI—Plasticity Index

PL—Plastic Limit

PPE—Personal Protective Equipment

PSP—Pierce Steel Planking

PVC—Polyvinylchloride

SA—Soil-Asphalt

SC—Soil-Cement

SL—Soil-Lime

SUBMAT—Submersible Matting System

TSPWG—Tri-Service Pavements Working Group

UFC—Unified Facilities Criteria

US—United States

USACE—US Army Corps of Engineers

USAF—United States Air Force

USCS—Unified Soil Classification System

USDA—US Department of Agriculture

USDOT—US Department of Transportation

WBDG—Whole Building Design Guide

Office Symbols

AF/A4—DCS/Logistics, Engineering & Force Protection

AF/A4C—Director of Civil Engineers

Terms

Air Force Civil Engineer Center (AFCEC)—Headquartered at Joint Base San Antonio-Lackland, AFCEC is a 1,900-person primary subordinate unit, assigned to Air Force Materiel Command and attached to the Air Force Installation and Mission Support Center, responsible for providing responsive, flexible full-spectrum installation engineering services. AFCEC missions include facility investment planning, design and construction, operations support, real property management, readiness, energy support, environmental compliance and restoration, and audit assertions, acquisition and program management. The unit conducts its operations at more than 75 locations worldwide.

Base—1. A locality from which operations are projected or supported. 2. An area or locality containing installations which provide logistic or other support. 3. Home airfield or home carrier.

Base Development—The acquisition, development, expansion, improvement, construction and/or replacement of the facilities and resources of a location to support forces.

Beddown—A location at which a deploying unit operates during a contingency. It is usually, but not always, in the area of responsibility.

Contingency—A situation requiring military operations in response to natural disasters, terrorists, subversives, or as otherwise directed by appropriate authority to protect US interest.

Contingency Location—A non-enduring location outside of the United States that supports and sustains operations during named and unnamed contingencies or other operations as directed by appropriate authority and is categorized by mission life-cycle requirements as initial, temporary, or semi-permanent.

Initial Contingency Location—A contingency location occupied by a force in immediate response to a named or unnamed contingency operation and characterized by austere infrastructure and limited services with little or no external support except through Service organic capabilities.

Theater of Operations—An operational area defined by the geographic combatant commander for the conduct or support of specific military operations. Also called TO.

Unexploded Explosive Ordnance—Explosive ordnance which has been primed, fused, armed or otherwise prepared for action, and which has been fired, dropped, launched, projected, or placed in such a manner as to constitute a hazard to operations, installations, personnel, or material and remains unexploded either by malfunction or design or for any other cause.

Attachment 2

ENGINEER REACHBACK AND OTHER LINKS

Table A2.1. Useful Organizational and Product Links.

Organization and Products Links
Air Force Civil Engineer Center (AFCEC): https://www.afcec.af.mil/
CE DASH (AFCEC Technical Support Portal): https://usaf.dps.mil/teams/CEDASH/scripts/homepage/home.aspx
CE Playbooks: https://www.ceplaybooks.com.
Air Force Publications and Forms: https://www.e-publishing.af.mil/
My Learning (Learning Management System): https://lms-jets.cce.af.mil/moodle/login/index.php
Air Force Design Guides (AFDG): https://www.wbdg.org/ffc/af-afcec
Whole Building Design Guide (WBDG): https://www.wbdg.org/
Pavement-Transportation Computer Assisted Structural Engineering (PCASE) Software: https://transportation.erdc.dren.mil/triservice/softwarePCASE.aspx
US Army Corp of Engineers Official Publications, http://www.publications.usace.army.mil/Home.aspx
Unified Facilities Criteria (UFC): https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc
USACE Reach-back Operations Center (UROC): https://uroc.usace.army.mil
Army Publications and Forms: https://armypubs.army.mil/

Attachment 3

BORROW PIT MATERIALS

A3.1. Borrow Pits. Borrow pits usually have unconsolidated material (sand, silt, clay and gravel that is neither stratified nor bound together) that can be removed easily by common earthmoving machinery, generally without blasting. When exploring potential sites, soil samples are usually taken using augers or by digging a test pit. Under hasty conditions, earthmoving equipment may be the preferred option to reach the underlying soil.

A3.1.1. Generally, engineers attempt to develop borrow pits near rivers, streams and related features (erosion or deposits) since they are easy to identify and are normally accessible.

A3.1.2. Alluvial fan deposits of gravel, sand, and silt that forms where flowing water interacts with mountains, hills, and steep canyon walls can provide large amounts of aggregate. Gravel or sandy gravel obtained from alluvial gravel pits is often very clean and free of clay and humus, making it particularly desirable. Bank or hill gravel pits produce clayey gravel or clayey, sandy gravel. This type of material is desirable for surfacing because of its binding qualities.

A3.1.3. In arid and semiarid regions, blowing winds can erode, transport, and deposit unconsolidated sediments. In mountainous regions and continentally glaciated regions, fluvial-glacial deposits can provide large quantities of quality aggregate, therefore, their presence should not necessarily be discounted in preference to fluvial materials.

A3.2. Natural Aggregate Materials. Natural materials from borrow pits may include any combination of the following:

A3.2.1. Gravels. An unconsolidated, natural accumulation of typically rounded rock fragments resulting from erosion, consisting predominately of particles larger than sand such as boulders, cobbles, pebbles, granules, or any combination of these fragments.

A3.2.2. Sands. Sands are unconsolidated or moderately consolidated sedimentary deposits consisting of medium-grained clastics. The material is most commonly composed of quartz. When the term sand is used without qualification, a siliceous composition is implied, but the particles may be of any mineral composition of rock or mineral fragments.

A3.2.3. Silts. Silts are unconsolidated or moderately unconsolidated sedimentary deposits consisting essentially of fine-grained clastics.

A3.2.4. Clays. Clay is a naturally occurring material composed primarily of fine-grained minerals.

A3.2.5. **Table A3.1** lists the primary use and level of operation needed to obtain or extract natural pit materials.

Table A3.1. Excavation and Use of Natural Borrow Pit Materials.

Borrow Pit Material	Primary Use	Operation
Soil, sand, and Gravel	Subgrades, base course, or fill	Medium and light mechanical
Gravel, coarse sand, and clay	Base course, surfacing, or fill	Medium and light mechanical

A3.3. Miscellaneous Pits. Miscellaneous pits; sometimes referred to as dumps may consist of mine tailings, slag, cinder, or similar material which may sometimes be used as surface aggregate.

Attachment 4

COMMON ROCK CLASSIFICATIONS AND PROPERTIES

A4.1. Rock Categories. Rocks are generally classified into three categories: igneous, sedimentary, and metamorphic. These rocks are usually composed of several different minerals. For example, granite is a granular igneous rock composed primarily of quartz and potassium feldspar, but portions of other minerals will likely be present.

A4.1.1. Igneous Rocks. Igneous rocks were formed by the release and cooling of molten rock material from within the earth's mantle. However, the mineral and chemical composition, texture, and mode of occurrence of igneous rocks can vary greatly.

A4.1.2. Sedimentary Rocks. Sedimentary rock results from the consolidation of loose sediment that has accumulated in layers, either by mechanical, chemical, or organic means. Clastic rock consists of mechanically formed fragments of older rock transported from its source and deposited in water or from air or ice. Chemical rock (such as gypsum) is formed by precipitation from solution. Organic rock (e.g., certain limestones) have the remains or secretions of plants or animals.

A4.1.3. Metamorphic Rocks. Metamorphic rock is any rock derived from preexisting rocks by mineralogical, chemical, or structural changes, essentially in the solid state, in response to marked changes in temperature, pressure, shearing stress, and chemical environment, generally at depth in the Earth's crust.

A4.2. Classifications and Properties. The engineering properties of available rocks determine their suitability for use in road and drainage development. See **Figure A4.1** through **Figure A4.7** for a brief look at common rock descriptions, classifications and properties. In all cases, rock properties can and will vary significantly within the same rock grouping. Some rocks may be good aggregates for concrete or asphalt production, while others are more suited as a base course or subbase for road construction, or for use as riprap. **Table A4.1** lists various rock types and their suitability as a base course and subbase aggregate.

Figure A4.1. Common Rock Classifications and Properties (Part 1).

Igneous	Properties
Granite  (Image Copyrighted 2004, Andrew Alden, used with permission)	An igneous rock with an even texture. Granite is gray, pink, or red, with crushing strength ranging from 15,000 to 30,000 pounds per square inch. Unweathered granites are strong and durable rock suitable for bridge piers, sea walls, and foundations of buildings. Its chief defect is that when heated and chilled, the quartz and feldspar grains expand or contract at different rates sometimes causing the rock surface to crumble or peel.
Diorite 	Diorites are a family of rocks that resemble dark granite and found most often in sills, dikes, and small stocks. Unweathered diorites are strong and durable and have an average compressive strength of 28,000 pounds per square inch.
Gabbro 	Gabbro is a dark gray, green, or black granular rock similar in appearance to diorites. Like granite, gabbro is found in batholiths, but it also forms small stocks, dikes, sills, and volcanic necks. Gabbro which is a durable construction material for all purposes, has a high degree of compressive strength (average is 26,000 per square inch), and low absorbability. Gabbro is chiefly used for road materials.

Figure A4.2. Common Rock Classifications and Properties (Part 2).

Felsites 	Felsites are a group of very dense, fine-grained, extrusive igneous rocks. They have dull, stony textures and are composed of quartz and feldspars. Colors range from light or medium gray to pink, brown, yellow, purplish, and light green. Weathering causes Felsites to become brown, rusty and crumble, eventually breaking down completely to become clay. Felsites are generally used as concrete aggregate.
Basalts  (Image Copyrighted 2006, Andrew Alden, used with permission)	Basalts are a group of very dense fine-grained igneous rocks whose colors range from black to dark gray to green to purplish. All basalts contain a great deal of lime, magnesium, and iron. They are a fine-grained equivalent of Gabbro and are closely related to the andesites.
Sedimentary	**Properties**
Limestone  (Image Copyrighted 2008, Andrew Alden, used with permission)	Any rock that contains more than 50 percent calcium carbonate in the form of calcite is considered limestone. When pure, limestones are white or colored, but they are usually colored gray to black by carbon or stained buff, yellow, red, or brown by iron oxides. Crushed limestone is used in the manufacture of Portland cement.

Figure A4.3. Common Rock Classifications and Properties (Part 3).

Dolomite  (Image Copyrighted 2006, Andrew Alden, used with permission)	Sometimes similar in appearance and usage to limestone, dolomite is a calcium magnesium carbonate of varying proportions.
Chert and Flint  (Image Copyrighted 2005, Andrew Alden, used with permission)	Siliceous sediments are usually found in limestone and shale. They are very hard and difficult to drill. Chert can be used as a satisfactory road material.
Rock Salt 	Rock salt is very abundant, soluble in water, and very soft. Rock salt deposits are of no value as a construction material.

Figure A4.4. Common Rock Classifications and Properties (Part 4).

Conglomerate  (Image Copyrighted 2009, Andrew Alden, used with permission)	Conglomerates are composed of cemented gravel of varying sizes.
Breccia  (Image Copyrighted 2008, Andrew Alden, used with permission)	A variety of conglomerate, Breccia is composed of cemented angular fragments and may be used for road material if properly graded or crushed to size. It is usually susceptible to rapid weathering and consequent weakness.
Diamictite  (Image Copyrighted 2008, Andrew Alden, used with permission)	Diamictite is a rock made up of poorly sorted rock fragments of various types and sizes and mixed with finer grained sedimentary material. It is normally a good source of material for earth dams and embankments, but usually not suitable for concrete and bituminous aggregates.

Figure A4.5. Common Rock Classifications and Properties (Part 5).

Sandstone  (Image Copyrighted 2008, Andrew Alden, used with permission)	Sandstone consists of small grains (1/16 millimeter to 2 millimeters) that have been cemented together to form rock. The color of sandstone depends on the nature of the cement. Iron oxides give the red, yellow, and brown shades. Sandstone that splits easily into even slabs is known as flagstone. Flagstone is commonly used as a decorative building material.
Siltstone  (Image Copyrighted 2007, Andrew Alden, used with permission)	Siltstone is similar to sandstone, but composed mainly of cemented particles that are between 1/256 and 1/16 millimeter in diameter.
Shale 	Shales and clays are made up of clay minerals, various oxides, silica, fine particles of ordinary minerals, and a greater or lesser amount of colloidal and organic material.

Figure A4.6. Common Rock Classifications and Properties (Part 6).

Metamorphic	Properties
Gneiss  (Image Copyrighted 2008, Andrew Alden, used with permission)	Gneiss is a banded rock of granite composition, containing quartz, feldspar, and mica. Their banded structure enables the rock to be split into essentially parallel surfaces allowing its use in the construction of tough walls and some road surfaces.
Schists  (Image Copyrighted 2006, Andrew Alden, used with permission)	Schist has much finer texture than gneiss and possesses a well-marked cleavage. Unlike gneisses, their bands are mineralogically alike causing treacherous rock slips in quarries, rock cuts, and in tunnels if unsupported on steep or vertical faces.
Slate  (Image Copyrighted 2006, Andrew Alden, used with permission)	Slate is a fine grained, hard, and dense rock. It splits easily into thin layers that cut across bedding planes. The most important feature of slate is its cleavage, which makes it valuable for roofing. Although not recommended, it can be used as a road material if necessary.

Figure A4.7. Common Rock Classifications and Properties (Part 7).

Quartzite  (Image Copyrighted 2008, Andrew Alden, used with permission)	Primarily, quartzite is metamorphosed sandstone. Other minerals are present as impurities and give the rock a pink, brownish, or red brick color. Similar in appearance to grainy limestone, quartzite is much harder. Quartzite is not used as building stone due to shattering during jointing, but when crushed, becomes good material for concrete work, railroad ballast, and roadwork.
Marble  (Image Copyrighted 2008, Andrew Alden, used with permission)	Marble is the result of the metamorphism of limestone and dolomite. When crushed and used as an aggregate, marble has the same value as limestone.

Table A4.1. Rock Suitability as a Base Course and Subbase Aggregate.

Classification	Rock Type	Use as a Base Course or Subbase
Igneous	Granite	Good
	Gabbrodiorite	Excellent
	Basalt	Excellent
	Felsite	Fair-good
Sedimentary	Conglomerate	Poor
	Sandstone	Fair-good
	Shale	Poor
	Limestone	Good
	Dolomite	Good
	Chert	Poor-fair
Metamorphic	Gneiss	Good
	Schist	Poor-fair
	Slate	Poor
	Quartzite	Fair-good
	Marble	Fair